

34. Ethernet MAC (RMAC)

34.1 Overview

Ethernet MAC (RMAC) is an Ethernet controller that conforms to the definition of the MAC (Media Access Control) layer for Ethernet in the IEEE 802.3 standard. By connecting with a physical-layer LSI chip (PHY-LSI) that complies with the standard, RMAC can transmit and receive Ethernet (IEEE 802.3) frames. RMAC has a single MAC layer interface.

Table 34.1 lists the RMAC specifications and Figure 34.1 shows a block diagram.

Table 34.1 RMAC specifications

Item	Description
Protocol	Flow control conforming with the IEEE 802.3x standard
	Priority-based Flow Control with 802.1Qbb
	Interspersing Express Traffic (preemption) conforming with IEEE 802.3br
	Support for IEEE 1588 time synchronization, Support for IEEE AS and IEEE AS-Rev acceleration
Transfer speed	Supports transfer at 10,100,1000, 2500Mbps
Mode	Full-duplex mode
	Energy Efficient Ethernet (Low Power Idle)
Interface	Supports the IEEE 802.3 standard MII (Media Independent Interface), GMII (Gigabit Media Independent Interface)
	PHY MDIO programming Interface
	Supports internal register interface (32bit data width) Supports internal data interface (64bit data width)
Original function	Configurable frame data padding A variety of flexible RX address filtering modes Configurable CRC pass through for transmit and receive Network statistic counter (IEEE 802.3, ROMN RFC2819)
Magic Packet™	Detection of Magic Packet™*1 and output of a detected signal

Note 1. Magic Packet™ is a trademark of Advanced Micro Devices, Inc.

Table 34.2 References

Title	Contents
IEEE Std 802.3™-2008	Carrier sense multiple access with Collision Detection (CSMA/CD) Access Method and Physical Layer Specifications
IEEE 802.3az™-2010	Media Access Control Parameters, Physical Layers, and Management Parameters for Energy-Efficient Ethernet
IEEE Std 802.3br™-2016	Specification and Management Parameters for Interspersing Express Traffic
P802.1AS-Rev™	Timing and Synchronization for Time Sensitive Applications
P802.1Qbu™	Frame Preemption
IEEE Std 1588™-2008	Precision Clock Synchronization Protocol for Networked Measurement and Control Systems
IEEE Std 802.1BA™-2011	Audio Video Bridging (AVB) Systems
IEEE802.1Qbb	Priority-based Flow Control

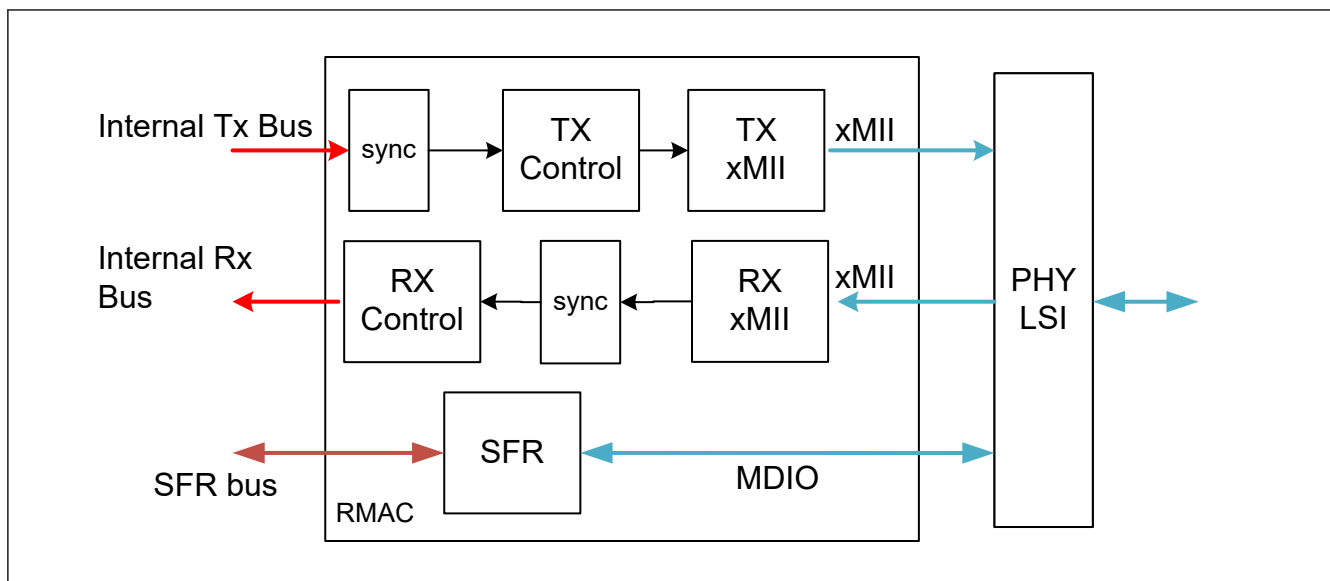


Figure 34.1 RMAC block diagram

Table 34.3 RAC functional blocks

Block name	Function
SFR	Contains all the RMAC SFRs and the bridging system to access PHY registers. RMAC registers are on clk.
Rx Control	Consists of the receive FSM and Data path logic, which disassembles the received Ethernet frame. It controls functions like checking and removing FCS, Destination Address filtering, Flow Control etc.
Tx Control	Consists of the transmission FSM and Data path logic, which facilitates the transmitted Ethernet frame. It controls functions like adding the Preamble and SFD, adding PAD bytes, calculating the FCS etc.
Rx xMII / Tx xMII	Provides the connectivity to the PHY. It supports several types of xMII (MII, GMII).
RMAC	IP described in this documentation.
Sync	Synchronized between bus clock domain and mii clock domain.

34.2 Pin Function

34.2.1 Pin Description

34.2.1.1 PHY Interfaces

PHY interface supports several Media Interfaces. Each media interface has its own interface signals.

PHY interface are described in “IEEE Std 802.3™-2008”

IPG (inter packet gap) of RMAC is defined as follows.

MII TYPE	Tx/Rx	IPG
GMII	Tx	Fixed 12 byte
	Rx	Minimum 12 byte
MII	Tx	Fixed 12 byte
	Rx	Minimum 12 byte

34.2.1.2 MDIO Interface

See MDIO standard “IEEE Std 802.3™-2018”.

34.2.1.3 MHD TX Timestamp Capture Interface

If a timestamp capture is request to RMAC on TX side (by SW or HW), this interface is used to transmit the timestamp. This interface is based on a handshake, the interface time diagram is described in the figure bellow.

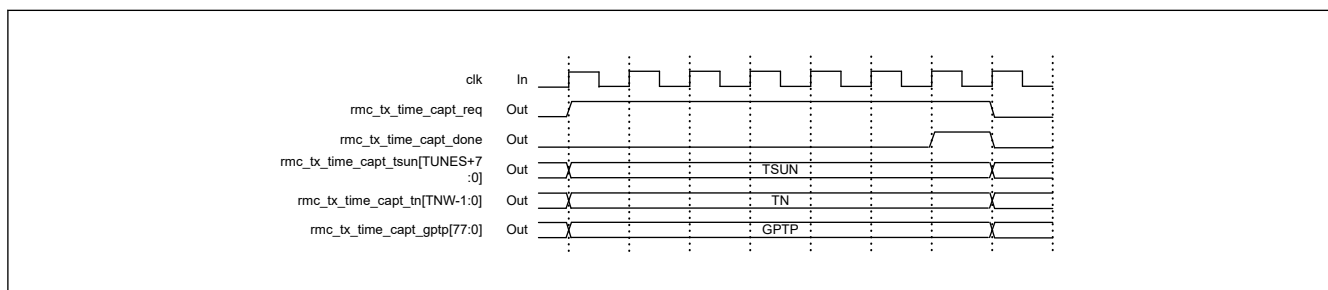


Figure 34.2 TX capture interface time chart

34.3 RMAC Register List

Table 34.4 RMAC registers (1 of 3)

Offset address	Register name	Register symbol	Write access mode
0x0000	PHY Station Management Register	MPSM	D, C, O
0x0004	PHY Interfaces Configuration Register	MPIC	C
0x0008	PHY Interfaces Monitoring Register	MPIM	—
0x0010	I/O Configuration Register	MIOC	D, C, O
0x0020	Transmission Frame Format Configuration Register	MTFFC	C
0x0024	Transmission Pause or PFC Frame Configuration Register	MTPFC	C
0x0028	Transmission Pause or PFC Frame Configuration Register 2	MTPFC2	C, O
0x0030 + 0x04 × t	Transmission Pause or PFC Frame Configuration Register 3 for Priority Group t	MTPFC3t	C
0x0080	Reception General Configuration Register	MRGC	C
0x0084	Reception MAC Address Configuration Register 0	MRMAC0	C
0x0088	Reception MAC Address Configuration Register 1	MRMAC1	C
0x008C	Reception Address Filter Configuration Register	MRAFC	C, O
0x0090	Reception Storm Configuration for E-Frames Register	MRSCE	C, O
0x0094	Reception Storm Configuration for P-Frames Register	MRSCP	C, O
0x0098	Reception Storm Counter Configuration Register	MRSCC	D, C, O
0x009C	Reception Frame Size Configuration for E-Frames Register	MRFSCC	C
0x00A0	Reception Frame Size Configuration for P-Frames Register	MRFSCP	C
0x00A4	Timestamp Reception Configuration Register	MTRC	C
0x00AC	Reception Pause or PFC Frame Monitoring Register	MRPFM	—
0x0100 + 0x04 × t	PTP Filtering Register Configuration Register t	MPFCt	C
0x0180	Link Verification Configuration Register	MLVC	O
0x0184	Energy Efficient Ethernet Configuration Register	MEEEC	D, C, O
0x0188	Loopback Configuration Register	MLBC	C
0x0200	Error Interrupt Status Register	MEIS	D, C, O

Table 34.4 RMAC registers (2 of 3)

Offset address	Register name	Register symbol	Write access mode
0x0204	Error Interrupt Enable Register	MEIE	D, C, O
0x0208	Error Interrupt Disable Register	MEID	D, C, O
0x0210	Monitoring Interrupt Status Register 0	MMIS0	D, C, O
0x0214	Monitoring Interrupt Enable Register 0	MMIE0	D, C, O
0x0218	Monitoring Interrupt Disable Register 0	MMID0	D, C, O
0x0220	Monitoring Interrupt Status Register 1	MMIS1	D, C, O
0x0224	Monitoring Interrupt Enable Register 1	MMIE1	D, C, O
0x0228	Monitoring Interrupt Disable Register 1	MMID1	D, C, O
0x0230	Monitoring Interrupt Status Register 2	MMIS2	D, C, O
0x0234	Monitoring Interrupt Enable Register 2	MMIE2	D, C, O
0x0238	Monitoring Interrupt Disable Register 2	MMID2	D, C, O
0x0300	Manual Pause Frame Transmit Counter Register	MMPFTCT	—
0x0304	Automatic Pause Frame Transmit Counter Register	MAPFTCT	—
0x0308	Pause Frame Receive Counter Register	MPFRCT	—
0x030C	False Carrier Indication Counter Register	MFCICT	—
0x0310	Energy Efficient Ethernet Counter Register	MEEECT	—
0x0320 + 0x04 × j	Manual PFC Frame Transmit Counter Register	MMPCFTCTt	—
0x0330 + 0x04 × j	Automatic PFC Frame Transmit Counter Register	MAPCFTCTt	—
0x0340 + 0x04 × j	PFC Frame Receive Counter Register	MPCFRCTt	—
0x0360	Receive Overflow Counter Register	MROVFC	—
0x0408	Received Good Frame Counter E-Frames Register	MRGFCE	—
0x040C	Received Good Frame Counter P-Frames Register	MRGFCP	—
0x0410	Received Good Broadcast Frame Counter Register	MRBFC	—
0x0414	Received Good Multicast Frame Counter Register	MRMFC	—
0x0418	Received Good Unicast Frame Counter Register	MRUFC	—
0x041C	Received PHY Error Frame Count Register	MRPEFC	—
0x0420	Received Nibble Error Frame Count Register	MRNEFC	—
0x0424	Received FCS/mCRC Error Frame Count Register	MRFMEFC	—
0x0428	Received Final Fragment Missing Error Frame Count Register	MRFFMEFC	—
0x042C	Received C-Fragment Count Error Frame Count Register	MRCFCEFC	—
0x0430	Received Fragment Count Error Frame Count Register	MRFCEFC	—
0x0434	Received RMAC Filter Error Frame Count Register	MRRCFEFC	—
0x0438	Received Frame Count Register	MRFC	—
0x043C	Received Good Undersize Error Frame Count Register	MRGUEFC	—
0x0440	Received Bad Undersize Error Frame Count Register	MRBUEFC	—
0x0444	Received Good Oversize Error Frame Count Register	MRGOEFC	—
0x0448	Received Bad Oversize Error Frame Count Register	MRBOEFC	—
0x044C	Received Byte Counter E-Frames Upper Side Register	MRXBCEU	—
0x0450	Received Byte Counter E-Frames Lower Side Register	MRXBCEL	—
0x0454	Received Byte Counter P-Frames Upper Side Register	MRXBCPU	—

Table 34.4 RMAC registers (3 of 3)

Offset address	Register name	Register symbol	Write access mode
0x0458	Received Byte Counter P-Frames Lower Side Register	MRXBCPL	—
0x0508	Transmitted Good Frame Counter E-Frames Register	MTGFCE	—
0x050C	Transmitted Good Frame Counter P-Frames Register	MTGFCP	—
0x0510	Transmitted Broadcast Frame Counter Register	MTBFC	—
0x0514	Transmitted Multicast Frame Counter Register	MTMFC	—
0x0518	Transmitted Unicast Frame Counter Register	MTUFC	—
0x051C	Transmitted Error Frame Counter Register	MTEFC	—
0x0520	Transmitted Byte Counter E-Frames Upper Side Register	MTXBCEU	—
0x0524	Transmitted Byte Counter E-Frames Lower Side Register	MTXBCEL	—
0x0528	Transmitted Byte Counter P-Frames Upper Side Register	MTXBCPU	—
0x052C	Transmitted Byte Counter P-Frames Lower Side Register	MTXBCPL	—

- Note:
- Base address: RMAC0 = 0x403C_B000, RMAC1 = 0x403C_D000, RMAC0_NS = 0x503C_B000, RMAC1_NS = 0x503C_D000
 - All registers can be read in any mode.
 - Write Access Mode:
 - Any: Register can be accessed in any mode.
 - D: Register can be accessed in DISABLE mode.
 - C: Register can be accessed in CONFIG mode.
 - O: Register can be accessed in OPERATION mode.

34.4 RMAC Register Descriptions

34.4.1 RMAC Function Registers

34.4.1.1 MPSM : PHY Station Management Register

Base address: $\text{RMACm} = 0x403C_B000 + 0x2000 \times m$ ($m = 0, 1$)
 $\text{RMACm_NS} = 0x503C_B000 + 0x2000 \times m$ ($m = 0, 1$)

Offset address: 0x0000

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	PRD[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	POP[1:0]		PRA[4:0]				PDA[4:0]				MFF		—	PSME	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PSME*2*3*4	PHY Station Management Enable 0: PHY register access is not on progress. 1: PHY register access is requested and on progress.	R/W*1
1	—	This bit is read as 0.	R

Bit	Symbol	Function	R/W
2	MFF	Management Frame Format 0: Normal management frame format defined in Clause 22 of IEEE802.3 Describe "MDIO" in this document. 1: Extension management frame format defined in Clause 45 of IEEE802.3 Describe "eMDIO" in this document.	R/W
7:3	PDA[4:0]	PHY Device Address These bits specify following value according to MPSM.MFF. See 8.5 for more details. • Case of MPSM.MFF = 0: The PHY Address field • Case of MPSM.MFF = 1: The PHY Address field	R/W
12:8	PRA[4:0]	PHY Register Address These bits specify following value according to MPSM.MFF. See 8.5 for more details. • Case of MPSM.MFF = 0: Register Address • Case of MPSM.MFF = 1: Device address	R/W
14:13	POP[1:0]	PHY Operation Code These bits specify the operation code. See 8.5 for more details. These bits specify following value according to MPSM.MFF. • Case of MPSM.MFF = 0 0 1: Write Frame 1 0: Read Frame Others: Reserved (0 0 or 1 1 should not be set when MPSM.MFF = 0) • Case of MPSM.MFF = 1 0 0: Address Frame 0 1: Write Frame 1 0: Post-read-increment-address frame 1 1: Read Frame	R/W
15	—	This bit is read as 0.	R
31:16	PRD[15:0]	PHY Register Data These bits specify following value according to MPSM.MFF. See 8.5 for more details. • Case of MPSM.MFF = 0: Data field of the PHY register • Case of MPSM.MFF = 1: Data/Address field of the PHY register	R/W ^{*1}

Note 1. Read value differs from written value.

Note 2. Accessing to MPSM register while this bit is asserted has no effect.

Note 3. This bit cannot be set when MPIC.PSMCS = 0.

Note 4. Before resetting RMAC, this register should always be checked. If this register is set, no reset should be performed.

Provides access to the PHY-LSI internal registers via the MDIO.

PSME bit (PHY Station Management Enable)

Set this bit to access PHY registers.

[Setting condition]

SW: By writing 1 to this register. It starts phy access.

[Clearing condition]

HW: When PHY register access is completed, this bit returns automatically to 0.

PRD[15:0] bits (PHY Register Data)

Write operation: HW send register value to PHY in write operation.

Read operation: HW write read data to this register from PHY in read operation.

34.4.1.2 MPIC : PHY Interfaces Configuration Register

Base address: RMACm = 0x403C_B000 + 0x2000 × m (m = 0, 1)
 RMACm_NS = 0x503C_B000 + 0x2000 × m (m = 0, 1)

Offset address: 0x0004

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	PSMCT[2:0]			—	PSMHT[2:0]			PSMD P	PSMCS[6:0]						
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	PLSP P	PIPP	PIP	—	—	LSC[2:0]			PIS[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	PIS[2:0]	PHY Interface Select These bits select the xMII interface connected to the PHY. 0 0 0: MII 0 1 0: GMII 1 0 0: Reserved 1 0 1: Reserved 1 1 0: Reserved 1 1 1: Reserved	R/W
5:3	LSC[2:0]	Link Speed Configuration 0 0 0: 10mbps 0 0 1: 100mbps 0 1 0: 1gbps 0 1 1: Reserved 1 1 0: Reserved 1 1 1: Reserved	R/W
7:6	—	These bits are read as 0.	R
8	PIP	PHY Interrupt Polarity This bit sets the active level of the PHY interrupt pin (rmc_phy_int). For the active level, refer to the specifications of PHY-LSI to be connected. 0: PHY interrupt is active low (default). 1: PHY interrupt is active high.	R/W
9	PIPP	PHY Interrupt Pin Plugged 0: Unplugged 1: Plugged	R/W
10	PLSPP	PHY Link Status Pin Plugged 0: Unplugged 1: Plugged	R/W
15:11	—	These bits are read as 0.	R
22:16	PSMCS[6:0] ^{*2}	PHY Station Management Clock Selection 0000000: MDC clock is off 0000001: clk / ((1+1)*2) 0000010: clk / ((2+1)*2) ⋮ 1111110: clk / ((126+1)*2) 1111111: clk / ((127+1)*2)	R/W ^{*1}
23	PSMDP ^{*2}	PHY Station Management Disable Preamble 0: PHY Station Management Preamble enabled 1: PHY Station Management Preamble disabled	R/W ^{*1}
26:24	PSMHT[2:0] ^{*2*3}	PHY Station Management Hold Time Adjustment 0 0 0: No adjusted hold time (i.e. change MDO at the rising edge of MDC) 0 0 1: 1 clk cycle extra hold time ⋮ 1 1 1: 7 clk cycle extra hold time	R/W ^{*1}

Bit	Symbol	Function	R/W
27	—	This bit is read as 0.	R
30:28	PSMCT[2:0]*2*3	PHY Station Management Capture Time Adjustment The configured value will shift the data capture timing of MDI with respect to the rising edge of MDC clock. 0 0 0: No adjusted capture time (i.e. capture MDI at the rising edge of MDC) 0 0 1: Capture before 1 clk cycle : 1 1 1: Capture before 7 clk cycle	R/W*1
31	—	This bit is read as 0.	R

Note 1. Read value differs from written value.

Note 2. SW: This register cannot be set if MPSM.PSME is set.

Note 3. SW: The configured value must be less than MDC clock cycle (less than MPIC.PSMCS*2)

Configures MDIO and xMII interface.

PSMCS[6:0] bits (PHY Station Management Clock Selection)

These bits configure the cycle of MDC clock. MDC clock is divided from clk clock. The duty cycle of MDC clock is 50%.

See to section 8.4 for detail.

PSMDP bit (PHY Station Management Disable Preamble)

This bit could cut the preamble pattern (sequence of 32 contiguous bits). When it is asserted, the preamble pattern is skipped and the management frame starts with the ST (Start of Frame) pattern.

PSMHT[2:0] bits (PHY Station Management Hold Time Adjustment)

The configured value will shift the data change timing of MDO with respect to the rising edge of MDC clock.

34.4.1.3 MPIM : PHY Interfaces Monitoring Register

Base address: RMACm = 0x403C_B000 + 0x2000 × m (m = 0, 1)
RMACm_NS = 0x503C_B000 + 0x2000 × m (m = 0, 1)

Offset address: 0x0008

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	LPIA	PLS
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PLS	PHY Link Status Flag 0: PHY link signal is at the low level. 1: PHY link signal is at the high level.	R
1	LPIA	LPI Active This bit indicates that LPI is active on RX side. 0: No LPI 1: LPI active	R
31:2	—	These bits are read as 0.	R

PLS bit (PHY Link Status Flag)

This bit indicates PHY link state. It could be used only when the link status pin (rmc_phy_link) is connected to the link signal from PHY-LSI.

For the active level, refer to the specifications of PHY-LSI to be connected.

LPIA bit (LPI Active)

[Setting condition]

HW: When LPI code is appeared on detected on xMII Rx data signals (rmc_rx_data_valid_in, rmc_rx_data_data_in[31:0], rmc_rx_data_err_in). For more details, see xMII specification.

[Clearing condition]

HW: When LPI code is not appeared on detected on xMII Rx data signals.

34.4.1.4 MIOC : I/O Configuration Register

Base address: RMACm = 0x403C_B000 + 0x2000 × m (m = 0, 1)
 RMACm_NS = 0x503C_B000 + 0x2000 × m (m = 0, 1)

Offset address: 0x0010

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	MIOC0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	MIOC0	I/O Configuration 0 0: Normal mode. clk_phy_tx and clk_phy_rx are provided. 1: Emergency clock recovery mode. clk_phy_tx and clk_phy_rx are provided free-run clock (clk_common). (So transmission is not possible in this state.)	R/W
31:1	—	These bits are read as 0.	R

Note: SW can write 1 to bit[0] when clock derivation from PHY is unexpectedly stopped.
 In this case, write 1 to bit[0] to flush all internal condition in RMAC. After waiting to flush and change DISABLE mode, write 0 to bit[0] and change RESET mode.

34.4.1.5 MTFFC : Transmission Frame Format Configuration Register

Base address: RMACm = 0x403C_B000 + 0x2000 × m (m = 0, 1)
 RMACm_NS = 0x503C_B000 + 0x2000 × m (m = 0, 1)

Offset address: 0x0020

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	FCM	DPAD
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	DPAD ^{*1*2}	Data Padding Disable Disables the automatic padding of too small transmit frames (less than 60 bytes) with no FCS If a too small transmit frame (less than 60 bytes) with no FCS is received when this bit is set, RMAC adds FCS and sends such a frame. 0: Padding is inserted. 1: Padding is not inserted.	R/W

Bit	Symbol	Function	R/W
1	FCM	Flow Control Mode This bit specifies the flow control mode 0: PAUSE 1: PFC	R/W
31:2	—	These bits are read as 0.	R

Note 1. HW: A Verify frame, a Pause frame and a PFC frame are not affected by this bit.

Note 2. HW: Padding is not executed when "fcs included(FI==1)" indication is applied by upper layer.

The register is used to configure the frame format to be transmitted.

34.4.1.6 MTPFC : Transmission Pause or PFC Frame Configuration Register

Base address: RMACm = 0x403C_B000 + 0x2000 × m (m = 0, 1)
 RMACm_NS = 0x503C_B000 + 0x2000 × m (m = 0, 1)

Offset address: 0x0024

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	PFRLV[4:0]					PFM	—	—	PFRT[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	PT[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	PT[15:0] ^{*1}	Pause Time These bits are common bit for PAUSE and PFC. 0x0000 Pause frame disable : 0x0001 512 × 1 bit-time : : 0xFFFF 512 × 65,535 bit-time :	R/W
23:16	PFRT[7:0] ^{*1*2}	Pause Frame Retransmission Time These bits are common bit for PAUSE and PFC. 0x00: 512 × 0 bit-time 0x01: 512 × 1 bit-time : 0xFF: 512 × 255 bit-time	R/W
25:24	—	These bits are read as 0.	R
26	PFM ^{*1}	Pause Frame Mode This bit is common bit for PAUSE and PFC. 0: Automatic pause frame. Sending a pause frame will be triggered by a hardware pause request. 1: Manual pause frame. Sending a Pause frame will be triggered by MTPFC2.MPFR or MTPFC2.MPFCFR.	R/W
31:27	PFRLV[4:0] ^{*1}	Pause or PFC Frame Retry Limit Value 0x00: No interrupt 0x01: Interrupt at 1 pause frame : 0x1F: Interrupt at 31 pause frames	R/W

Note 1. SW: This register is only writable in CONFIG mode

Note 2. SW: This register should be set to a smaller value than one of MTPFC.PT.

The register is used to configure the TIME parameter value for pause frames.

PT[15:0] bits (Pause Time)

These bits set the TIME parameter value for pause frames.

- The value of these bits is set to Time field of a PAUSE or PFC frame.
- If the value of these bits is 0, PAUSE or PFC is not sent.
- The unit for the setting is 512 bit-time. 1 bit-time is equal to 1ns in 1000 Mbps, 10 ns in 100 Mbps, 100 ns in 10 Mbps.

PFRT[7:0] bits (Pause Frame Retransmission Time)

These bits set the TIME parameter value of pause frame retransmission.

- A pause frame will be retransmitted when the previous frame has been sent more than $(PT-PFRT) \pm 10$ cycles bit-time ago.
- The unit for the setting is 512 bit-time. 1 bit-time is equal to 1ns in 1000 Mbps, 10 ns in 100 Mbps, 100 ns in 10 Mbps.

PFRLV[4:0] bits (Pause or PFC Frame Retry Limit Value)

Value of pause frame sent without pause frame request signal deassertion from which at which RMAC will emit an interrupt.

This bit is common bit for PAUSE and PFC.

The number of pause (pfc) frames in this field indicates the total number of transmission attempt.

34.4.1.7 MTPFC2 : Transmission Pause or PFC Frame Configuration Register 2

Base address: RMACm = 0x403C_B000 + 0x2000 × m (m = 0, 1)
 RMACm_NS = 0x503C_B000 + 0x2000 × m (m = 0, 1)

Offset address: 0x0028

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	MPFR	PFTTZ
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	MPFCFR1 to MPFCFR0		—	—	—	—	—	—	PFCTTZ1 to PFCTTZ0	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	PFCTTZ1 to PFCTTZ0 ^{*1*2}	PFC Frame n Transmission with TIME = 0 (n = 0, 1) 0: The transmission of PFC frames with TIME value zero is disabled. 1: The transmission of PFC frames with TIME value zero is enabled.	R/W
7:2	—	These bits are read as 0.	R
9:8	MPFCFR1 to MPFCFR0 ^{*3}	Manual PFC Frame n Request (n = 0, 1)	R/W
15:10	—	These bits are read as 0.	R
16	PFTTZ ^{*2}	Pause Frame Transmission with TIME = 0 0: The transmission of Pause frames with TIME value zero is disabled. 1: The transmission of Pause frames with TIME value zero is enabled.	R/W
17	MPFR ^{*3}	Manual Pause Frame Request	R/W
31:18	—	These bits are read as 0.	R

Note 1. SW: This bit has no effect for Manual PFC frame transmission.

Note 2. SW: This register is only writable in CONFIG mode.

Note 3. SW: This register is only writable in OPERATION mode.

The register is used to set the TIME parameter value for PFC frames.

PFCTTzn bits (PFC Frame n Transmission with TIME = 0 (n = 0, 1))

This bit enables the transmission of automatic PFC frames with TIME value zero.

It is transmitted when the MHD negated the PFC request in automatic PFC frame mode or when the software negates MTPFC2.

MPFCFR register in manual PFC frame mode and the remaining PFC time is more than PFRT.

MPFCFRn bits (Manual PFC Frame n Request (n = 0, 1))

Writing 1 to this bit request manual pfc frame.

If a pfc frame is requested so fast that the hardware did not have time to send the previous one (because a long frame is ongoing or because the software requests too many pfc frames), the hardware will ignore the pfc frame requests.

After completion of non-pause frame transmission the next attempt for PFC frame will be only when timer expires if request for PFC frame is still high.

[Setting condition]

SW: Writing 1 to this bit will set.

[Clearing condition]

SW: Writing 0 to this bit will clear.

HW: Going out of OPERATION mode will clear this register.

PFTTZ bit (Pause Frame Transmission with TIME = 0)

This bit is used for PAUSE frame only.

This bit enables the transmission of automatic Pause frames with TIME value zero. It is transmitted when the MHD negates the pause request in automatic pause frame mode or when the software negates MTPFC2.MPFR register in manual pause frame mode and the remaining pause time is more than PFRT.

This bit has no effect for Manual Pause frame transmission.

MPFR bit (Manual Pause Frame Request)

This bit is used for a PAUSE frame only.

Writing 1 to this bit requests a manual pause frame.

If a pause frame is requested so fast that the hardware has time to send the previous one (because a long frame is ongoing or because the software requests too many pause frames), the hardware will ignore the pause frame requests.

[Setting condition]

SW: Writing 1 to this bit will set the bit

[Clearing condition]

SW: Writing 0 to this bit will clear the bit.

HW: Going out of OPERATION mode will clear this register.

34.4.1.8 MTPFC3t : Transmission Pause or PFC Frame Configuration Register 3 for Priority Group t

Base address: RMACm = 0x403C_B000 + 0x2000 × m (m = 0, 1)
 RMACm_NS = 0x503C_B000 + 0x2000 × m (m = 0, 1)

Offset address: 0x0030 + 0x4 × t

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	PFCPG7 to PFCPG0							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	PFCPG7 to PFCPG0 ^{*1}	PFC Priority Enable n (n = 0 to 7) 0: Priority x is not assigned to priority group t. 1: Priority x is assigned to group t.	R/W
31:8	—	These bits are read as 0.	R

Note 1. SW: This register is only writable in CONFIG mode.

The register is used to set the TIME parameter value for PFC frames.

PFCPGn bits (PFC Priority Enable n (n = 0 to 7))

Function when MTFFC.FCM=1(PFC)

- When PFC frame transmission request (hardware request for automatic request, MTPFC2.MPFCFR for manual request) is set, PFC frame is generated. Then, the Field of Class-Enable vector is set to this register value.
- When all bits of PFCPGn = 0, the transmission request is ignored.

Function when MTFFC.FCM = 0 (PAUSE)

- PAUSE frame transmission request is set by hardware pause request or MTPFC2.MPFR.
- When PFCPG0 = 0, the transmission request is ignored.

34.4.1.9 MRGC : Reception General Configuration Register

Base address: RMACm = 0x403C_B000 + 0x2000 × m (m = 0, 1)
 RMACm_NS = 0x503C_B000 + 0x2000 × m (m = 0, 1)

Offset address: 0x0080

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	PFCRC7 to PFCRC0							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	RFCF E	MPDE	PFRT Z	PFRC	RCPT
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	RCPT	Receive CRC Pass Through This bit selects to pass FCS field of the received frame. The CRC is checked regardless of this configuration. 0: Correct FCS is not passed to the MHD. Incorrect FCS is passed to the MHD. 1: Both Correct and Incorrect FCS are passed to the MHD.	R/W
1	PFR ^{C*1*2}	Pause Frame Reception Control This bit enables the Flow Control (Pause frame detection) function. 0: Flow control for the receiving port is disabled. 1: Flow control for the receiving port is enabled.	R/W
2	PFR ^{TZ}	Pause or PFC Frame Reception with Time = 0 When this bit is set to 1, the reception of Pause or PFC frame with the Timer value 0 releases the transmission wait state. 0: Reception of Pause or PFC frames with the TIME value 0 is disabled. 1: Reception of Pause or PFC frames with the TIME value 0 is enabled.	R/W
3	MPDE	Magic Packet™ Detection Enable This bit enables Magic Packet™ detection by hardware to allow activation via the Ethernet connection. (Only for e-frames) 0: Magic Packet™ detection is disabled. 1: Magic Packet™ detection is enabled.	R/W
4	RFCFE	Reception Flow Control Forwarding Enable This bit enables the Flow Control Frame (PAUSE or PFC frame) forwarding function. 0: Flow Control Frame (PAUSE or PFC frame) is not forward to Rx MHD I/F. 1: Flow Control Frame (PAUSE or PFC frame) is forward to Rx MHD I/F.	R/W
15:5	—	These bits are read as 0.	R
23:16	PFCRC7 to PFCRC0 ^{*1*3}	PFC Frame Reception Control n (n = 0 to 7) This bit enables the Flow Control (Pause frame detection) function. 0: Flow control for the receiving priority level x (8) is disabled. 1: Flow control for the receiving priority level x (8) is enabled.	R/W
31:24	—	These bits are read as 0.	R

Note 1. HW: Only for e-frames

Note 2. SW: When any bits of MRGC.PFCRC is set, SW should not set this bit.

Note 3. SW: When MRGC.PFR is set, SW should not set this bit.

The register is used to configure the frame reception.

34.4.1.10 MRMAC0 : Reception MAC Address Configuration Register 0

Base address: RMAC_m = 0x403C_B000 + 0x2000 × m (m = 0, 1)
 RMAC_m_NS = 0x503C_B000 + 0x2000 × m (m = 0, 1)

Offset address: 0x0084

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	MAU[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	MAU[15:0]	MAC Address Upper Part These bits are used to set the 16 higher-order bits of the MAC address. For example, if the MAC address is 01-23-45-67-89-AB (hexadecimal), set H'0123 in this register.	R/W
31:16	—	These bits are read as 0.	R

Specifies the 16 higher-order bits of the 48-bit unicast MAC address. This register is also used as source address filed when the IP transmits a Pause frame or a PFC frame.

34.4.1.11 MRMAC1 : Reception MAC Address Configuration Register 1

Base address: $\text{RMACm} = 0x403C_B000 + 0x2000 \times m$ ($m = 0, 1$)
 $\text{RMACm_NS} = 0x503C_B000 + 0x2000 \times m$ ($m = 0, 1$)

Offset address: 0x0088

Bit position: 31

0

Bit field:

MAL[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	MAL[31:0]	MAC Address Lower Part These bits are used to set the 32 lower-order bits of the MAC address. For example, if the MAC address is 01-23-45-67-89-AB (hexadecimal), set H'456789AB in this register.	R/W

The register is to use to specify the 32 lower-order bits of the 48-bit unicast MAC address. This register is also used as source address filed when the IP transmits a Pause frame or a PFC frame.

34.4.1.12 MRAFC : Reception Address Filter Configuration Register

Base address: $\text{RMACm} = 0x403C_B000 + 0x2000 \times m$ ($m = 0, 1$)
 $\text{RMACm_NS} = 0x503C_B000 + 0x2000 \times m$ ($m = 0, 1$)

Offset address: 0x008C

Bit position: 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16

Bit field:

—	—	—	—	—	MSAR EP	NSAR EP	SDSF REP	NDAR EP	BCAC P	MCAC P	BSTE NP	MSTE NP	BCEN P	MCEN P	UCEN P
---	---	---	---	---	------------	------------	-------------	------------	-----------	-----------	------------	------------	-----------	-----------	-----------

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1

Bit position: 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bit field:

—	—	—	—	—	MSAR EE	NSAR EE	SDSF REE	NDAR EE	BCAC E	MCAC E	BSTE NE	MSTE NE	BCEN E	MCEN E	UCEN E
---	---	---	---	---	------------	------------	-------------	------------	-----------	-----------	------------	------------	-----------	-----------	-----------

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1

Bit	Symbol	Function	R/W
0	UCENE	Unicast Reception Enable E-Frames This bit enables the unicast filtering based on MAC address (MRMAC0/1). 0: Unicast reception disabled 1: Unicast reception enabled	R/W
1	MCENE	Multicast Reception Enable E-Frames This bit enables the multicast filtering. 0: Multicast reception disabled 1: Multicast reception enabled	R/W
2	BCENE	Broadcast Reception Enable E-Frames This bit enables the broadcast filtering. 0: Broadcast reception disabled 1: Broadcast reception enabled	R/W
3	MSTENE ^{*1}	Multicast Storm Filter Reception Enable E-Frames This bit enables the multicast storm filtering. When it is asserted, Multicast frames are rejected after the number of received multicast e-frames reaches the configured threshold (MRSCE.CMFE). 0: Multicast storm filter reception disabled 1: Multicast storm filter reception enabled	R/W

Bit	Symbol	Function	R/W
4	BSTENE* ²	Broadcast Storm Filter Reception Enable E-Frames This bit enables the broadcast storm filtering. When it is asserted, Broadcast frames are rejected after the number of received broadcast e-frames reaches to the configured threshold (MRSCE.CBFE). 0: Broadcast storm filter reception disabled 1: Broadcast storm filter reception enabled	R/W
5	MCACE* ³	Multicast Storm Autoclear E-Frames This bit enables auto-clear of multicast frame storm counter. When it is asserted, multicast frame storm counter is cleared by detecting a frame with non-multicast destination address. After the counter is cleared, the IP can receive multicast frame again. 0: Multicast Storm Autoclear disabled 1: Multicast Storm Autoclear enabled	R/W
6	BCACE	Broadcast Storm Autoclear E-Frames This bit enables auto-clear of broadcast frame storm counter. When it is asserted, broadcast frames storm counter is cleared by detecting a frame with non-broadcast destination address. After the counter is cleared, the IP can receive Broadcast frame again. 0: Broadcast Storm Autoclear disabled 1: Broadcast Storm Autoclear enabled	R/W
7	NDAREE	Null Destination Address Reception Enable E-Frames This bit enables the reception of e-frames with destination MAC address null. 0: disable 1: enable	R/W
8	SDSFREE	Same DA and SA Frames Reception Enable E-Frames This bit enables the reception of e-frame with equal destination address and source address. 0: disable 1: enable	R/W
9	NSAREE	Null Source Address Reception Enable E-Frames This bit enables the reception of e-frames with destination MAC address null. 0: disable 1: enable	R/W
10	MSAREE	Multicast Source Address Reception Enable E-Frames This bit enables the reception of e-frame with multicast and broadcast source address 0: disable 1: enable	R/W
15:11	—	These bits are read as 0.	R
16	UCENP	Unicast Reception Enable P-Frames This bit enables the unicast filtering based on MAC address (MRMAC0/1). 0: Unicast reception disabled 1: Unicast reception enabled	R/W
17	MCENP	Multicast Reception Enable P-Frames This bit enables the multicast filtering. 0: Multicast reception disabled 1: Multicast reception enabled	R/W
18	BCENP	Broadcast Reception Enable P-Frames This bit enables the broadcast filtering. 0: Broadcast reception disabled 1: Broadcast reception enabled	R/W
19	MSTENP* ¹	Multicast Storm Filter Reception Enable P-Frames This bit enables the multicast storm filtering. When it is asserted, Multicast frames are rejected after the number of received multicast p-frames reaches the configured threshold (MRSCP.CMFP). 0: Multicast storm filter reception disabled 1: Multicast storm filter reception enabled	R/W
20	BSTENP* ²	Broadcast Storm Filter Reception Enable P-Frames This bit enables the broadcast storm filtering. When it is asserted, Broadcast frames are rejected after the number of received broadcast p-frames reaches the configured threshold (MRSCP.CBFP). 0: Broadcast storm filter reception disabled 1: Broadcast storm filter reception enabled	R/W

Bit	Symbol	Function	R/W
21	MCACP*3	Multicast Storm Autoclear P-Frames This bit enables auto-clear of multicast frame storm counter. When it is asserted, multicast frame storm counter is cleared by detecting a frame with non-multicast destination address. After the counter is cleared, the bit can receive multicast frame again. 0: Multicast Storm Autoclear disabled 1: Multicast Storm Autoclear enabled	R/W
22	BCACP	Broadcast Storm Autoclear P-Frames This bit enables auto-clear of broadcast frame storm counter. When it is asserted, broadcast frames storm counter is cleared by detecting a frame with non-broadcast destination address. After the counter is cleared, the IP can receive Broadcast frame again 0: Broadcast Storm Autoclear disabled 1: Broadcast Storm Autoclear enabled	R/W
23	NDAREP	Null Destination Address Reception Enable P-Frames This bit enables the reception of p-frames with destination MAC address null. 0: disable 1: enable	R/W
24	SDSFREP	Same DA and SA Frames Reception Enable P-Frames This bit enables the reception of p-frame with equal destination address and source address. 0: disable 1: enable	R/W
25	NSAREP	Null Source Address Reception Enable P-Frames This bit enables the reception of p-frames with a destination MAC address null. 0: disable 1: enable	R/W
26	MSAREP	Multicast Source Address Reception Enable P-Frames This bit enables the reception of p-frame with multicast and broadcast source address 0: disable 1: enable	R/W
31:27	—	These bits are read as 0.	R

Note 1. SW: It is valid only when Multicast reception is enabled (MRAFC.MCENE=1).

Note 2. SW: It is valid only when Broadcast reception is enabled (MRAFC.BCENE=1)

Note 3. HW: Broadcast frames are not taken in account by multicast storm counter.

34.4.1.13 MRSCE : Reception Storm Configuration for E-Frames Register

Base address: RMACm = 0x403C_B000 + 0x2000 × m (m = 0, 1)
RMACm_NS = 0x503C_B000 + 0x2000 × m (m = 0, 1)

Offset address: 0x0090

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	CBFE[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	CMFE[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	CMFE[15:0]	Consecutive Multicast Frame Reception Count Setting for E-Frames These bits specify the upper-limitation of received consecutive multicast frames. After the number of received consecutive multicast frames reaches the set value, the subsequent multicast frames are discarded. 0x0000: 1 Multicast frame accepted until multicast frame counter reset 0x0001: 2 Multicast frame accepted until multicast frame counter reset ⋮ 0xFFFF: 65535 Multicast frames accepted until frame counter reset 0xFFFF: Reserved	R/W
31:16	CBFE[15:0]	Consecutive Broadcast Frame Reception Count Setting for E-Frames These bits specify the upper-limitation of received consecutive broadcast frames. After the number of received consecutive broadcast frames reaches the set value, the subsequent broadcast frames are discarded. 0x0000: 1 Broadcast frame accepted until broadcast frame counter reset 0x0001: 2 Broadcast frame accepted until broadcast frame counter reset ⋮ 0xFFFF: 65535 Broadcast frames accepted until broadcast frame counter reset 0xFFFF: Reserved	R/W

The register is to use to configure the number of times how many broadcast/multicast frames can be received consecutively.

34.4.1.14 MRSCP : Reception Storm Configuration for P-Frames Register

Base address: $RMACm = 0x403C_B000 + 0x2000 \times m$ ($m = 0, 1$)
 $RMACm_NS = 0x503C_B000 + 0x2000 \times m$ ($m = 0, 1$)

Offset address: 0x0094

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	CBFP[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	CMFP[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	CMFP[15:0]	Consecutive Multicast Frame Reception Count Setting for P-Frames These bits specify the upper-limitation of received consecutive multicast frames. After the number of received consecutive multicast frames reaches the set value, the subsequent multicast frames are discarded. 0x0000: 1 Multicast frame accepted until multicast frame counter reset 0x0001: 2 Multicast frame accepted until multicast frame counter reset ⋮ 0xFFFF: 65535 Multicast frames accepted until multicast frame counter reset 0xFFFF: Reserved	R/W
31:16	CBFP[15:0]	Consecutive Broadcast Frame Reception Count Setting for P-Frames These bits specify the upper-limitation of received consecutive Broadcast frames. After the number of received consecutive broadcast frames reaches the set value, the subsequent broadcast frames are discarded. 0x0000: 1 Broadcast frame accepted until broadcast frame counter reset 0x0001: 2 Broadcast frame accepted until broadcast frame counter reset ⋮ 0xFFFF: 65535 Broadcast frames accepted between two broadcast frame counter resets 0xFFFF: Reserved	R/W

Configures the number of times how many broadcast/multicast frames can be received consecutively.

34.4.1.15 MRSCC : Reception Storm Counter Configuration Register

Base address: RMACm = 0x403C_B000 + 0x2000 × m (m = 0, 1)
 RMACm_NS = 0x503C_B000 + 0x2000 × m (m = 0, 1)

Offset address: 0x0098

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	BSCP	MSCCP
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	BSCCE	MSCCE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	MSCCE	Multicast Storm Counter Clear E-Frames This bit requests to clear the multicast frames storm counter. After the counter is cleared, the IP can receive multicast frames again until the upper-limitation of received consecutive multicast frames (MRSCE.CMFE). It should be used with multicast Storm Autoclear disabled (MRAFC.MCACE = 0). Writing 1 to this bit clear the multicast storm counter	R
1	BSCCE	Broadcast Storm Counter Clear E-Frames This bit requests to clear the broadcast frames storm counter. After the counter is cleared, the IP can receive Broadcast frames again until the upper-limitation of received consecutive Broadcast frames (MRSCE.CBFE). It should be used with Broadcast Storm Autoclear disabled (MRAFC.BCACE = 0). Writing 1 to this bit clear the broadcast storm counter	R
15:2	—	These bits are read as 0.	R
16	MSCCP	Multicast Storm Counter Clear P-Frames This bit requests to clear the multicast frames storm counter. After the counter is cleared, the IP can receive multicast frames again until the upper-limitation of received consecutive multicast frames (MRSCP.CMFP). It should be used with multicast Storm Autoclear disabled (MRAFC.MCACP = 0). Writing 1 to this bit clears the multicast storm counter	R
17	BSCCP	Broadcast Storm Counter Clear P-Frames This bit requests to clear the broadcast frames storm counter. After the counter is cleared, the IP can receive broadcast frames again until the upper-limitation of received consecutive broadcast frames (MRSCP.CBFP). It should be used with Broadcast Storm Autoclear disabled (MRAFC.BCACP = 0). Writing 1 to this bit clears the broadcast storm counter	R
31:18	—	These bits are read as 0.	R

Register to clear Broadcast and Multicast Storm Counters.

34.4.1.16 MRFSC : Reception Frame Size Configuration for E-Frames Register

Base address: $\text{RMACm} = 0x403C_B000 + 0x2000 \times m$ ($m = 0, 1$)
 $\text{RMACm_NS} = 0x503C_B000 + 0x2000 \times m$ ($m = 0, 1$)

Offset address: 0x009C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	EMNS[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	EMXS[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	EMXS[15:0]	E-Frame Maximum Size A part of received frame which exceeds the set value are truncated. Don't set a value under 16 bytes. MRFSC.EMXS have to set 3FFFH or less.	R/W
31:16	EMNS[15:0]	E-Frame Minimum Size	R/W

Note: The size limitations concern the frame size RMAC outputs to an upper layer (e.g. MHD), and so the FCS removal impacts the register values.

34.4.1.17 MRFSCP : Reception Frame Size Configuration for P-Frames Register

Base address: $\text{RMACm} = 0x403C_B000 + 0x2000 \times m$ ($m = 0, 1$)
 $\text{RMACm_NS} = 0x503C_B000 + 0x2000 \times m$ ($m = 0, 1$)

Offset address: 0x00A0

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	PMNS[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	PMXS[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	PMXS[15:0]	P-Frame Maximum Size A part of received frame which exceeds the set value are truncated. Don't set a value under 16 bytes.	R/W
31:16	PMNS[15:0]	P-Frame Minimum Size	R/W

Note: The size limitations concern the frame size RMAC outputs to an upper layer (e.g. MHD), and so the FCS removal impacts the register values.

34.4.1.18 MTRC : Timestamp Reception Configuration Register

Base address: RMACm = 0x403C_B000 + 0x2000 × m (m = 0, 1)
 RMACm_NS = 0x503C_B000 + 0x2000 × m (m = 0, 1)

Offset address: 0x00A4

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	DTN[1:0]		TCTS P	TCTS E	TRDD P	TRDD E	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	TRHFME1 to TRHFME0	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	TRHFME1 to TRHFME0	Timestamp Reception Hardware Filter Match Enable n (n = 0 to 1) 0: disable 1: enable	R/W
23:2	—	These bits are read as 0.	R
24	TRDDE	Timestamp Reception Default Disable E-Frame A timestamp for every frame will be transmitted to the MHD. This timestamp will be taken on the default timer except when TRHFME is set and the hardware filter matches. 0: enable 1: disable	R/W
25	TRDDP	Timestamp Reception Default Disable P-Frame A timestamp for every frame will be transmitted to the MHD. This timestamp will be taken on the default timer. 0: enable 1: disable	R/W
26	TCTSE	Timestamp Capture on TX Side E-Frame The bit enables the timestamp capture on TX side for e-frames 0: disable 1: enable	R/W
27	TCTSP	Timestamp Capture on TX Side P-Frame The bit enables the timestamp capture on TX side for p-frames 0: disable 1: enable	R/W
29:28	DTN[1:0]	Default Timer Number The bit set Timer used for timestamp forwarding on rx mhd i/f when no timer is detected.	R/W
31:30	—	These bits are read as 0.	R

TRHFME_n bits (Timestamp Reception Hardware Filter Match Enable n (n = 0 to 1))

Setting bit i to 1 in this register will enable hardware filtering for timer i.

When bit i is set to 1, at least one of the bit i in MPFC_n registers should be set to one.

When hardware filter matches, the timestamp from the matched timer will be sent along with the frame.

34.4.1.19 MRPFM : Reception Pause or PFC Frame Monitoring Register

Base address: $\text{RMACm} = 0x403C_B000 + 0x2000 \times m$ ($m = 0, 1$)
 $\text{RMACm_NS} = 0x503C_B000 + 0x2000 \times m$ ($m = 0, 1$)

Offset address: $0x00AC$

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	PFCTCA7 to PFCTCA0							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	PTCA
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PTCA	Pause Time Counting Active This bit indicates that the Pause time of a received Pause frame has not been completed. 0: No pause state 1: Pause state	R
15:1	—	These bits are read as 0.	R
23:16	PFCTCA7 to PFCTCA0	PFC Time Counting Active n ($n = 0$ to 7) This bit indicates that the Pause time of a received PFC frame has not been completed. 0: No pause state 1: Pause state	R
31:24	—	These bits are read as 0.	R

34.4.1.20 MPFCT : PTP Filtering Register Configuration Register t ($t = 0$ to 15)

Base address: $\text{RMACm} = 0x403C_B000 + 0x2000 \times m$ ($m = 0, 1$)
 $\text{RMACm_NS} = 0x503C_B000 + 0x2000 \times m$ ($m = 0, 1$)

Offset address: $0x0100 + 0x4 \times t$

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	TEF1 to TEF0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	PFBV[7:0]								PFBN[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	PFBN[7:0]	PTP Filtering Byte Number 0x00: Byte number 0 will be filtered. ⋮ 0xFF: Byte number 255 will be filtered.	R/W
15:8	PFBV[7:0]	PTP Filtering Byte Value When the filtered byte has a value equal to this register, the byte will be eligible for the corresponding timers.	R/W
17:16	TEF1 to TEF0	Timer Enable for Filtering n ($n = 0$ to 1) Setting bit i to 1 in this register will apply the filter to timer i.	R/W
31:18	—	These bits are read as 0.	R

Note: This register is used for RX time domain selection.
When an overlapping configuration is done, HW will always choose the timer with the smaller number.

The register is used to configure the PTP frame filtering for time domain selection.

34.4.1.21 MLVC : Link Verification Configuration Register

Base address: $\text{RMACm} = 0x403C_B000 + 0x2000 \times m$ ($m = 0, 1$)
 $\text{RMACm_NS} = 0x503C_B000 + 0x2000 \times m$ ($m = 0, 1$)

Offset address: 0x0180

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	PLV
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	PASE	—	LVT[6:0]						
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
6:0	LVT[6:0]	Link Verification Timer These bits define the nominal wait time between verification attempts in unit of milliseconds. It means the time of retransmitting SMD-V frame. 000 0000: 1ms wait time 000 0001: 2ms wait time ⋮ 000 1001: 10ms wait time (default specified by IEEE 802.3) ⋮ 111 1111: 128ms wait time	R/W
7	—	This bit is read as 0.	R
8	PASE	Preemption Auto Response Enable This bit enables the automatic response to verify frames.	R/W
15:9	—	These bits are read as 0.	R
16	PLV	Preemption Link Verification 0: Preemption Link Verification not requested 1: Preemption Link Verification requested	R/W ^{*1}
31:17	—	These bits are read as 0.	R

Note: When PLV is set to 1, do not make write access continuously. Please wait completion of Link Verification Procedure for re-write access.

Note 1. Read value differs from written value.

PLV bit (Preemption Link Verification)

By asserting this bit, link verification is requested.

This bit requests the Link Verification process. When link verification is completed, this bit is cleared.

When Link Verification succeeded, Preemption status bit (MMIS0.LVSS) are set. While it failed, the status bit (MMIS0.LVFS) is set.

Going out of OPERATION mode will clear this register.

As described in 7.2.2, this bit should only be used in initialization.

SW set this bit when another transmission is not requested.

34.4.1.22 MEEEC : Energy Efficient Ethernet Configuration Register

Base address: $\text{RMACm} = 0x403C_B000 + 0x2000 \times m$ ($m = 0, 1$)
 $\text{RMACm_NS} = 0x503C_B000 + 0x2000 \times m$ ($m = 0, 1$)

Offset address: 0x0184

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	LPITR
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	LPITR ^{*1}	LPI Transmit Request This bit requests to enter Low Power Idle (LPI) mode. RMAC enters to LPI state after all ongoing transmissions have been completed. When this bit is negated, this IP leaves LPI state. LPI signal should be transmitted only when TX is disabled. Going out of OPERATION mode will clear this register. 0: LPI mode is not requested 1: LPI mode is requested	R/W
31:1	—	These bits are read as 0.	R

Note 1. SW: LPITR bit can only be set when transmission request to RMAC is stopped, and transmission request can only be allowed after LPITR is cleared.

It is prohibited to request transmission from HMD when LPITR bit is set.

34.4.1.23 MLBC : Loopback Configuration Register

Base address: $\text{RMACm} = 0x403C_B000 + 0x2000 \times m$ ($m = 0, 1$)
 $\text{RMACm_NS} = 0x503C_B000 + 0x2000 \times m$ ($m = 0, 1$)

Offset address: 0x0188

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	LBME
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	LBME	Loopback Mode Enable Transmit xMII signals are connected to reception xMII signals internally. 0: disable 1: enable	R/W
31:1	—	These bits are read as 0.	R

Note: When loopback mode is enabled, frame received by external PHY Interface are ignored.
If this bit is set, clk_phy_tx and clk_phy_rx must be synchronized with each other.

34.4.2 RMAC Counter Registers

Counter registers

Note: All counter registers in RMAC are read clear registers.

When counters reach their own maximum value, they stop.

34.4.2.1 MMPFTCT : Manual Pause Frame Transmit Counter Register

Base address: $\text{RMACm} = 0x403C_B000 + 0x2000 \times m$ ($m = 0, 1$)
 $\text{RMACm_NS} = 0x503C_B000 + 0x2000 \times m$ ($m = 0, 1$)

Offset address: 0x0300

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	MPFTC[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	MPFTC[15:0]	Manual Pause frame Transmit Counter	R
31:16	—	These bits are read as 0.	R

MPFTC[15:0] bits (Manual Pause frame Transmit Counter)

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Reading this register clears it.

[Increment condition]

HW: Incremented by 1 when manual Pause frame is transmitted.

34.4.2.2 MAPFTCT : Automatic Pause Frame Transmit Counter Register

Base address: $\text{RMACm} = 0x403C_B000 + 0x2000 \times m$ ($m = 0, 1$)
 $\text{RMACm_NS} = 0x503C_B000 + 0x2000 \times m$ ($m = 0, 1$)

Offset address: 0x0304

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	APFTC[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	APFTC[15:0]	Automatic pause Frame Counter	R
31:16	—	These bits are read as 0.	R

APFTC[15:0] bits (Automatic pause Frame Counter)

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Reading this register clears it.

[Increment condition]

HW: Incremented by 1 when automatic Pause frame is transmitted.

34.4.2.3 MPFRCT : Pause Frame Receive Counter Register

Base address: $\text{RMACm} = 0x403C_B000 + 0x2000 \times m$ ($m = 0, 1$)
 $\text{RMACm_NS} = 0x503C_B000 + 0x2000 \times m$ ($m = 0, 1$)

Offset address: 0x0308

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	PFRC[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	PFRC[15:0] ^{*1}	Pause Frame Receive Counter	R
31:16	—	These bits are read as 0.	R

Note 1. SW: Set condition is only available when Flow Control is enabled in reception. (MRGC.PFRC = 1)

PFRC[15:0] bits (Pause Frame Receive Counter)

[Clearing condition]

HW: Being in RESET mode will clear this register

SW: Reading this register clears it.

[Increment condition]

HW: Incremented by 1 when automatic Pause frame is received.

34.4.2.4 MFCICT : False Carrier Indication Counter Register

Base address: $\text{RMACm} = 0x403C_B000 + 0x2000 \times m$ ($m = 0, 1$)
 $\text{RMACm_NS} = 0x503C_B000 + 0x2000 \times m$ ($m = 0, 1$)

Offset address: 0x030C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	FCIC[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	FCIC[15:0]	False Carrier Indication Counter	R
31:16	—	These bits are read as 0.	R

Note: This register is counter of "False Carrier Indication" event. This event goes through the synchronizer between clk and clk_phy_rx. If the event occurs continuously, there is a possibility of miss counting.

FCIC[15:0] bits (False Carrier Indication Counter)

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Reading this register clears it.

[Increment condition]

HW: Incremented by 1 when False Carrier Indication pattern is detected.

34.4.2.5 MEEECT : Energy Efficient Ethernet Counter Register

Base address: RMACm = 0x403C_B000 + 0x2000 × m (m = 0, 1)
 RMACm_NS = 0x503C_B000 + 0x2000 × m (m = 0, 1)

Offset address: 0x0310

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	EEERC[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	EEERC[15:0]	Energy Efficient Ethernet Receive Counter	R
31:16	—	These bits are read as 0.	R

EEERC[15:0] bits (Energy Efficient Ethernet Receive Counter)

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Reading this register clears it.

[Increment condition]

HW: Incremented by 1 when LPI signal is detected on xMII reception bus.

34.4.2.6 MMPCFTCTt : Manual PFC Frame Transmit Counter Register (t = 0, 1)

Base address: RMACm = 0x403C_B000 + 0x2000 × m (m = 0, 1)
 RMACm_NS = 0x503C_B000 + 0x2000 × m (m = 0, 1)

Offset address: 0x0320 + 0x4 × t

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	MPCFCTC[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	MPCFCTC[15:0]	Manual PFC frame Transmit Counter	R
31:16	—	These bits are read as 0.	R

MPCFCTC[15:0] bits (Manual PFC frame Transmit Counter)

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Reading this register clears it.

[Increment condition]

HW: Incremented by 1 when manual PFC frames is transmitted.

34.4.2.7 MAPCFTCTt : Automatic PFC Frame Transmit Counter Register (t = 0, 1)

Base address: $\text{RMACm} = 0x403C_B000 + 0x2000 \times m$ (m = 0, 1)
 $\text{RMACm_NS} = 0x503C_B000 + 0x2000 \times m$ (m = 0, 1)

Offset address: $0x0330 + 0x4 \times t$

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	APCFCTC[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	APCFCTC[15:0]	Automatic PFC Frame Counter	R
31:16	—	These bits are read as 0.	R

APCFCTC[15:0] bits (Automatic PFC Frame Counter)

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Reading this register clears it.

[Increment condition]

HW: Incremented by 1 when manual PFC frames is transmitted.

34.4.2.8 MPCFRCTt : PFC Frame Receive Counter Register (t = 0, 1)

Base address: $\text{RMACm} = 0x403C_B000 + 0x2000 \times m$ (m = 0, 1)
 $\text{RMACm_NS} = 0x503C_B000 + 0x2000 \times m$ (m = 0, 1)

Offset address: $0x0340 + 0x4 \times t$

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	PCFCRC[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	PCFCRC[15:0] ^{*1}	PFC Frame Receive Counter	R
31:16	—	These bits are read as 0.	R

Note 1. SW: Set condition is only available when Flow Control is enabled in reception (MRGC.PFRC = 1).

PCFCRC[15:0] bits (PFC Frame Receive Counter)

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Reading this register clears it.

[Increment condition]

HW: Incremented by 1 when manual PFC frames is received.

34.4.2.9 MROVFC : Receive Overflow Counter Register

Base address: $\text{RMACm} = 0x403C_B000 + 0x2000 \times m$ ($m = 0, 1$)
 $\text{RMACm_NS} = 0x503C_B000 + 0x2000 \times m$ ($m = 0, 1$)

Offset address: 0x0360

Bit position: 31

0

Bit field:

ROVFC[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	ROVFC[31:0]	Receive overflow counter	R

ROVFC[31:0] bits (Receive overflow counter)

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Reading this register clears it.

[Increment condition]

HW: Incremented by 1 when frame receive overflow occurred (when set condition for MEIS.REOES or MEIS.RPOES are met).

34.4.2.10 MRXBCEU : Received Byte Counter E-Frames Upper Side Register

Base address: $\text{RMACm} = 0x403C_B000 + 0x2000 \times m$ ($m = 0, 1$)
 $\text{RMACm_NS} = 0x503C_B000 + 0x2000 \times m$ ($m = 0, 1$)

Offset address: 0x044C

Bit position: 31

0

Bit field:

RBNEU[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	RBNEU[31:0]	Received byte number E-frames upper side When the RMAC does not pass the FCS to the upper layer, FCS is not taken in account in total byte calculation. Read before reading MRXBCEL.	R

RBNEU[31:0] bits (Received byte number E-frames upper side)

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Reading this register clears it.

[Increment condition]

Incremented each time the received data of e-Frame is sent to the rx mhd interface regardless error is detected.

34.4.2.11 MRXBCEL : Received Byte Counter E-Frames Lower Side Register

Base address: $\text{RMACm} = 0x403C_B000 + 0x2000 \times m$ ($m = 0, 1$)
 $\text{RMACm_NS} = 0x503C_B000 + 0x2000 \times m$ ($m = 0, 1$)

Offset address: 0x0450

Bit position: 31

0

Bit field:

RBNEL[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	RBNEL[31:0]	Received byte number E-frames lower side When the RMAC does not pass the FCS to the upper layer, FCS is not taken in account in total byte calculation. Read after reading MRXBCEU.	R

RBNEL[31:0] bits (Received byte number E-frames lower side)

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: None. (Reading MRXBCEL register holds it.) Increment condition

[Increment condition]

Incremented each time the received data of e-frame is sent to the rx mhd interface regardless the error detection.

34.4.2.12 MRXBCPU : Received Byte Counter P-Frames Upper Side Register

Base address: $\text{RMACm} = 0x403C_B000 + 0x2000 \times m$ ($m = 0, 1$)
 $\text{RMACm_NS} = 0x503C_B000 + 0x2000 \times m$ ($m = 0, 1$)

Offset address: 0x0454

Bit position: 31

0

Bit field:

RBNPU[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	RBNPU[31:0]	Received byte number P-frames upper side If the RMAC does not pass the FCS to the upper layer, FCS is not taken in account in total byte calculation. Read before reading MRXBCPL.	R

RBNPU[31:0] bits (Received byte number P-frames upper side)

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Reading this register clears it.

[Increment condition]

Incremented each time the received data of p-frame is sent to the rx mhd interface regardless the error detection.

34.4.2.13 MRXBCPL : Received Byte Counter P-Frames Lower Side Register

Base address: $\text{RMACm} = 0x403C_B000 + 0x2000 \times m$ ($m = 0, 1$)
 $\text{RMACm_NS} = 0x503C_B000 + 0x2000 \times m$ ($m = 0, 1$)

Offset address: 0x0458

Bit position: 31

0

Bit field:

RBNPL[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	RBNPL[31:0]	Received byte number P-frames lower side When the RMAC does not pass the FCS to the upper layer, FCS is not taken in account in total byte calculation. Read after reading MRXBCPU.	R

RBNPL[31:0] bits (Received byte number P-frames lower side)

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: None. (Reading MRXBCPL register holds it.)

[Increment condition]

Incremented each time the received data of p-frame is sent to the rx mhd interface regardless the error detection.

34.4.2.14 MRGFCE : Received Good Frame Counter E-Frames Register

Base address: $\text{RMACm} = 0x403C_B000 + 0x2000 \times m$ ($m = 0, 1$)
 $\text{RMACm_NS} = 0x503C_B000 + 0x2000 \times m$ ($m = 0, 1$)

Offset address: 0x0408

Bit position: 31

0

Bit field:

RGFNE[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	RGFNE[31:0]	Received good frame number E-frames	R

RGFNE[31:0] bits (Received good frame number E-frames)

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Reading this register clears it.

[Increment condition]

Incremented when the reception of e-frame is completed without error.

34.4.2.15 MRGFCP : Received Good Frame Counter P-Frames Register

Base address: $\text{RMACm} = 0x403C_B000 + 0x2000 \times m$ ($m = 0, 1$)
 $\text{RMACm_NS} = 0x503C_B000 + 0x2000 \times m$ ($m = 0, 1$)

Offset address: 0x040C

Bit position: 31 0

Bit field: GFNP[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	GFNP[31:0]	Received good frame number P-frames	R

GFNP[31:0] bits (Received good frame number P-frames)

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Reading this register clears it.

[Increment condition]

Incremented when the reception of p-frame is completed without error.

34.4.2.16 MRBFC : Received Good Broadcast Frame Counter Register

Base address: $\text{RMACm} = 0x403C_B000 + 0x2000 \times m$ ($m = 0, 1$)
 $\text{RMACm_NS} = 0x503C_B000 + 0x2000 \times m$ ($m = 0, 1$)

Offset address: 0x0410

Bit position: 31 0

Bit field: RBFN[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	RBFN[31:0]	Received good broadcast frame number	R

RBFN[31:0] bits (Received good broadcast frame number)

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Reading this register clears it.

[Increment condition]

Incremented when destination address field of broadcast frame is sent to rx mhd interface without error.

34.4.2.17 MRMFC : Received Good Multicast Frame Counter Register

Base address: $\text{RMACm} = 0x403C_B000 + 0x2000 \times m$ ($m = 0, 1$)
 $\text{RMACm_NS} = 0x503C_B000 + 0x2000 \times m$ ($m = 0, 1$)

Offset address: 0x0414

Bit position: 31 0

Bit field: RMFN[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	RMFN[31:0]	Received good multicast frame number	R

RMFN[31:0] bits (Received good multicast frame number)

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Reading this register clears it.

[Increment condition]

Incremented when destination address field of multicast frame is sent to rx mhd interface without error.

34.4.2.18 MRUFC : Received Good Unicast Frame Counter Register

Base address: $\text{RMACm} = 0x403C_B000 + 0x2000 \times m$ ($m = 0, 1$)
 $\text{RMACm_NS} = 0x503C_B000 + 0x2000 \times m$ ($m = 0, 1$)

Offset address: 0x0418

Bit position: 31

0

Bit field:

RUFN[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	RUFN[31:0]	Received good unicast frame number	R

RUFN[31:0] bits (Received good unicast frame number)

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Reading this register clears it.

[Increment condition]

Incremented when destination address field of unicast frame is sent to rx mhd interface without error.

34.4.2.19 MRPEFC : Received PHY Error Frame Count Register

Base address: $\text{RMACm} = 0x403C_B000 + 0x2000 \times m$ ($m = 0, 1$)
 $\text{RMACm_NS} = 0x503C_B000 + 0x2000 \times m$ ($m = 0, 1$)

Offset address: 0x041C

Bit position:

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16

Bit field:

—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit position:

15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bit field:

RPEFN[15:0]

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
15:0	RPEFN[15:0]	Received PHY error frame number	R
31:16	—	These bits are read as 0.	R

RPEFN[15:0] bits (Received PHY error frame number)

[Clearing condition]

SW: Reading this register clears it.

Incremented when received data is sent to rx mhd interface with phy error.

Offset address: 0x0420

[illegible]

Bit	Symbol	Function	R/W
15:0	RNEFN[15:0]	Received nibble error frame number	R
31:16	—	These bits are read as 0.	R

Incremented when received data is sent to rx mhd interface with nibble error.

Offset address: 0x0424

[illegible]

Bit	Symbol	Function	R/W
31:0	RFMEFN[31:0]	Received FCS/mCRC error frame number	R

Incremented when received data is sent to rx mhd interface with FCS/mCRC error.

34.4.2.22 MRFFMEFC : Received Final Fragment Missing Error Frame Count Register

Base address: $RMACm = 0x403C_B000 + 0x2000 \times m$ ($m = 0, 1$)
 $RMACm_NS = 0x503C_B000 + 0x2000 \times m$ ($m = 0, 1$)

Offset address: 0x0428

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	RFFMEFN[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	RFFMEFN[15:0]	Received final fragment missing error frame number	R
31:16	—	These bits are read as 0.	R

RFFMEFN[15:0] bits (Received final fragment missing error frame number)

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Reading this register clears it.

[Increment condition]

Incremented when received data is sent to rx mhd interface with final fragment error.

34.4.2.23 MRCFCEFC : Received C-Fragment Count Error Frame Count Register

Base address: $RMACm = 0x403C_B000 + 0x2000 \times m$ ($m = 0, 1$)
 $RMACm_NS = 0x503C_B000 + 0x2000 \times m$ ($m = 0, 1$)

Offset address: 0x042C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	RCFCEFN[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	RCFCEFN[15:0]	Received C-fragment count error frame number	R
31:16	—	These bits are read as 0.	R

RCFCEFN[15:0] bits (Received C-fragment count error frame number)

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Reading this register clears it.

[Increment condition]

Incremented when received data is sent to rx mhd interface with C-fragment count error.

34.4.2.24 MRFCEFC : Received Fragment Count Error Frame Count Register

Base address: $\text{RMACm} = 0x403C_B000 + 0x2000 \times m$ ($m = 0, 1$)
 $\text{RMACm_NS} = 0x503C_B000 + 0x2000 \times m$ ($m = 0, 1$)

Offset address: 0x0430

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	RFCEFN[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	RFCEFN[15:0]	Received fragment count error frame number	R
31:16	—	These bits are read as 0.	R

RFCEFN[15:0] bits (Received fragment count error frame number)

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Reading this register clears it.

[Increment condition]

Incremented when received data is sent to rx mhd interface with fragment count error.

34.4.2.25 MRRCFEFC : Received RMAC Filter Error Frame Count Register

Base address: $\text{RMACm} = 0x403C_B000 + 0x2000 \times m$ ($m = 0, 1$)
 $\text{RMACm_NS} = 0x503C_B000 + 0x2000 \times m$ ($m = 0, 1$)

Offset address: 0x0434

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	RRCFEFN[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	RRCFEFN[15:0]	Received RMAC filter error frame number	R
31:16	—	These bits are read as 0.	R

RRCFEFN[15:0] bits (Received RMAC filter error frame number)

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Reading this register clears it.

[Increment condition]

Incremented when received data is sent to rx mhd interface with RMAC filtered.

34.4.2.26 MRFC : Received Frame Count Register

Base address: $\text{RMACm} = 0x403C_B000 + 0x2000 \times m$ ($m = 0, 1$)
 $\text{RMACm_NS} = 0x503C_B000 + 0x2000 \times m$ ($m = 0, 1$)

Offset address: 0x0438

Bit position: 31

0

Bit field:

RFN[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	RFN[31:0]	Received frame number	R

RFN[31:0] bits (Received frame number)

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Reading this register clears it.

[Increment condition]

Incremented when received frame is sent to rx mhd interface.

34.4.2.27 MRGUEFC : Received Good Undersize Error Frame Count Register

Base address: $\text{RMACm} = 0x403C_B000 + 0x2000 \times m$ ($m = 0, 1$)
 $\text{RMACm_NS} = 0x503C_B000 + 0x2000 \times m$ ($m = 0, 1$)

Offset address: 0x043C

Bit position: 31

0

Bit field:

RUEFN[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	RUEFN[31:0]	Received good undersize error frame number	R

RUEFN[31:0] bits (Received good undersize error frame number)

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Reading this register clears it.

[Increment condition]

Incremented when received data is sent to rx mhd interface with undersize error but without fcs/mcrc error.

34.4.2.28 MRBUEFC : Received Bad Undersize Error Frame Count Register

Base address: $\text{RMACm} = 0x403C_B000 + 0x2000 \times m$ ($m = 0, 1$)
 $\text{RMACm_NS} = 0x503C_B000 + 0x2000 \times m$ ($m = 0, 1$)

Offset address: 0x0440

Bit position: 31

0

Bit field:

RUEFN[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	RUEFN[31:0]	Received bad undersize error frame number	R

RUEFN[31:0] bits (Received bad undersize error frame number)

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Reading this register clears it.

[Increment condition]

Incremented when received data is sent to rx mhd interface with undersize error and fcs/mcrc error.

34.4.2.29 MRGOEFC : Received Good Oversize Error Frame Count Register

Base address: $RMACm = 0x403C_B000 + 0x2000 \times m$ ($m = 0, 1$)
 $RMACm_NS = 0x503C_B000 + 0x2000 \times m$ ($m = 0, 1$)

Offset address: 0x0444

Bit position: 31

0

Bit field:

RGOEFN[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	RGOEFN[31:0]	Received good oversize error frame number	R

RGOEFN[31:0] bits (Received good oversize error frame number)

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Reading this register clears it.

[Increment condition]

Incremented when received data is sent to rx mhd interface with oversize error, but without fcs/mcrc error.

34.4.2.30 MRBOEFC : Received Bad Oversize Error Frame Count Register

Base address: $RMACm = 0x403C_B000 + 0x2000 \times m$ ($m = 0, 1$)
 $RMACm_NS = 0x503C_B000 + 0x2000 \times m$ ($m = 0, 1$)

Offset address: 0x0448

Bit position: 31

0

Bit field:

RBOEFN[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	RBOEFN[31:0]	Received bad oversize error frame number	R

RBOEFN[31:0] bits (Received bad oversize error frame number)

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Reading this register clears it.

[Increment condition]

Incremented when received data is sent to rx mhd interface with oversize error and fcs/mcrc error.

34.4.2.31 MTXBCEU : Transmitted Byte Counter E-Frames Upper Side Register

Base address: $RMACm = 0x403C_B000 + 0x2000 \times m$ ($m = 0, 1$)
 $RMACm_NS = 0x503C_B000 + 0x2000 \times m$ ($m = 0, 1$)

Offset address: 0x0520

Bit position: 31

0

Bit field:

TBNEU[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	TBNEU[31:0]	Transmitted byte number E-frames upper side	R

Note: If the FCS is not passed to the upper layer FCS is not taken in account in total byte calculation.

TBNEU[31:0] bits (Transmitted byte number E-frames upper side)

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Reading this register clears it.

[Increment condition]

Increment every time transmit data for e-frame is received on tx mhd i/f.

Read before reading MTXBCEL.

34.4.2.32 MTXBCEL : Transmitted Byte Counter E-Frames Lower Side Register

Base address: $RMACm = 0x403C_B000 + 0x2000 \times m$ ($m = 0, 1$)
 $RMACm_NS = 0x503C_B000 + 0x2000 \times m$ ($m = 0, 1$)

Offset address: 0x0524

Bit position: 31

0

Bit field:

TBNEL[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	TBNEL[31:0]	Transmitted byte number E-frames lower side	R

Note: If the FCS is not passed to the upper layer FCS is not taken in account in total byte calculation.

TBNEL[31:0] bits (Transmitted byte number E-frames lower side)

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: None. (Reading MTXBCEL register holds it.)

[Increment condition]

Increment every time transmit data for e-frame is received on tx mhd i/f.

Read before reading MTXBCEU.

34.4.2.33 MTXBCPU : Transmitted Byte Counter P-Frames Upper Side Register

Base address: $RMACm = 0x403C_B000 + 0x2000 \times m$ ($m = 0, 1$)
 $RMACm_NS = 0x503C_B000 + 0x2000 \times m$ ($m = 0, 1$)

Offset address: 0x0528

Bit position: 31

0

Bit field:

TBNPU[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	TBNPU[31:0]	Transmitted byte number P-frames upper side	R

Note: If the FCS is not passed to the upper layer, FCS is not taken in account in total byte calculation.

TBNPU[31:0] bits (Transmitted byte number P-frames upper side)

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Reading this register clears it.

[Increment condition]

Increment every time transmit data for p-frame is received on tx mhd i/f.

Read before reading MTXBCPL.

34.4.2.34 MTXBCPL : Transmitted Byte Counter P-Frames Lower Side Register

Base address: $RMACm = 0x403C_B000 + 0x2000 \times m$ ($m = 0, 1$)
 $RMACm_NS = 0x503C_B000 + 0x2000 \times m$ ($m = 0, 1$)

Offset address: 0x052C

Bit position: 31

0

Bit field:

TBNPL[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	TBNPL[31:0]	Transmitted byte number P-frames lower side	R

Note: If the FCS is not passed to the upper layer, FCS is not taken in account in total byte calculation.

TBNPL[31:0] bits (Transmitted byte number P-frames lower side)

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: None. (Reading MTXBCPL register holds it.)

[Increment condition]

Increment every time transmit data for p-frame is received on tx mhd i/f.

Read before reading MTXBCPU.

34.4.2.35 MTGFCE : Transmitted Good Frame Counter E-Frames Register

Base address: $\text{RMACm} = 0x403C_B000 + 0x2000 \times m$ ($m = 0, 1$)
 $\text{RMACm_NS} = 0x503C_B000 + 0x2000 \times m$ ($m = 0, 1$)

Offset address: 0x0508

Bit position: 31

0

Bit field:

TGFNE[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	TGFNE[31:0]	Transmitted good frame number E-frames	R

TGFNE[31:0] bits (Transmitted good frame number E-frames)

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Reading this register clears it.

[Increment condition]

Increment when transmission of e-frame is completed on tx mhd i/f without error.

34.4.2.36 MTGFCP : Transmitted Good Frame Counter P-Frames Register

Base address: $\text{RMACm} = 0x403C_B000 + 0x2000 \times m$ ($m = 0, 1$)
 $\text{RMACm_NS} = 0x503C_B000 + 0x2000 \times m$ ($m = 0, 1$)

Offset address: 0x050C

Bit position: 31

0

Bit field:

TGFNP[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	TGFNP[31:0]	Transmitted good frame number P-frames	R

TGFNP[31:0] bits (Transmitted good frame number P-frames)

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Reading this register clears it.

[Increment condition]

Increment when transmission of p-frame is completed on tx mhd i/f without error.

34.4.2.37 MTBFC : Transmitted Broadcast Frame Counter Register

Base address: $\text{RMACm} = 0x403C_B000 + 0x2000 \times m$ ($m = 0, 1$)
 $\text{RMACm_NS} = 0x503C_B000 + 0x2000 \times m$ ($m = 0, 1$)

Offset address: 0x0510

Bit position: 31

0

Bit field:

TBFN[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	TBFN[31:0]	Transmitted broadcast frame number	R

TBFN[31:0] bits (Transmitted broadcast frame number)

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Reading this register clears it.

[Increment condition]

Increment when destination address field of broadcast frame is sent on tx mhd i/f.

34.4.2.38 MTMFC : Transmitted Multicast Frame Counter Register

Base address: RMACm = 0x403C_B000 + 0x2000 × m (m = 0, 1)
 RMACm_NS = 0x503C_B000 + 0x2000 × m (m = 0, 1)

Offset address: 0x0514

Bit position: 31

0

Bit field:

TMFN[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	TMFN[31:0]	Transmitted multicast frame number	R

TMFN[31:0] bits(Transmitted multicast frame number)

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Reading this register clears it.

[Increment condition]

Increment when destination address field of multicast frame is sent on tx mhd i/f.

34.4.2.39 MTUFC : Transmitted Unicast Frame Counter Register

Base address: RMACm = 0x403C_B000 + 0x2000 × m (m = 0, 1)
 RMACm_NS = 0x503C_B000 + 0x2000 × m (m = 0, 1)

Offset address: 0x0518

Bit position: 31

0

Bit field:

TUFN[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	TUFN[31:0]	Transmitted unicast frame number	R

TUFN[31:0] bits (Transmitted unicast frame number)

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Reading this register clears it.

[Increment condition]

Increment when destination address field of unicast frame is sent on tx mhd i/f.

34.4.2.40 MTEFC : Transmitted Error Frame Counter Register

Base address: RMACm = 0x403C_B000 + 0x2000 × m (m = 0, 1)
 RMACm_NS = 0x503C_B000 + 0x2000 × m (m = 0, 1)

Offset address: 0x051C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	TEFN[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	TEFN[15:0]	Transmitted error frame number	R
31:16	—	These bits are read as 0.	R

TEFN[15:0] bits (Transmitted error frame number)

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Reading this register clears it.

[Increment condition]

Increment when frame transmission is completed on tx mhd interface with error.

34.4.3 RMAC Interrupt Registers

34.4.3.1 MEIS : Error Interrupt Status Register

Base address: RMACm = 0x403C_B000 + 0x2000 × m (m = 0, 1)
 RMACm_NS = 0x503C_B000 + 0x2000 × m (m = 0, 1)

Offset address: 0x0200

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	FOES	FUES	FFS	RPOO MS	FRCE S	CFCE S	FFME S	FCMC ES	PNAE S	PDES	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	CTLES1 to CTLES0	—	RPOE S	REOE S	FCES	BFES	TBCIS	TCES	FCDS	PFRR OS	PRES	—	—	TSLs
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	TSLs	Transmission Stream Lost Status Flag When using Cut through, this flag set condition may be due to the state ([COMA] CAEIS0.BPOPS) where all Buffer pointers are used. Please use the Watermark function so that Buffer Pool Out of Pointer Status does not occur. If it gets this error with Buffer Pool Out of Pointer Status using Cut through (it may get hung-up in the transmit state), initialize the state with Emergency reset flow [TOP].	R/W ^{*1}
1	—	This bit is read as 0.	R
2	PRES	Pause or PFC Frame Reception Error Status Flag	R/W ^{*1}
3	PFRR OS	Pause or PFC Frame Retransmit Retry Over Status Flag	R/W ^{*1}

Bit	Symbol	Function	R/W
4	FCDS	False Carrier Detection Status Flag	R/W ^{*1}
5	TCES	TX CRC Error Status Flag	R/W ^{*1}
6	TBCIS	TX Bad CRC Insertion Status Flag	R/W ^{*1}
7	BFES	Bad Fragment Error Status Flag When C-fragment is received at very first time before any valid S-fragment is received or C-fragment is received after frame termination is occurred before any valid S-fragment is received, then RMAC will discard the fragment and this error status flag will not be set. For Error status flag to get set, RMAC should have received one valid S-fragment before error condition (bad fragment error) has occurred.	R/W ^{*1}
8	FCES	Frame Count Error Status Flag When C-fragment is received at very first time before any valid S-fragment is received or C-fragment is received after frame termination is occurred before any valid S-fragment is received, then RMAC will discard the fragment and this error status flag will not be set. For Error status flag to get set, RMAC should have received one valid S-fragment before error condition (frame count error) has occurred.	R/W ^{*1}
9	REOES ^{*2}	E-Frame Overflow Error Status Flag When frame reception is already ongoing and buffer overflow occurs i.e. CABPPCM.RPC = 0, RMAC adds one byte padding to the reception error Frames to terminate the protocol internally. When new frame is received, during buffer overflow i.e. CABPPCM.RPC = 0, the incoming frame will be rejected completely. 0: No overflow error 1: Overflow error occurred during e-frame reception.	R/W ^{*1}
10	RPOES ^{*2}	P-Frame Overflow Error Status Flag When frame reception is already ongoing and buffer overflow occurs i.e. CABPPCM.RPC = 0, RMAC adds one byte padding to the reception error Frames to terminate the protocol internally. When new frame is received, during buffer overflow i.e. CABPPCM.RPC = 0, the incoming frame will be rejected completely. 0: No overflow error 1: Overflow error occurred during p-frame reception.	R/W ^{*1}
11	—	This bit is read as 0.	R
13:12	CTLES1 to CTLES0	Captured Timestamp Lost Error Status Flag n (n = 0 to 1)	R/W ^{*1}
19:14	—	These bits are read as 0.	R
20	PDES	PHY Data Error Status Flag	R/W ^{*1}
21	PNAES	PHY Nibble Alignment Error Status Flag	R/W ^{*1}
22	FCMCES	FCS/mCRC Error Status Flag	R/W ^{*1}
23	FFMES	Final Fragment Missing Error Status Flag	R/W ^{*1}
24	CFCES	C Fragment Count Error Status Flag	R/W ^{*1}
25	FRCES	Fragment Count Error Status Flag	R/W ^{*1}
26	RPOOMS	Reception Partially out of Operation Mode Status Flag This error has nothing to do with partial rejection due to buffer overflow. E-Frames can only be fragmented/partialized by buffer overflows and it is not affected by partial out of operation.	R/W ^{*1}
27	FFS	Frame Filtered Status Flag	R/W ^{*1}
28	FUES	Undersize Error Status Flag	R/W ^{*1}
29	FOES	Oversize Error Status Flag	R/W ^{*1}
31:30	—	These bits are read as 0.	R

Note: This interrupt register is mapped to rmc_meis_int interrupt wire.

Note 1. Read value differs from written value.

Note 2. HW: Pulse is generated per 1 frame although multiple deassertion is generated.

TSLs bit (Transmission Stream Lost Status Flag)

[Setting condition]

HW: when MHD do not provide the transmitted data timely. In this case an invalid FCS is transmitted. (only for e-frames)

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Writing 1 to this register will clear it.

PRES bit (Pause or PFC Frame Reception Error Status Flag)

[Setting condition]

HW: When Pause or PFC frame is received with lower than minimum pause/pfc frame length.

Minimum PAUSE Length: 18Byte (MRGC.RCPT==0) or 22 Byte ((MRGC.RCPT==1)

Minimum PFC Length: 34Byte (MRGC.RCPT==0) or 38 Byte ((MRGC.RCPT==1)

HW: When PAUSE/PFC frame is received with fcs error in MRGC.RFCFE==0.

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Writing 1 to this register will clear it.

PFRROS bit (Pause or PFC Frame Retransmit Retry Over Status Flag)

[Setting condition]

HW: when the total number of Pause (pfc) transmit count reaches the configured upper limit and Pause time expires.

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Writing 1 to this register will clear it.

FCDS bit (False Carrier Detection Status Flag)

[Setting condition]

HW: When PHY-LSI has detected a false carrier on the line in case of MII and GMII.

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Writing 1 to this register will clear it.

TCES bit (TX CRC Error Status Flag)

[Setting condition]

HW: When CRC error occurred.

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Writing 1 to this register will clear it.

TBCIS bit (TX Bad CRC Insertion Status Flag)

[Setting condition]

HW: A bad CRC is inserted from the upper layer (e.g. MHD).

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Writing 1 to this register will clear it.

BFES bit (Bad Fragment Error Status Flag)

[Setting condition]

HW: C fragment has been detected without receiving S fragment.

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Writing 1 to this register will clear it.

FCES bit (Frame Count Error Status Flag)

[Setting condition]

When S fragment has been received in the wrong order.

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Writing 1 to this register will clear it.

REOES bit (E-Frame Overflow Error Status Flag)

[Setting condition]

HW: The acknowledgment signal was not received successfully from an upper layer (e.g. MHD) during e-frame reception.

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Writing 1 to this register will clear it.

RPOES bit (P-Frame Overflow Error Status Flag)

[Setting condition]

HW: The acknowledgment signal was not received successfully from an upper layer (e.g. MHD) during p-frame reception.

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Writing 1 to this register will clear it.

CTLESn bits (Captured Timestamp Lost Error Status Flag n (n = 0 to 1))

[Setting condition]

HW: When captured timestamp is lost.

This bit is fixed to 0

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Writing 1 to this register will clear it.

[Error recovery]

No recovery. Timestamp is lost.

PDES bit (PHY Data Error Status Flag)

[Setting condition]

HW: When frame is received with error.

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Writing 1 to this register will clear it.

PNAES bit (PHY Nibble Alignment Error Status Flag)

[Setting condition]

HW: When residual-bit is received.

[Clearing condition]

SW: Writing 1 to this register will clear it.

FCMCES bit (FCS/mCRC Error Status Flag)

[Setting condition]

HW: When unexpected FCS/mCRC are received.

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Writing 1 to this register will clear it.

FFMES bit (Final Fragment Missing Error Status Flag)

[Setting condition]

HW: When an S-fragment is received even though the previous fragment has been terminated by a mCRC.

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Writing 1 to this register will clear it.

CFCES bit (C Fragment Count Error Status Flag)

[Setting condition]

HW: When a C fragment not corresponding to the current frame is received For example: S0-C0-C0-C1.

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Writing 1 to this register will clear it.

FRCES bit (Fragment Count Error Status Flag)

[Setting condition]

HW: When a C-fragment is received with a bad fragment count field.

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Writing 1 to this register will clear it.

RPOOMS bit (Reception Partially out of Operation Mode Status Flag)

[Setting condition]

HW: When a frame is being received while RMAC is moving out of operation mode (suspending).

1. Initial or intermediate fragment of p-frame is receiving.
2. E-frame is receiving when initial p-frame is already received and last fragment of p-frame is not received.

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Writing 1 to this register will clear it.

FFS bit (Frame Filtered Status Flag)

[Setting condition]

HW: When frame did not pass the RMAC filters

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Writing 1 to this register will clear it.

FUES bit (Undersize Error Status Flag)

[Setting condition]

HW: When too short frame is received. (When the received frame size is shorter than the setting in MRFSC.EMNS and MRFSC.PMNS)

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Writing 1 to this register will clear it.

FOES bit (Oversize Error Status Flag)

[Setting condition]

HW: When too big frame is received. (When the received frame size is bigger than the setting in MRFSC.EMXS and MRFSC.PMXS)

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Writing 1 to this register will clear it.

Table 34.5 Transmission error list

Error type	Event	Status Bit
TX Bad CRC Insertion Status	A packet has been received for transmission by RMAC with an error and a wrong FCS has been inserted.	MEIS.TBCIS
TX CRC Error Status	A packet which included a presupposed valid FCS had a faulty FCS.	MEIS.TCES
Pause frame Retransmit Retry Over Status	A Pause or PFC retransmit count reaches the configured upper limit(MTPFC.APFRV) and Pause time expires.	MEIS.PFRROS
Transmission Stream Lost Status	An e-frame data has not been provided on time to RMAC.	MEIS.TSLS

Table 34.6 Reception error list

No	Error type	Event	Frame status bit	Status register
1	PHY data error	Received a data with error.	Bit 0	MEIS.PDES
2	PHY nibble alignment error	Received residual-bit.	Bit 1	MEIS.PNAES
3	FCS/mCRC error	Received unexpected FCS field value.	Bit 2	MEIS.FCMCES
4	Final fragment missing error	Occurs when an S-frame is received even though the previous fragment has been terminated by a mCRC.	Bit 3	MEIS.FFMES
5	C fragment count error	Occurs when a C fragment not corresponding to the current frame is received. For example: S0-C0-C0-C1	Bit 4	MEIS.CFCES
6	Fragment count error	Occurs when a C-frame is received with a bad fragment count field.	Bit 5	MEIS.FRCES
7	Reception partially out of operation mode	Occurs when a frame is being received while RMAC is moving out of operation mode.	Bit 6	MEIS.RPOOMS
8	Frame filtered	The frame did not pass the RMAC filters.	Bit 7	MEIS.FFS
9	Undersize error	The frame is too small.	Bit 8	MEIS.FUES
10	Oversize error	The frame is too big.	Bit 9	MEIS.FOES
11	Buffer overflow error	Occurs when the rx done signal has been deasserted during frame reception.	Bit 12	MEIS.REOES MEIS.RPOES
12	Frame Count Error Status	Occurs when a wrong S-frame order is received. For example: S0-S1-S2-S1	—	MEIS.FCES
13	Bad Fragment Error Status	Occurs when a C fragment has been detected without receiving S fragment.	—	MEIS.BFES
14	False Carrier Detection Status	Occurs when PHY-LSI has detected a false carrier on the line.	—	MEIS.FCDS

34.4.3.2 MEIE : Error Interrupt Enable Register

Base address: RMACm = 0x403C_B000 + 0x2000 × m (m = 0, 1)
 RMACm_NS = 0x503C_B000 + 0x2000 × m (m = 0, 1)

Offset address: 0x0204

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	FOEE	FUEE	FFE	RPOO ME	FRCE E	CFCE E	FFME E	FCMC EE	PNAE E	PDEE	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	CTLEE1 to CTLEE0		—	RPOE E	REOE E	FCEE	BFEE	TBCIE	TCEE	FCDE	PFRR OE	PREE	—	TSLE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	TSLE	Transmission Stream Lost Enable	R/W ¹
1	—	This bit is read as 0.	R
2	PREE	Pause or PFC Frame Reception Error Enable	R/W ¹
3	PFRR OE	Pause or PFC Frame Retransmit Retry Over Enable	R/W ¹
4	FCDE	False Carrier Detection Enable	R/W ¹
5	TCEE	Tx CRC Error Enable	R/W ¹
6	TBCIE	TX Bad CRC Insertion Enable	R/W ¹
7	BFEE	Bad Fragment Error Enable	R/W ¹
8	FCEE	Frame Count Error Enable	R/W ¹
9	REOEE	E-Frame Overflow Error Enable	R/W ¹
10	RPOEE	P-Frame Overflow Error Enable	R/W ¹
11	—	This bit is read as 0.	R
13:12	CTLEE1 to CTLEE0	Captured Timestamp Lost Error Enable n (n = 0 to 1)	R/W ¹
19:14	—	These bits are read as 0.	R
20	PDEE	PHY Data Error Enable	R/W ¹
21	PNAEE	PHY Nibble Alignment Error Enable	R/W ¹
22	FCMCEE	FCS/mCRC Error Enable	R/W ¹
23	FFMEE	Final Fragment Missing Error Enable	R/W ¹
24	CFCEE	C Fragment Count Error Enable	R/W ¹
25	FRCEE	Fragment Count Error Enable	R/W ¹
26	RPOOME	Reception Partially out of Operation Mode Enable	R/W ¹
27	FFE	Frame Filtered Enable	R/W ¹
28	FUEE	Undersize Error Enable	R/W ¹
29	FOEE	Oversize Error Enable	R/W ¹
31:30	—	These bits are read as 0.	R

Note 1. Read value differs from written value.

TSLE bit (Transmission Stream Lost Enable)

[Setting condition]

Writing 1 to one of this bit will set it.

[Clearing condition]

Writing 1 to MEID.TSLD will clear this bit.

PREE bit (Pause or PFC Frame Reception Error Enable)

[Setting condition]

Writing 1 to one of this bit will set it.

[Clearing condition]

Writing 1 to MEID.PRED will clear this bit.

PFRROE bit (Pause or PFC Frame Retransmit Retry Over Enable)

[Setting condition]

Writing 1 to one of this bit will set it.

[Clearing condition]

Writing 1 to MEID.PFRROD will clear this bit.

FCDE bit (False Carrier Detection Enable)

[Setting condition]

Writing 1 to one of this bit will set it.

[Clearing condition]

Writing 1 to MEID.FCDD will clear this bit.

TCEE bit (Tx CRC Error Enable)

[Setting condition]

Writing 1 to one of this bit will set it.

[Clearing condition]

Writing 1 to MEID.TCED will clear this bit.

TBCIE bit (TX Bad CRC Insertion Enable)

[Setting condition]

Writing 1 to one of this bit will set it.

[Clearing condition]

Writing 1 to MEID.TBCID will clear this bit.

BFEE bit (Bad Fragment Error Enable)

[Setting condition]

Writing 1 to one of this bit will set it.

[Clearing condition]

Writing 1 to MEID.BFED will clear this bit.

FCEE bit (Frame Count Error Enable)

[Setting condition]

Writing 1 to one of this bit will set it.

[Clearing condition]

Writing 1 to MEID.FCED will clear this bit.

REOOE bit (E-Frame Overflow Error Enable)

[Setting condition]

Writing 1 to one of this bit will set it.

[Clearing condition]

Writing 1 to MEID.REOED will clear this bit.

RPOEE bit (P-Frame Overflow Error Enable)

[Setting condition]

Writing 1 to one of this bit will set it.

[Clearing condition]

Writing 1 to MEID.RPOED will clear this bit.

CTLEEn bits (Captured Timestamp Lost Error Enable n (n = 0 to 1))

[Setting condition]

Writing 1 to one of this bit will set it.

[Clearing condition]

Writing 1 to MEID.CTLED will clear this bit.

PDEE bit (PHY Data Error Enable)

[Setting condition]

Writing 1 to one of this bit will set it.

[Clearing condition]

Writing 1 to MEID.PDED will clear this bit.

PNAEE bit (PHY Nibble Alignment Error Enable)

[Setting condition]

Writing 1 to one of this bit will set it.

[Clearing condition]

Writing 1 to MEID.PNAED will clear this bit.

FCMCEE bit (FCS/mCRC Error Enable)

[Setting condition]

Writing 1 to one of this bit will set it.

[Clearing condition]

Writing 1 to MEID.FCMCED will clear this bit.

FFMEE bit (Final Fragment Missing Error Enable)

[Setting condition]

Writing 1 to one of this bit will set it.

[Clearing condition]

Writing 1 to MEID.FFMED will clear this bit.

CFCEE bit (C Fragment Count Error Enable)

[Setting condition]

Writing 1 to one of this bit will set it.

[Clearing condition]

Writing 1 to MEID.CFCED will clear this bit.

FRCEE bit (Fragment Count Error Enable)

[Setting condition]

Writing 1 to one of this bit will set it.

[Clearing condition]

Writing 1 to MEID.FRCEd will clear this bit.

RPOOME bit (Reception Partially out of Operation Mode Enable)

[Setting condition]

Writing 1 to one of this bit will set it.

[Clearing condition]

Writing 1 to MEID.RPOOMD will clear this bit.

FFE bit (Frame Filtered Enable)

[Setting condition]

Writing 1 to one of this bit will set it.

[Clearing condition]

Writing 1 to MEID.FFD will clear this bit.

FUEE bit (Undersize Error Enable)

[Setting condition]

Writing 1 to one of this bit will set it.

[Clearing condition]

Writing 1 to MEID.FUED will clear this bit.

FOEE bit (Oversize Error Enable)

[Setting condition]

Writing 1 to one of this bit will set it.

[Clearing condition]

SW: Writing 1 to MEID.FOED will clear this bit.

34.4.3.3 MEID : Error Interrupt Disable Register

Base address: RMACm = 0x403C_B000 + 0x2000 × m (m = 0, 1)
 RMACm_NS = 0x503C_B000 + 0x2000 × m (m = 0, 1)

Offset address: 0x0208

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	FOED	FUED	FFD	RPOOMD	FRCEd	CFCEd	FFMEd	FCMCED	PNAEd	PDED	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	CTLED1 to CTLED0	—	RPOEd	REOEd	FCED	BFED	TBCID	TCED	FCDD	PFRR OD	PRED	—	—	TSLD
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	TSLD	Transmission Stream Lost Disable Writing 1 to this register will clear MEIE.TSLE register.	R/W ¹
1	—	This bit is read as 0.	R
2	PRED	Pause or PFC Frame Reception Error Disable Writing 1 to this register will clear MEIE.PREE register.	R/W ¹
3	PFRR OD	Pause or PFC Frame Retransmit Retry Over Disable Writing 1 to this register will clear MEIE.PFRROE register.	R/W ¹

Bit	Symbol	Function	R/W
4	FCDD	False Carrier Detection Disable Writing 1 to this register will clear MEIE.FCDE register.	R/W ^{*1}
5	TCED	Tx CRC Error Disable Writing 1 to this register will clear MEIE.TCEE register.	R/W ^{*1}
6	TBCID	TX Bad CRC Insertion Disable Writing 1 to this register will clear MEIE.TBCIE register.	R/W ^{*1}
7	BFED	Bad Fragment Error Disable Writing 1 to this register will clear MEIE.BFEE register.	R/W ^{*1}
8	FCED	Frame Count Error Disable Writing 1 to this register will clear MEIE.FCEE register.	R/W ^{*1}
9	REOED	E-Frame Overflow Error Disable Writing 1 to this register will clear MEIE.REOEE register.	R/W ^{*1}
10	RPOED	P-Frame Overflow Error Disable Writing 1 to this register will clear MEIE.RPOEE register.	R/W ^{*1}
11	—	This bit is read as 0.	R
13:12	CTLED1 to CTLED0	Captured Timestamp Lost Error Disable n (n = 0 to 1) Writing 1 to this register will clear MEIE.CTLEE register.	R/W ^{*1}
19:14	—	These bits are read as 0.	R
20	PDED	PHY Data Error Disable Writing 1 to this register will clear MEIE.PDEE register.	R/W ^{*1}
21	PNAED	PHY Nibble Alignment Error Disable Writing 1 to this register will clear MEIE.PNAEE register.	R/W ^{*1}
22	FCMCED	FCS/mCRC Error Disable Writing 1 to this register will clear MEIE.FCMCEE register.	R/W ^{*1}
23	FFMED	Final Fragment Missing Error Disable Writing 1 to this register will clear MEIE.FFMEE register.	R/W ^{*1}
24	CFCED	C Fragment Count Error Disable Writing 1 to this register will clear MEIE.CFCEE register.	R/W ^{*1}
25	FRCED	Fragment Count Error Disable Writing 1 to this register will clear MEIE.FCEE register.	R/W ^{*1}
26	RPOOMD	Reception Partially out of Operation Mode Disable Writing 1 to this register will clear MEIE.RPOOME register.	R/W ^{*1}
27	FFD	Frame Filtered Disable Writing 1 to this register will clear MEIE.FFE register.	R/W ^{*1}
28	FUED	Undersize Error Disable Writing 1 to this register will clear MEIE.FUEE register.	R/W ^{*1}
29	FOED	Oversize Error Disable Writing 1 to this register will clear MEID.FOEE register.	R/W ^{*1}
31:30	—	These bits are read as 0.	R

Note 1. Read value differs from written value.

34.4.3.4 MMIS0 : Monitoring Interrupt Status Register 0

Base address: $\text{RMACm} = 0x403C_B000 + 0x2000 \times m$ ($m = 0, 1$)
 $\text{RMACm_NS} = 0x503C_B000 + 0x2000 \times m$ ($m = 0, 1$)

Offset address: 0x0210

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	VFRS	LVFS	LVSS	PIDS	PLSCS
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PLSCS ^{*2*3}	PHY Link Signal Change Status Flag	R/W ^{*1}
1	PIDS ^{*2*4}	PHY Interrupt Detection Status Flag	R/W ^{*1}
2	LVSS ^{*2*5}	Link Verification Succeed Status Flag	R/W ^{*1}
3	LVFS ^{*2*5}	Link Verification Failed Status Flag	R/W ^{*1}
4	VFRS ^{*2*5}	Verify Frame Reception Status Flag	R/W ^{*1}
31:5	—	These bits are read as 0.	R

Note: This interrupt register is mapped to rmc_mmis0_int interrupt wire.

Note 1. Read value differs from written value.

Note 2. SW: Writing 0 to this bit has no effect.

Note 3. SW: To check the current link state, refer to the link status pin state bit in the PHY link status register (MPIM.PLS).

Note 4. HW: This flag could be asserted unintentionally before configuring PHY interrupt polarity (MPIC.PIP) into the right value. In the phase, this flag should be ignored.

Note 5. It's no effect when suspending, e.g. an upper layer (e.g. MHD) is not in OPERATION state.

PLSCS bit (PHY Link Signal Change Status Flag)

[Setting condition]

HW: When the value of link status pin (rmc_phy_link port) is changed.

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Writing 1 to this bit will clear it.

PIDS bit (PHY Interrupt Detection Status Flag)

[Setting condition]

HW: PHY interrupt pin (rmc_phy_int port) is active level.

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Writing 1 to this bit will clear it.

LVSS bit (Link Verification Succeed Status Flag)

[Setting condition]

HW: When Link Verification process succeed.

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Writing 1 to this bit will clear it.

LVFS bit (Link Verification Failed Status Flag)

[Setting condition]

HW: When Link Verification process failed. It means that the remote node is not able to handle Preemptable frames.

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Writing 1 to this bit will clear it.

VFRS bit (Verify Frame Reception Status Flag)

[Setting condition]

HW: When valid verify frame has been received.

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Writing 1 to this bit will clear it.

34.4.3.5 MMIE0 : Monitoring Interrupt Enable Register 0

Base address: RMACm = 0x403C_B000 + 0x2000 × m (m = 0, 1)
 RMACm_NS = 0x503C_B000 + 0x2000 × m (m = 0, 1)

Offset address: 0x0214

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	VFRE	LVFE	LVSE	PIDE	PLSC E
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PLSCE	PHY Link Signal Change Enable Writing 1 to this register will set it. Writing 0 to this register has no effect.	R/W ¹
1	PIDE	PHY Interrupt Detection Enable Writing 1 to this register will set it. Writing 0 to this register has no effect.	R/W ¹
2	LVSE	Link Verification Succeed Enable Writing 1 to this register will set it. Writing 0 to this register has no effect.	R/W ¹
3	LVFE	Link Verification Failed Enable Writing 1 to this register will set it. Writing 0 to this register has no effect.	R/W ¹
4	VFRE	Verify Frame Reception Enable Writing 1 to this register will set it. Writing 0 to this register has no effect.	R/W ¹
31:5	—	These bits are read as 0.	R

Note 1. Read value differs from written value.

34.4.3.6 MMID0 : Monitoring Interrupt Disable Register 0

Base address: RMACm = 0x403C_B000 + 0x2000 × m (m = 0, 1)
 RMACm_NS = 0x503C_B000 + 0x2000 × m (m = 0, 1)

Offset address: 0x0218

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	VFRD	LVFD	LVSD	PIDD	PLSCD
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PLSCD	PHY Link Signal Change Disable Writing 1 to this register will clear the corresponding bit in the corresponding enable register. Writing 0 to this register has no effect.	R/W *1
1	PIDD	PHY Interrupt Detection Disable Writing 1 to this register will clear the corresponding bit in the corresponding enable register. Writing 0 to this register has no effect.	R/W *1
2	LVSD	Link Verification Succeed Disable Writing 1 to this register will clear the corresponding bit in the corresponding enable register. Writing 0 to this register has no effect.	R/W *1
3	LVFD	Link Verification Failed Disable Writing 1 to this register will clear the corresponding bit in the corresponding enable register. Writing 0 to this register has no effect.	R/W *1
4	VFRD	Verify Frame Reception Disable Writing 1 to this register will clear the corresponding bit in the corresponding enable register. Writing 0 to this register has no effect.	R/W *1
31:5	—	These bits are read as 0.	R

Note 1. Read value differs from written value.

34.4.3.7 MMIS1 : Monitoring Interrupt Status Register 1

Base address: RMACm = 0x403C_B000 + 0x2000 × m (m = 0, 1)
 RMACm_NS = 0x503C_B000 + 0x2000 × m (m = 0, 1)

Offset address: 0x0220

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	PPRACS	PAACS	PWACS	PRACS
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PRACS*2	PHY Read Access Completed Status Flag	R/W*1
1	PWACS*2	PHY Write Access Completed Status Flag	R/W*1
2	PAACS*2	PHY Address Access Completed Status Flag	R/W*1
3	PPRACS*2	PHY Post-Read Access Completed Status Flag	R/W*1

Bit	Symbol	Function	R/W
31:4	—	These bits are read as 0.	R

Note: This interrupt register is mapped to rmc_mdio_int interrupt wire.

Note 1. Read value differs from written value.

Note 2. Writing 0 to this bit has no effect.

PRACS bit (PHY Read Access Completed Status Flag)

[Setting condition]

HW: When PHY read access has completed. See section 8.4.

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Writing 1 to this bit will clear it.

PWACS bit (PHY Write Access Completed Status Flag)

[Setting condition]

HW: When PHY write access has completed. See section 8.4.

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Writing 1 to this bit will clear it.

PAACS bit (PHY Address Access Completed Status Flag)

[Setting condition]

HW: When PHY address access has completed. See section 8.4.

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Writing 1 to this bit will clear it.

PPRACS bit (PHY Post-Read Access Completed Status Flag)

[Setting condition]

HW: When PHY post-read access has completed. See section 8.4.

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Writing 1 to this bit will clear it.

34.4.3.8 MMIE1 : Monitoring Interrupt Enable Register 1

Base address: RMACm = 0x403C_B000 + 0x2000 × m (m = 0, 1)
 RMACm_NS = 0x503C_B000 + 0x2000 × m (m = 0, 1)

Offset address: 0x0224

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	PPRA CE	PAAC E	PWAC E	PRAC E
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PRACE ^{*2}	PHY Read Access Completed Enable	R/W ^{*1}
1	PWACE ^{*2}	PHY Write Access Completed Enable	R/W ^{*1}
2	PAACE ^{*2}	PHY Address Access Completed Enable	R/W ^{*1}
3	PPRACE ^{*2}	PHY Post-Read Access Completed Enable	R/W ^{*1}
31:4	—	These bits are read as 0.	R

Note 1. Read value differs from written value.

Note 2. SW: Writing 0 to this bit has no effect.

PRACE bit (PHY Read Access Completed Enable)

[Setting condition]

HW: Writing 1 to this bit will set it.

[Clearing condition]

HW: Writing 1 to MMID1.PRACD register will clear this bit.

PWACE bit (PHY Write Access Completed Enable)

[Setting condition]

HW: Writing 1 to this bit will set it.

[Clearing condition]

HW: Writing 1 to MMID1.PWACD register will clear this bit.

PAACE bit (PHY Address Access Completed Enable)

[Setting condition]

HW: Writing 1 to this bit will set it.

[Clearing condition]

HW: Writing 1 to MMID1.PAACD register will clear this bit.

PPRACE bit (PHY Post-Read Access Completed Enable)

[Setting condition]

HW: Writing 1 to this bit will set it.

[Clearing condition]

HW: Writing 1 to MMID1.PPRACD register will clear this bit.

34.4.3.9 MMID1 : Monitoring Interrupt Disable Register 1

Base address: RMACm = 0x403C_B000 + 0x2000 × m (m = 0, 1)
 RMACm_NS = 0x503C_B000 + 0x2000 × m (m = 0, 1)

Offset address: 0x0228

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	PPRA CD	PAAC D	PWAC D	PRAC D
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PRACD ^{*1}	PHY Read Access Completed Disable Writing 1 to one of this bit will clear MMIE1.PRACE register.	W
1	PWACD ^{*1}	PHY Write Access Completed Disable Writing 1 to one of this bit will clear MMIE1.PWACE register.	W
2	PAACD ^{*1}	PHY Address Access Completed Disable Writing 1 to one of this bit will clear MMIE1.PAACE register.	W
3	PPRACD ^{*1}	PHY Post-Read Access Completed Disable Writing 1 to one of this bit will clear MMIE1.PPRACE register.	W
31:4	—	These bits are read as 0.	R

Note 1. Writing 0 to this register has no effect.

34.4.3.10 MMIS2 : Monitoring Interrupt Status Register 2

Base address: RMACm = 0x403C_B000 + 0x2000 × m (m = 0, 1)
 RMACm_NS = 0x503C_B000 + 0x2000 × m (m = 0, 1)

Offset address: 0x0230

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	LPIDIS	LPIAIS	MPDIS
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	MPDIS ^{*2*3*4*5}	Magic Packet Detection Interrupt Status Flag	R/W ^{*1}
1	LPIAIS ^{*2*5}	LPI Assertion Interrupt Status Flag	R/W ^{*1}
2	LPIDIS ^{*2*5}	LPI De-Assertion Interrupt Status Flag	R/W ^{*1}
31:3	—	These bits are read as 0.	R

Note:

- This interrupt register is mapped to rmc_mp_int, rmc_lpi_start_int and rmc_lpi_stop_int interrupt wire.
- This register's value goes through the synchronizer between clk and clk_phy_rx.
 The result of write access is reflected to each SFR bit after "5clk + 4clk_phy_rx" in Worst Case in RMAC.
 Do not make write access continuously. Please wait following time for re-write access.

Note 1. Read value differs from written value.

Note 2. SW: Writing 0 to this bit has no effect.

Note 3. HW: Magic Packet in Preemptable frames is ignored.

Note 4. HW: CRC of magic packet is not checked.

Note 5. HW: It's no effect when suspending.

MPDIS bit (Magic Packet Detection Interrupt Status Flag)

[Setting condition]

HW: When a Magic Packet has been detected.

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Writing 1 to this bit will clear it.

LPIAIS bit (LPI Assertion Interrupt Status Flag)

[Setting condition]

HW: When LPI request is detected. This register is continuously asserted when LPI request continues.

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Writing 1 to this bit will clear it.

LPIDIS bit (LPI De-Assertion Interrupt Status Flag)

[Setting condition]

HW: When LPI request deassertion.

[Clearing condition]

HW: Being in RESET mode will clear this register.

SW: Writing 1 to this bit will clear it.

34.4.3.11 MMIE2 : Monitoring Interrupt Enable Register 2

Base address: RMACm = 0x403C_B000 + 0x2000 × m (m = 0, 1)
 RMACm_NS = 0x503C_B000 + 0x2000 × m (m = 0, 1)

Offset address: 0x0234

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	LPIDIE	LPIAIE	MPDIE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	MPDIE	Magic Packet™ Detection Interrupt Enable Writing 1 to this register will set it. Writing 0 to this register has no effect.	R/W ¹
1	LPIAIE	LPI Assertion Interrupt Enable Writing 1 to this register will set it. Writing 0 to this register has no effect.	R/W ¹
2	LPIDIE	LPI De-Assertion Interrupt Enable Writing 1 to this register will set it. Writing 0 to this register has no effect.	R/W ¹
31:3	—	These bits are read as 0.	R

Note: This register's value goes through the synchronizer between clk and clk_phy_rx.
 The result of write access is reflected to each SFR bit after "5clk + 4clk_phy_rx" in Worst Case in RMAC.
 Do not make write access continuously. Please wait following time for re-write access.

Note 1. Read value differs from written value.

34.4.3.12 MMID2 : Monitoring Interrupt Disable Register 2

Base address: RMACm = 0x403C_B000 + 0x2000 × m (m = 0, 1)
 RMACm_NS = 0x503C_B000 + 0x2000 × m (m = 0, 1)

Offset address: 0x0238

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	LPIDID	LPIAID	MPDID
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	MPDID	Magic Packet™ Detection Interrupt Disable Writing 1 to this register will clear the corresponding bit in the corresponding enable register. Writing 0 to this register has no effect.	W
1	LPIAID	LPI Assertion Interrupt Disable Writing 1 to this register will clear the corresponding bit in the corresponding enable register. Writing 0 to this register has no effect.	W
2	LPIDID	LPI De-Assertion Interrupt Disable Writing 1 to this register will clear the corresponding bit in the corresponding enable register. Writing 0 to this register has no effect.	W
31:3	—	These bits are read as 0.	R

Note: This register's value goes through the synchronizer between clk and clk_phy_rx.
 The result of write access is reflected to each SFR bit after "5clk + 4clk_phy_rx" in Worst Case in RMAC.
 Do not make write access continuously. Please wait following time for re-write access.

34.5 Register Utilizations

34.5.1 Operation Modes

Operational mode is controlled. Please refer to IPS. RMAC received a mode control signal for controlling operational mode of RMAC.

34.5.1.1 Frame Handling

In transmission, this IP transmits all frames from an upper module (e.g. MHD) when this IP is in operation state, e.g. when MHD is in OPERATION state. This IP receives an e-frame and a p-frame from the MHD.

In reception, this IP only receives frames from PHY when this IP is in operation state, e.g. when MHD is in OPERATION state., and outputs them to the upper module (e.g. MHD). In other modes, this IP discards all received frames from PHY internally, and does not output them to the upper module.

When the system supports preemption and the Mode change is executed, this IP of the Rx side behaves like the following figures.

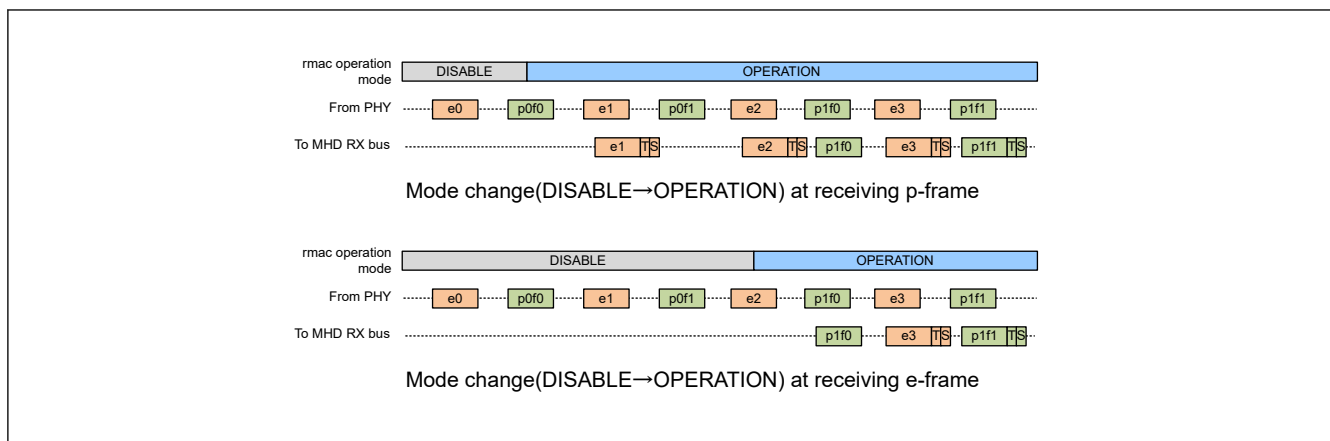


Figure 34.3 Mode change (DISABLE→OPERATION) on Rx

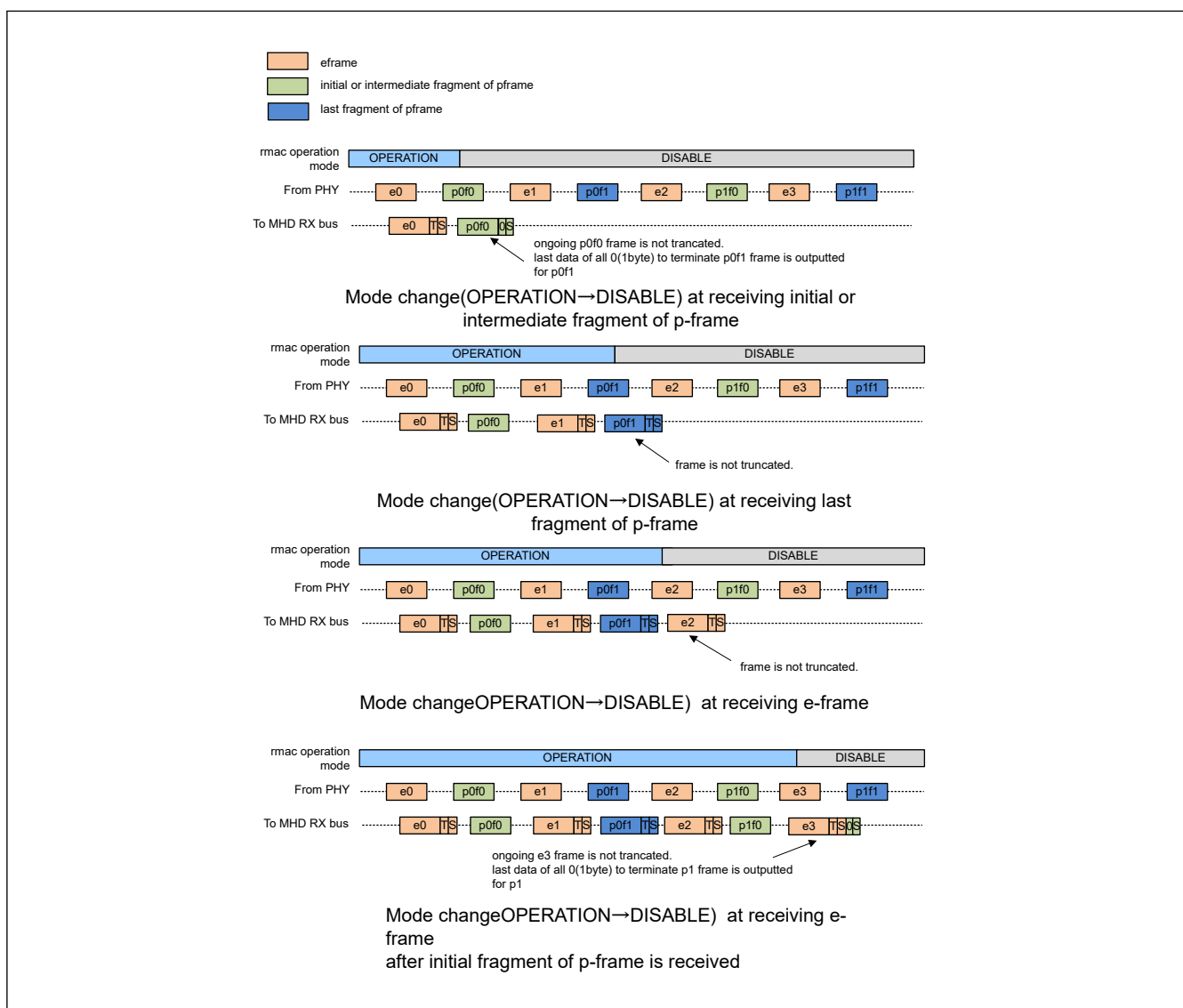


Figure 34.4 Mode change (OPERATION→DISABLE) on Rx

34.5.2 Software Flows

Restrictions:

SW: Please follow to the flow in this section.

34.5.2.1 Software Flow Legend

Software flow legend is described in [Figure 34.5](#).

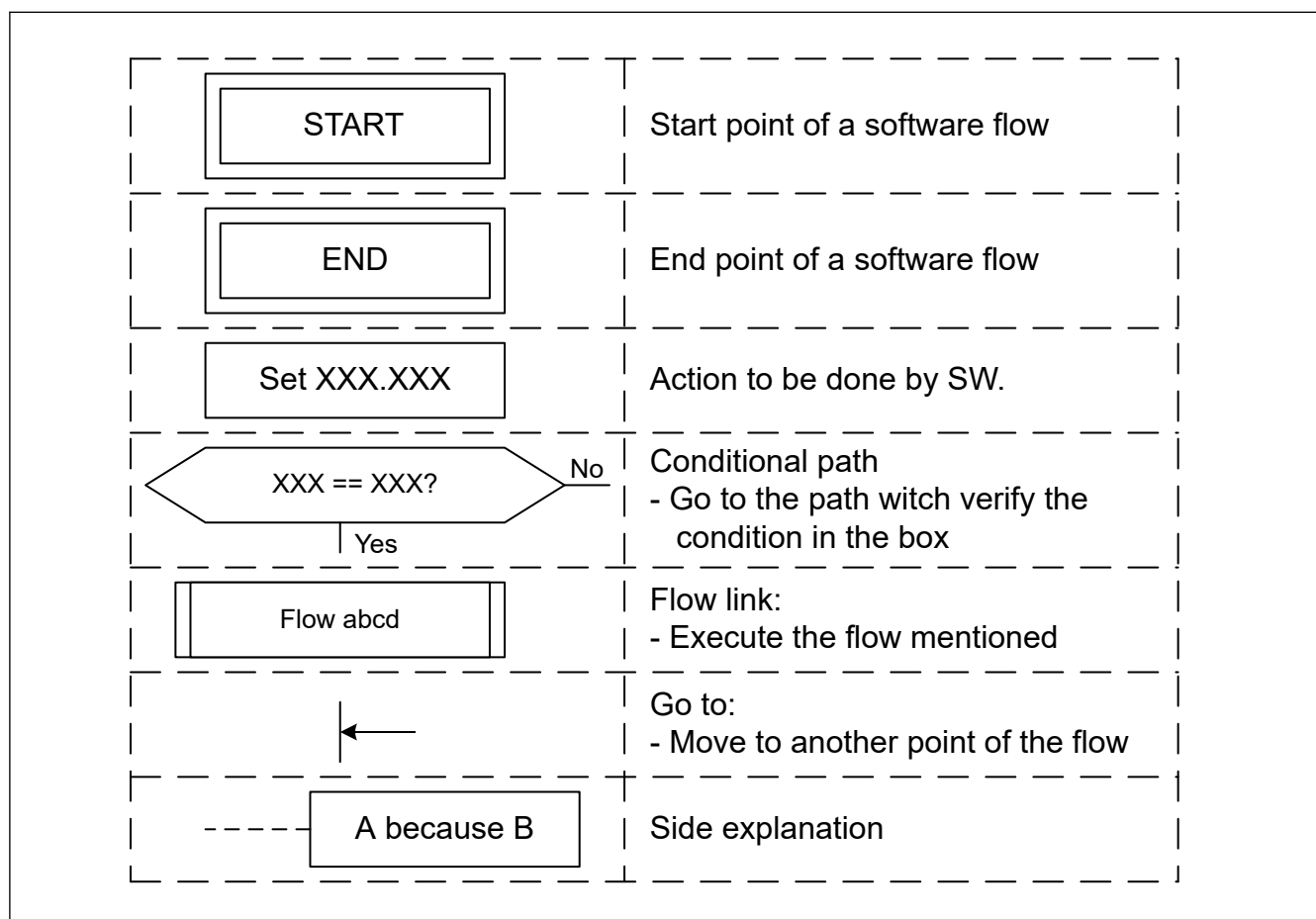


Figure 34.5 Software flow legend

34.5.2.2 Initialization Flow

The figure below shows the initialization flow.

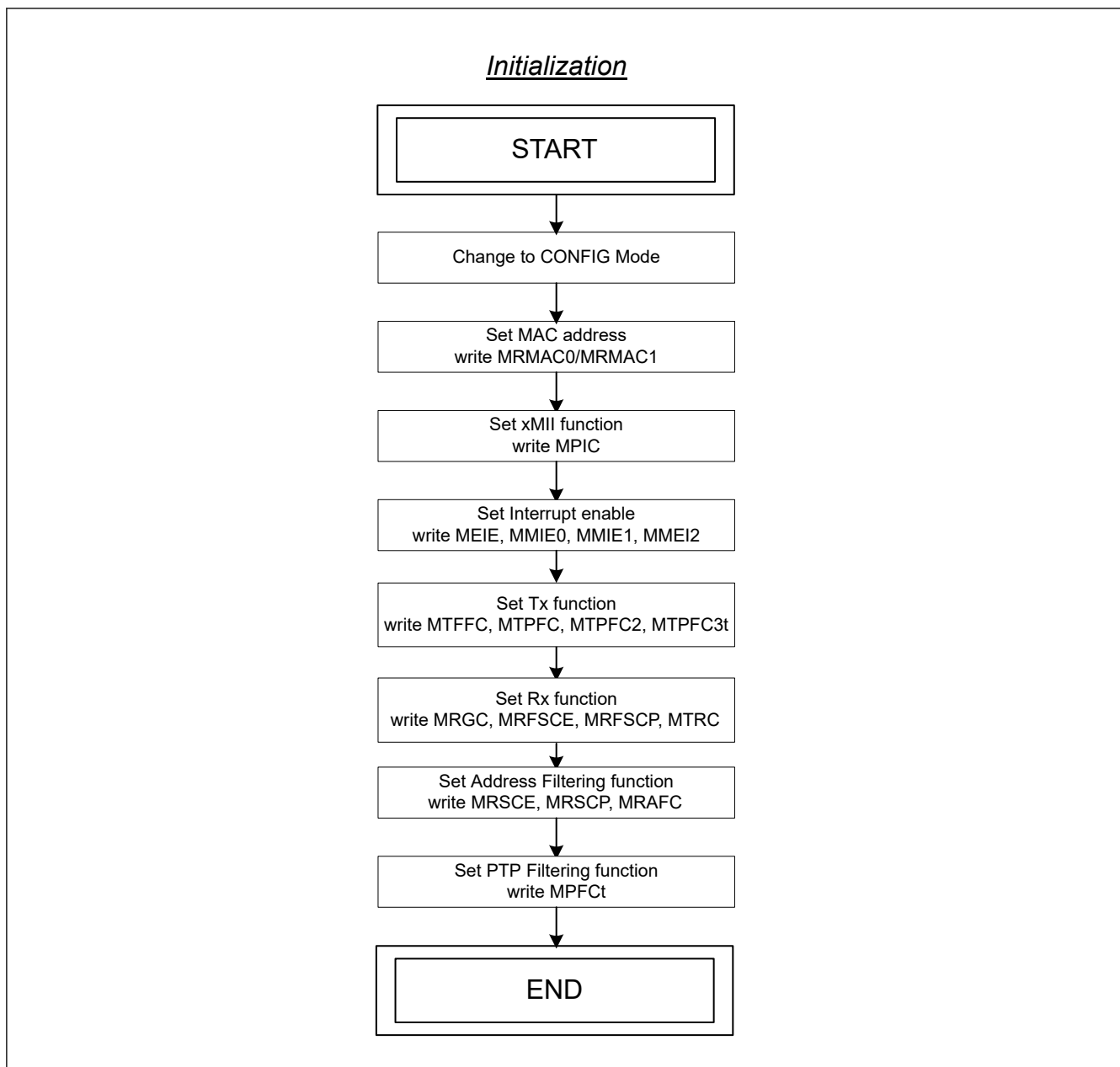


Figure 34.6 Initialization flow

34.5.2.3 Interrupt Handling Flow

Interrupt handling flow can be used in any mode except RESET mode.

The figure below shows the interrupt handling flow.

Restrictions:

This flow is not usable in RESET modes.

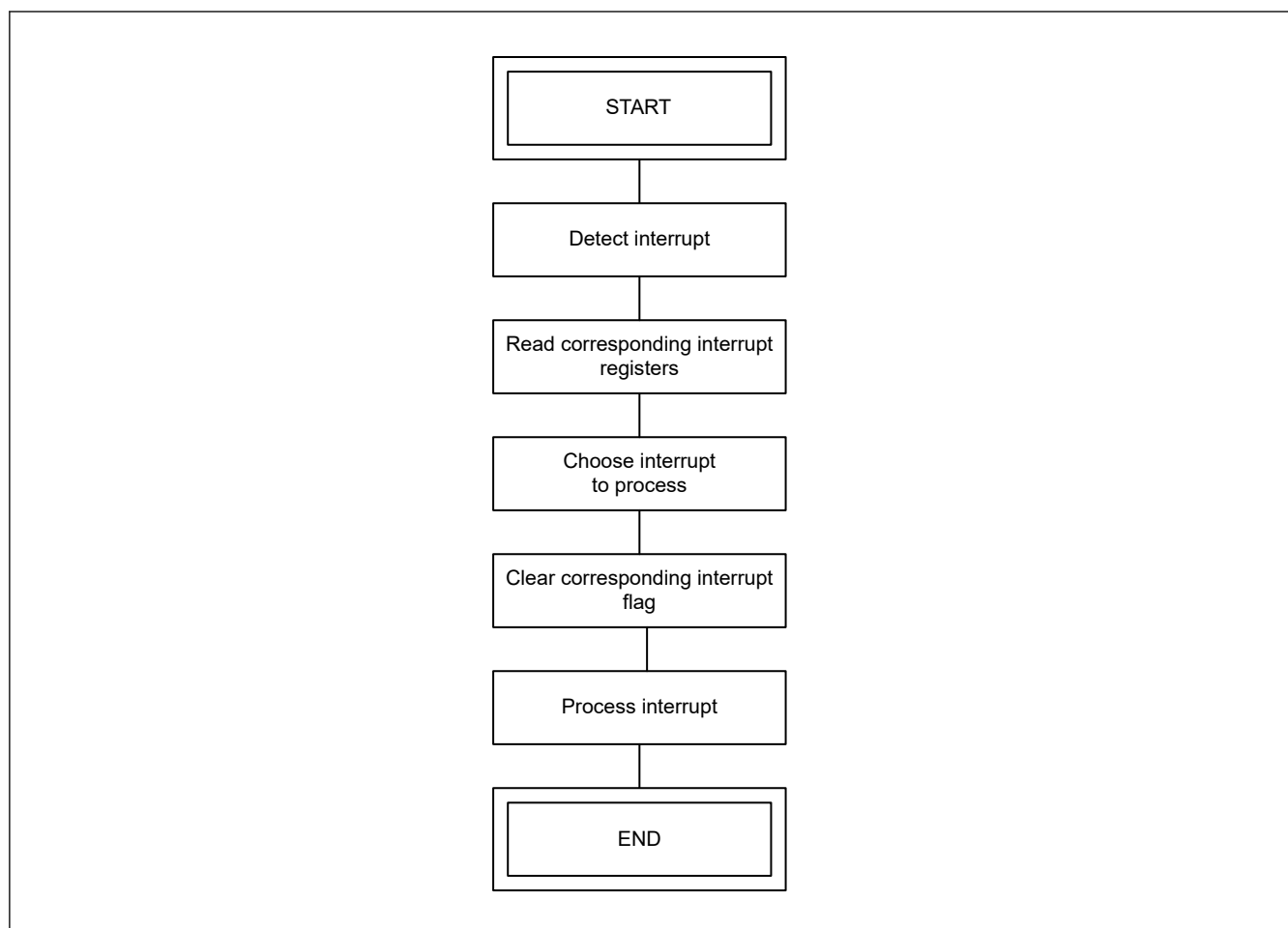


Figure 34.7 Interrupt handling flow

34.5.2.4 PHY MDIO and eMDIO Access Flows

The figure below shows the PHY MDIO and eMDIO access flow.

Restrictions:

This flow is not usable in RESET modes.

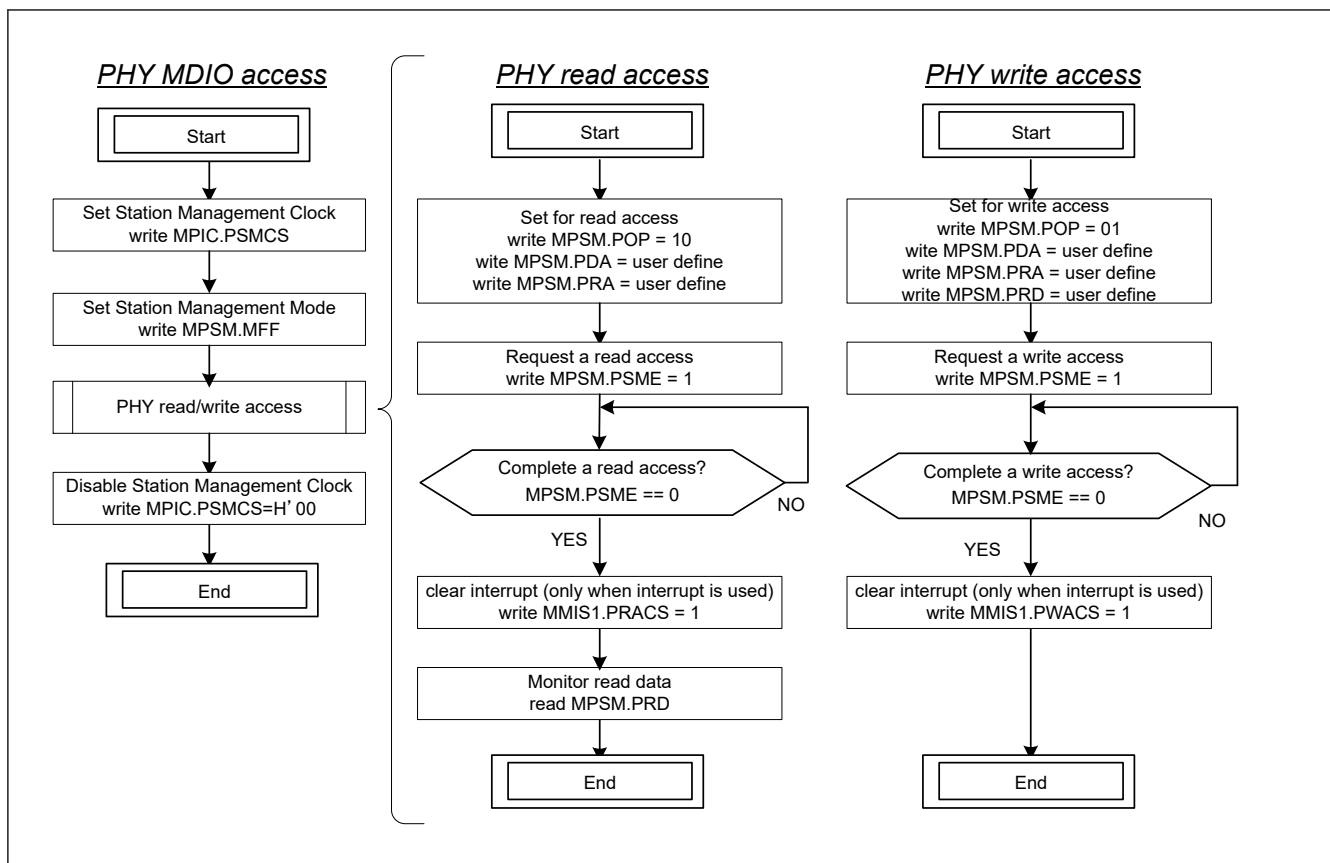


Figure 34.8 PHY MDIO access flow

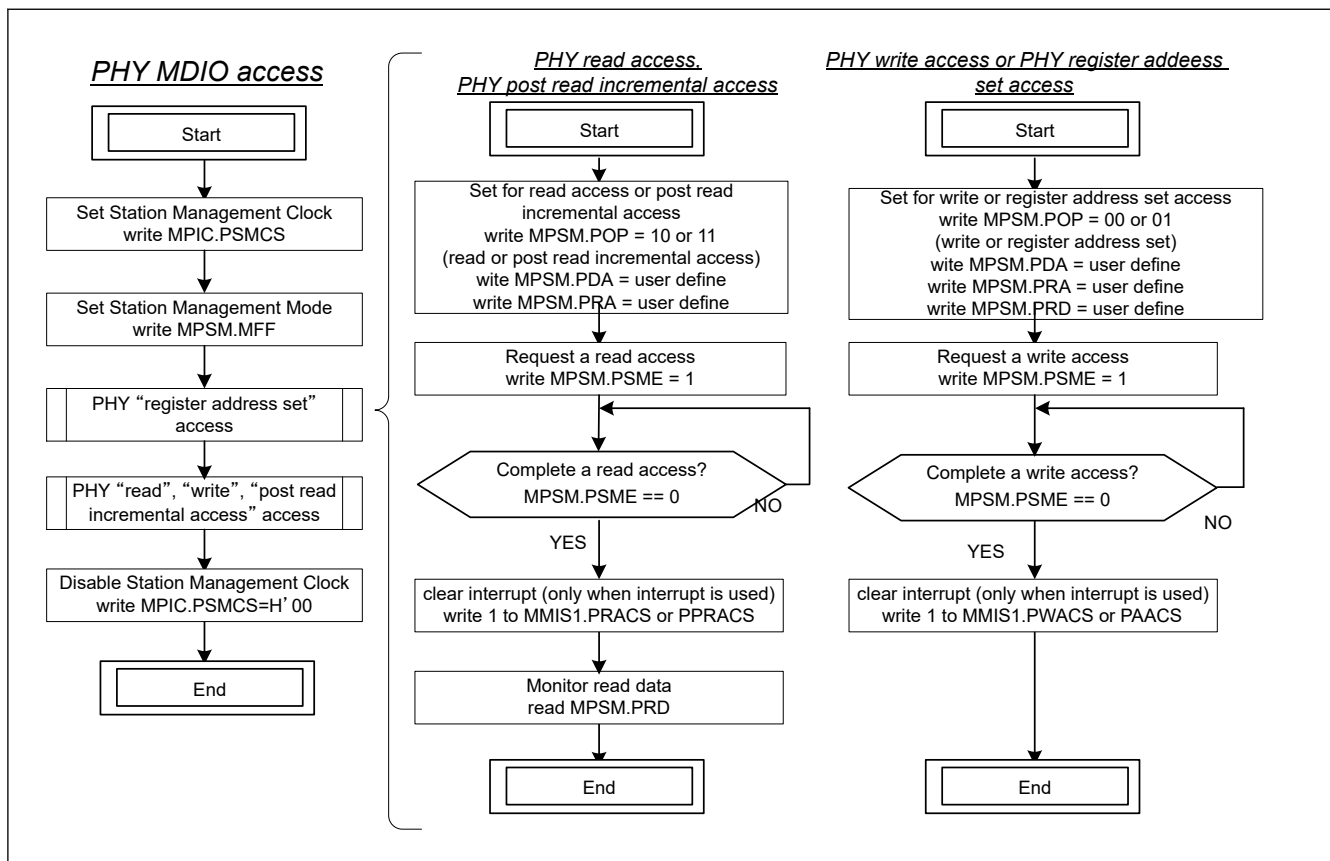


Figure 34.9 PHY eMDIO access flow

34.5.2.5 LPI Transition for Tx Port

Figure 34.10 shows the LPI transition flow. When resumed from LPI, the transmission should be requested until PHY wake-up completed.

Restrictions:

This flow is not usable in RESET modes.

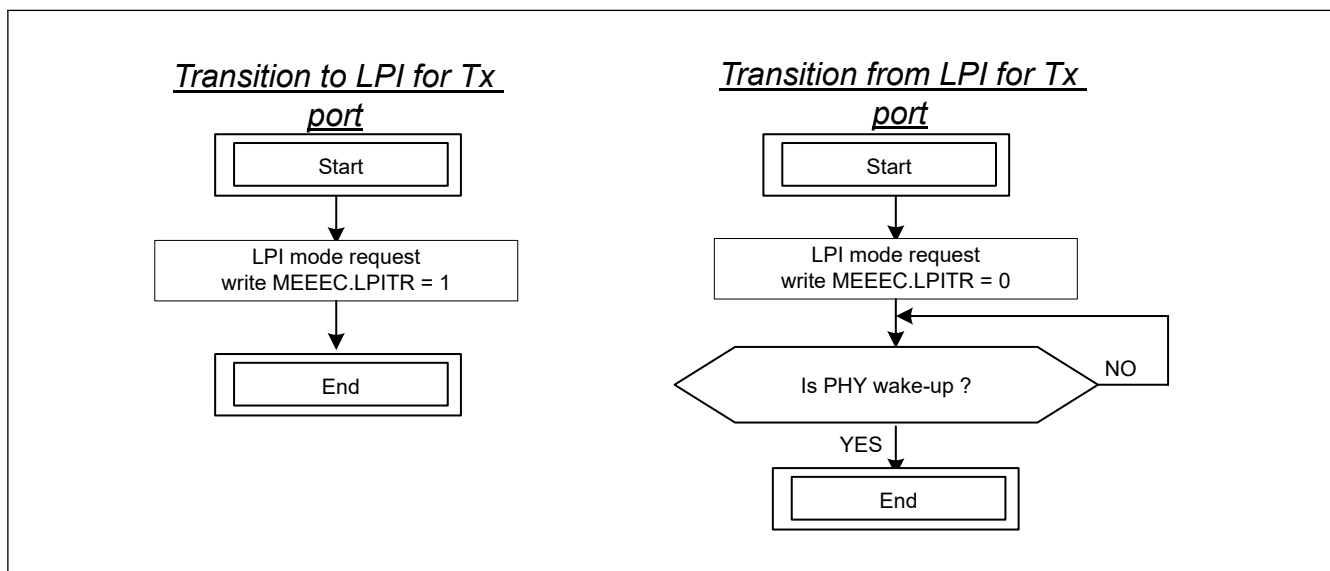


Figure 34.10 LPI transition for Tx port

34.5.2.6 LINK Verification

The figure below shows Link Verification flow.

Restrictions:

This flow is usable in OPERATIONAL mode.

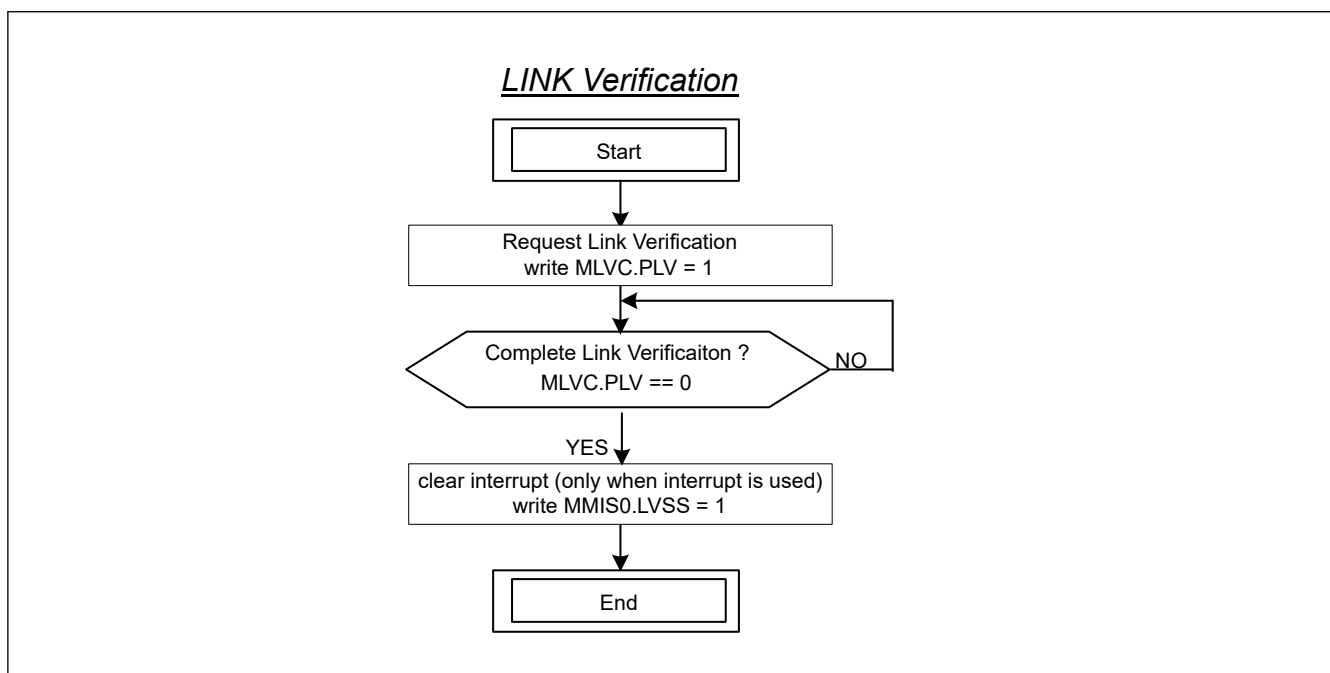


Figure 34.11 Link Verification flow

34.5.2.7 Called Software Flows

The flows described in this section can only be called from other flows thanks to a “flow link” box (Figure 34.5) and cannot be used alone.

34.5.3 Register Writable Without Software Flow

This section describes registers that have not been described so far. These registers can be changed dynamically. (However, it is necessary that the initial settings such as the clock enabling have been completed.)

34.6 Functional Details

34.6.1 Preemption

“Preemption” is the function specified in IEEE802.3.br. This function separates an ethernet frame to several fragments like Figure 34.12. This figure shows an example of preemption of 3 fragments. Preemption for only two fragments or over three fragments is possible.

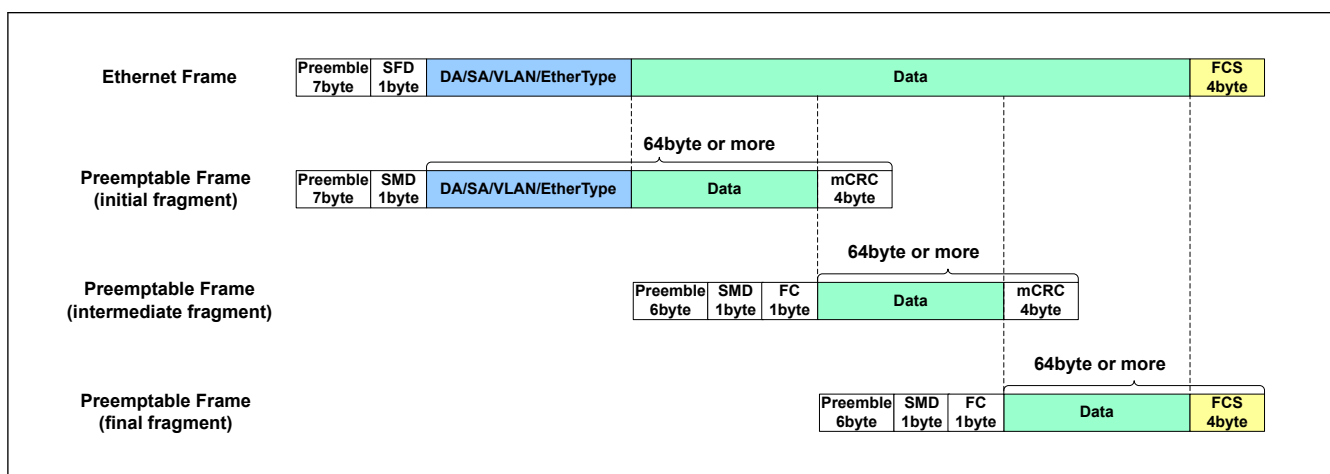


Figure 34.12 Outline of preemption

This function is handled by this IP and an upper module (e.g. MHD).

On the Tx Side, This IP only generates the fields of “Preamble”, “SMD”, “FC” and “mCRC” of the Preemptable frame complying with the rule of IEEE802.3.br. The upper module (e.g. MHD) has to separate a transmit frame appropriately.

On the Rx Side, this IP removes the fields of “Preamble”, “SMD”, “FC” and “mCRC” of the Preemptable frame and outputs the fields of “DA/SA/VLAN/EtherType”, “Data” and “FCS” (To output FCS part whether or not it is selectable by MRGC.RCPT) to upper module. The upper module (e.g. MHD) has to re-assemble a received frame.

“Preemption” is effective to prevent decreasing of throughput for high priority frame by transferring of low priority frames. Figure 34.13 shows one of examples. The software does not care the length of Low priority frames.

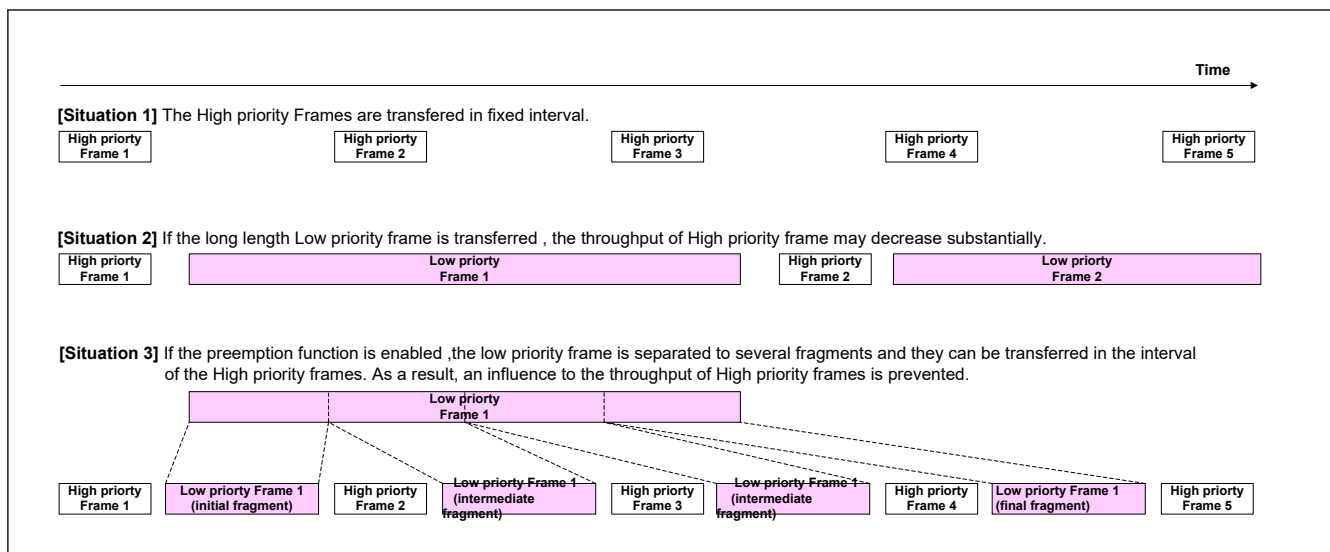


Figure 34.13 Effective of Preemption

To use “Preemption”, the Link Verification process should be executed on the system level.

Note: This IP does not hold an information whether the preemption function is valid or not in the system.

34.6.1.1 Link Verification

This IP supports Link Verification process of [Figure 34.14](#).

By setting MLVC.PASE this IP answers to a Verify frame by a Response frame to authorize the linked IP that preemption is possible. When a valid Verify frame is received, MMIS.VFRS is asserted as an interrupt.

When MLVC.PLV is set to 1, this IP starts to transmit Verify frame. The transmission attempts 3 times with the configured interval (MLVC.LVT) until Response frame is received from remote node. If the received Response frame is valid, MMIS.LVSS is asserted as an interrupt.

If not receiving Response frame after attempted 3 times, MMIS.LVFS is asserted as an interrupt.

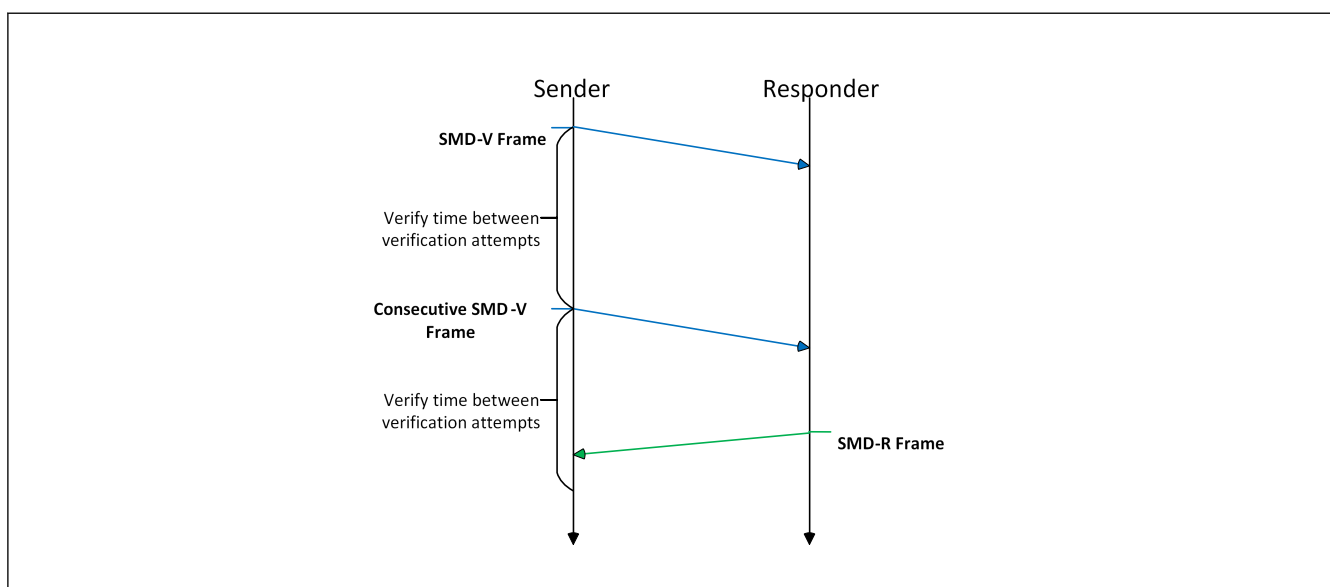


Figure 34.14 Example of link verification flow

34.6.2 Flow Control

This IP supports the following two types of Flow Control functions.

1. Flow Control specified in IEEE 802.3 for a full-duplex operation using a PAUSE frame.

2. Priority Based Flow Control specified in IEEE 802.1Qbb for a full-duplex operation using a PFC frame.

34.6.2.1 Flow Control by PAUSE Frame

This Flow Control is only for a full-duplex operation and this consists of Transmission and Reception of a PAUSE frame.

34.6.2.1.1 Transmission of PAUSE Frame

1. While MTFFC.FCM = 0 and MTPFC.PT[15:0] ≠ 0, the transmission function of a PAUSE frame is enabled. This IP transmits a PAUSE frame as described in Figure 34.15 by asserting a transmission request. The value of MTPFC.PT[15:0] is used for “PAUSE_TIME” field, and MRMAC0 and MRMAC1 are used for “SA” field.

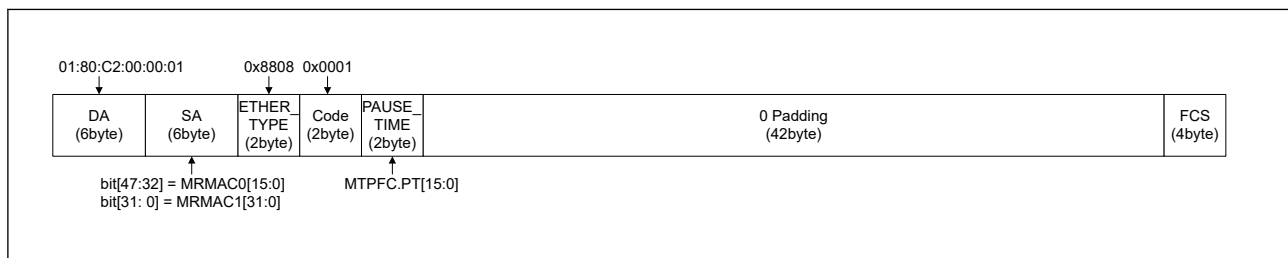


Figure 34.15 PAUSE frame format

- Auto PAUSE Frame
When MTPFC.PFM = 0, this IP uses a hardware pause transmission request as a transmit request. After transmitting the PAUSE Frame, MAPFTCT is incremented.
- Manual PAUSE Frame
When MTPFC.PFM = 1, this IP uses MTPFC2.MPFR as a transmit request. After transmitting the PAUSE Frame, MMPFTCT is incremented.

When the transmit request of the PAUSE frame is asserted while the IP is transmitting the other frame, the PAUSE frame is transmitted after the transmitting frame.

2. While the transmit request is asserted, a PAUSE frame is continuously re-transmitted at a fixed interval of time configured by MTPFC.PFRT. If the number of re-transmission reaches the value of MTPFC.PFRLV, this IP sets MEIS.PFRROS to 1 for an interrupt. Even if the interrupt is detected, this IP continues re-transmission of a PAUSE frame while the transmit request is asserted.
3. After deasserting the transmission request, this IP stops a transmission of PAUSE frame. At that time, when MTPFC2.PFTTZ = 1 and an internal counter to count a pausing-period exceeds the value of MTPFC.PFRT (*), this IP transmits one more a PAUSE frame with “PAUSE_TIME” field as 0.

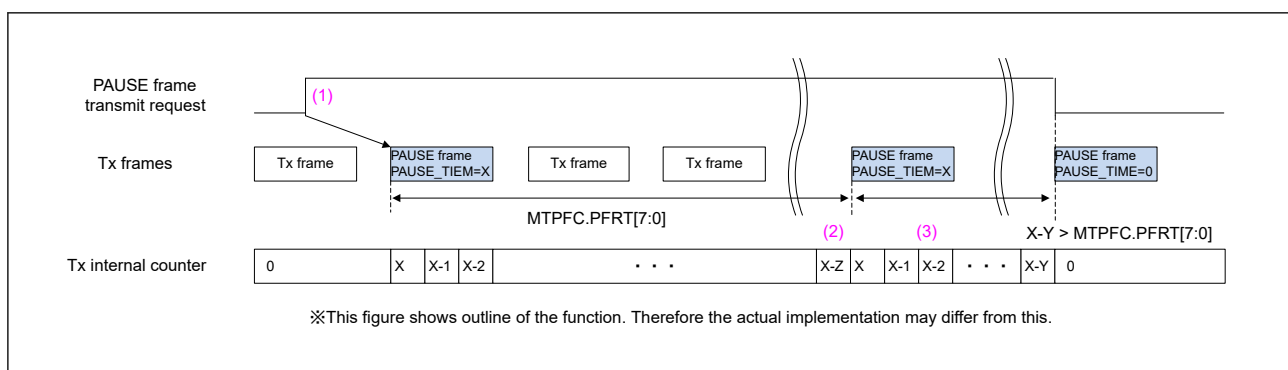


Figure 34.16 Transmission of PAUSE frame

34.6.2.1.2 Reception of PAUSE Frame

1. While MRGC.PFRC = 1, this IP receives a PAUSE frame and it starts internally counting a pausing-period specified by “PAUSE_TIME” field of a received PAUSE frame. While counting the pausing-period, notification of pause frame reception to an upper layer (e.g. MHD) and MRPFC.PTCA is asserted. And any transmissions except for a

Verify_packet, a Respond_packet, and a PAUSE frame are inhibited. When this IP receives a PAUSE frame while transmitting any frames, transmissions are inhibited after the transmission completion.

- When a PAUSE frame with “PAUSE_TIME $\neq$ 0” is received again before expiring the internal pausing-period, the internal counter is updated to that value.
- When the internal pausing-period is expired, notification of pause frame reception to an upper layer (e.g. MHD) and MRPFM.PTCA is deasserted.
- When MRGC.PFRTZ = 1 and PAUSE frame with “PAUSE_TIME = 0” is received before expiring the internal pause-period, the internal counter is cleared. Then notification of pause frame reception to an upper layer (e.g. MHD) and MRPFM.PTCA is deasserted.

In any case, MPFRCT is incremented after receiving a PAUSE frame in MRGC.PFRC = 1. And the received PAUSE frames are discarded internally.

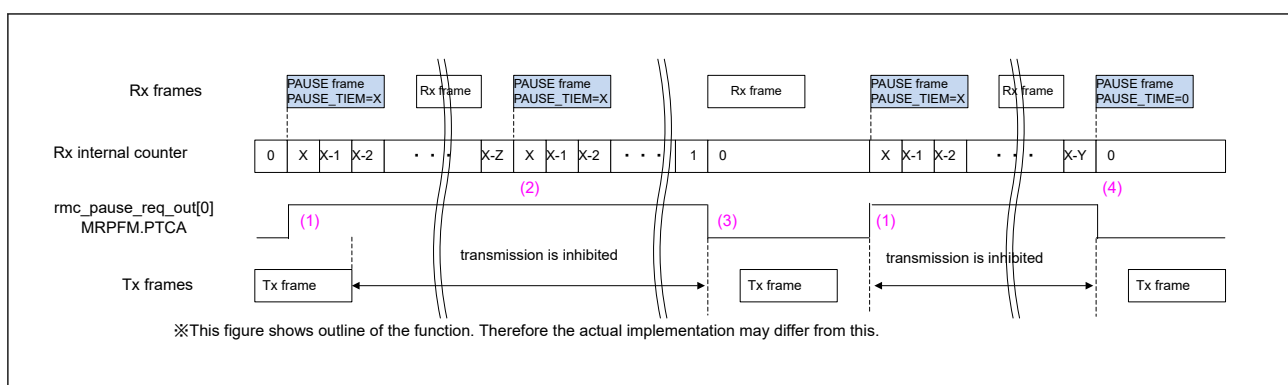


Figure 34.17 Reception of PAUSE frame

34.6.2.2 Flow Control by PFC Frame

This Flow Control is for a full-duplex operation and this consists of Transmission and Reception of a PFC frame.

34.6.2.2.1 Transmission of PFC Frame

When MTFFC.FCM = 1 and MTPFC.PT[15:0] $\neq$ 0, the transmission function of a PFC frame is enabled.

This IP treats several priorities as one priority-group to transmission of the PFC frame. The number of priority-group is 2. Each priority-group is configured by MTPFC30 and MTPFC31.

This IP transmits a PFC frame as described in Figure 34.18 for each priority-group by asserting a transmission request. The value of MTPFC.PT[15:0] is used for all “PAUSE_TIME” fields. “Priority_Enable” field value is different for each priority-group and the values of MTPFC30.PFCPG and MTPFC31.PFCPG are used for these. And MRMAC0 and MRMAC1 are used for “SA” field. When MTPFC30.PFCPG (or MTPFC31.PFCPG) is set to all 0, this IP does not transmit a PFC frame for the priority-group 0 (or priority-group 1) even when a transmission request is asserted.

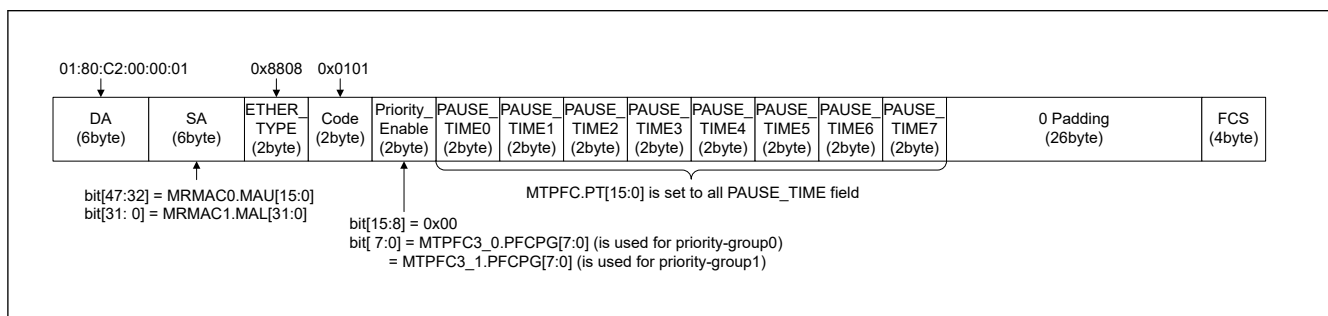


Figure 34.18 PFC frame Format

• Auto PFC Frame

When MTPFC.PFM = 0, this IP uses a hardware pause transmission request as a transmit request. The bit[0] is for priority-group0 and the bit[1] is for priority-group1. MAPCFTCT0 and MAPCFTCT1 are incremented respectively after transmitting a PFC Frame of each priority-group.

- Manual PFC Frame

When $MTPFC.PFM = 1$, this IP uses $MTPFC2.MPFCFRn$ as a transmit request. The bit[0] is for priority-group 0 and the bit[1] is for priority-group 1. $MMPCFTCT0$ and $MMPCFTCT1$ are incremented respectively after transmitting a PFC Frame of each priority-group.

If several transmit requests are asserted at same time, one PFC frame which has merged priority_enable information is transmitted.

If several transmit requests are asserted while transmitting other frames, an internal counter is cleared and one PFC frame which has merged priority_enable information is transferred.

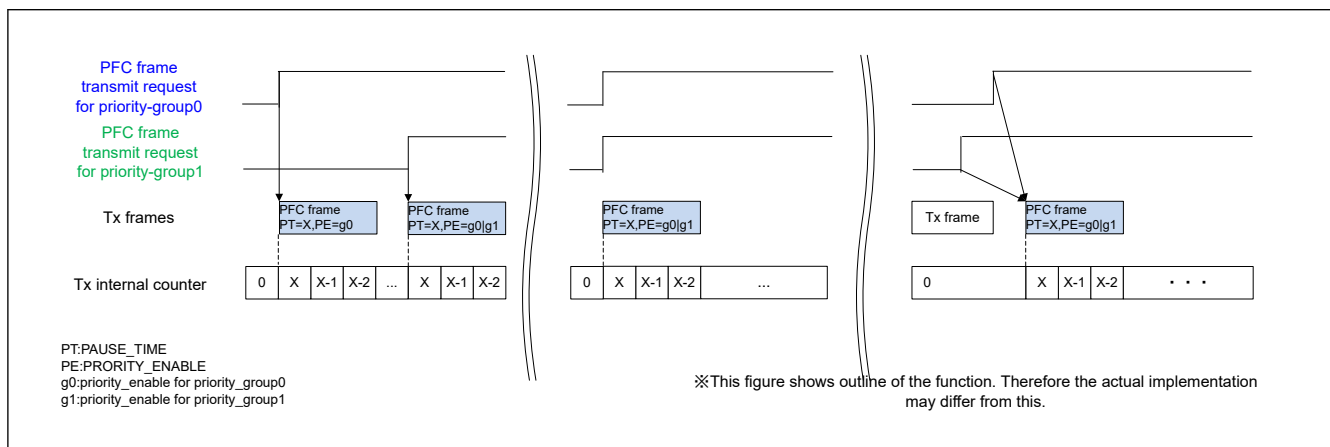


Figure 34.19 Transmitting PFC frame (part 1)

While the transmit request is asserted, a PFC frame corresponding to that request is continuously re-transmitted at a fixed interval configured by $MTPFC.PFRT$. When the number of re-transmissions for one of priority-groups reaches the value of $MTPFC.PFRLV$, this IP sets $MEIS.PFRROS$ to 1 for an interrupt. Even when the interrupt is detected, this IP continues re-transmission of a PFC frame while the transmit request is asserted.

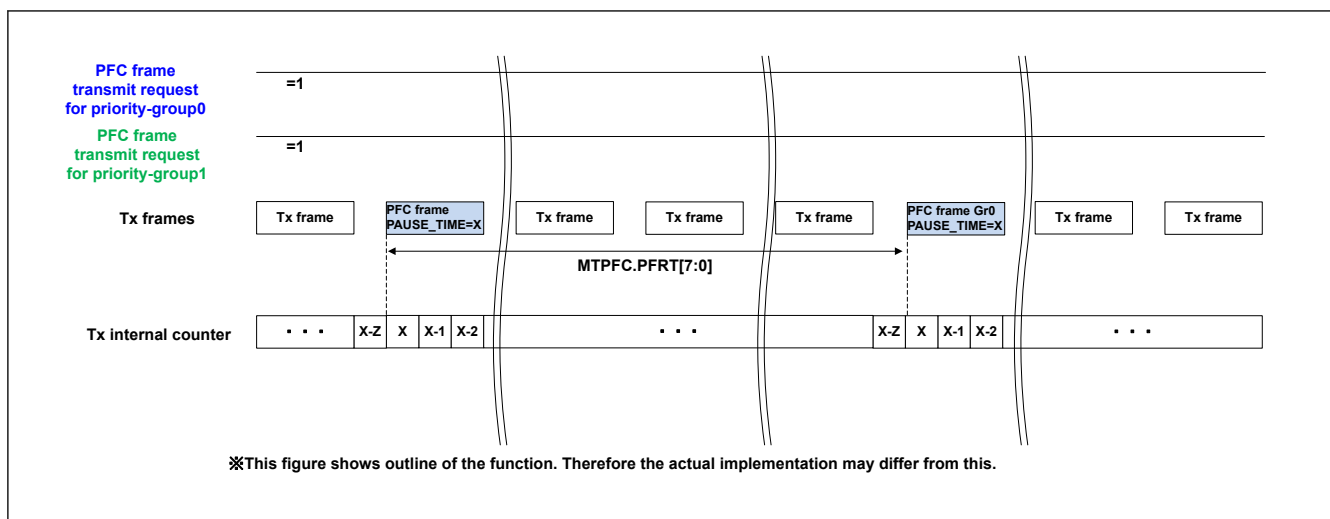


Figure 34.20 Transmitting PFC frame (part 2)

After deasserting the transmission request, this IP stops the transmission of the PFC frame corresponding to that request. At that time, if the corresponding bit of $MTPFC2.PFCTTZn$ for deasserted priority-group request is 1 and an internal counter to count the pausing-period exceeds $MTPFC.PFRT$ (*), this IP transmits one more PFC frame with all "PAUSE_TIME" fields as 0 for that priority-group.

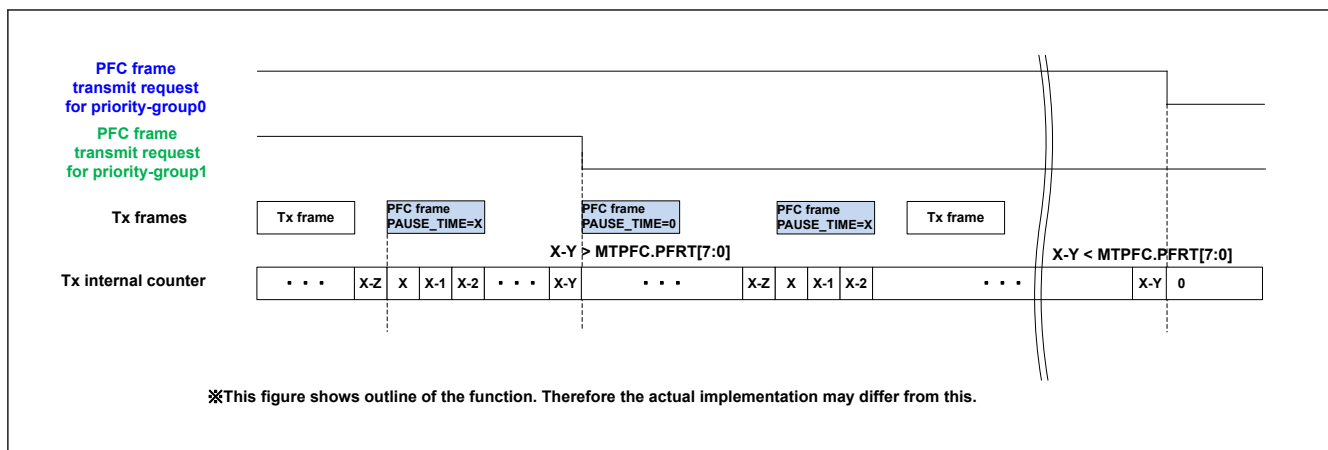


Figure 34.21 Transmitting PFC frame (part 3)

34.6.2.2.2 Reception of PFC Frame

This IP treats several priorities specified in received PFC frames. The number of supported priorities is 8.

1. When this IP receives a PFC frame with a priority-enable field which is corresponding to a priority configured by MRGC.PFCRCn, it starts internally counting a pausing-period for each priority respectively specified by each "PAUSE_TIME" fields of a received PFC frame. During counting the pausing-period, notification of PFC frame reception to an upper layer (e.g. MHD) and MRPFM.PFCTCAn corresponding to each priority are asserted. The bit number is corresponding to priority number.
2. When PFC frame with "PAUSE_TIME ≠ 0" with corresponding to valid "priority_enable" field is received again before expiring the internal pausing-period, the internal counter for that priority is updated to that value.
3. When MRGC.PFRTZ = 1 and PFC frame with "PAUSE_TIME = 0" with corresponding to valid "priority_enable" field is received before expiring the internal pause-period, the internal counter for that priority is cleared and notification of pause frame reception to an upper layer (e.g. MHD) and MRPFM.PFCTCAn for that priority is deasserted.
4. When the internal pausing-period is expired, notification of pause frame reception to an upper layer (e.g. MHD) and MRPFM.PFCTCAn for that priority are deasserted.

In any case, MPCFRCT0 to MPCFRCT7 are incremented respectively after receiving a PFC frame for enabled priority by MRGC.PFCRCn. And all received PFC frames are discarded internally.

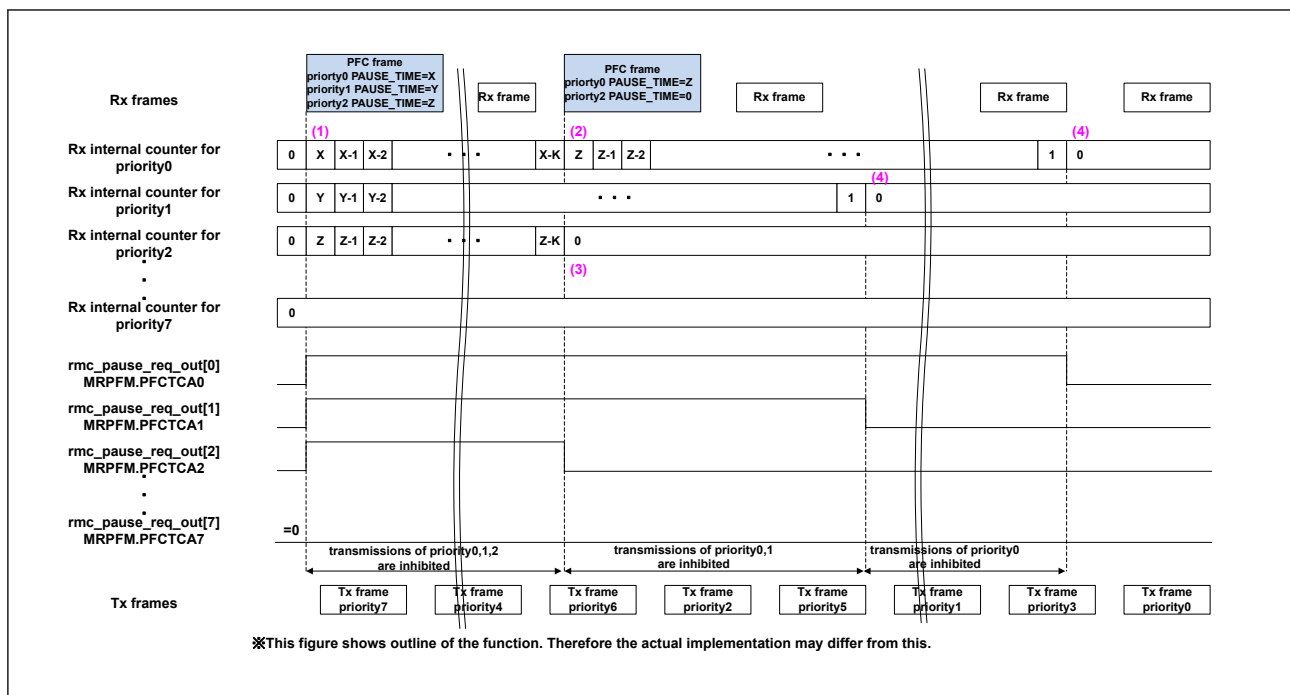


Figure 34.22 Reception of PAUSE frame

34.6.3 Timestamping

This IP supports time synchronization for IEEE 1588. It supports time synchronization accuracy in the sub-microsecond range.

34.6.3.1 Timestamp Capture

This IP can capture timestamps on both TX and RX side. gPTP timer value includes two 62bit timers, bit[61:0] is a Timer0, bit[123:62] is a Timer1.

[Timestamp capture on TX side]

“Timestamp capture” and “Timer number” fields in [GWCA] control this function for both e-frame and p-frame. These are configured by a Tx Descriptor. If “Timestamp capture” =1, the timer specified by “Timer number” field is captured.

The captured timestamp is sent to an upper module(e.g. MHD) with “Timestamp unique number” in [GWCA] via MHD Tx timestamp capture Interface.

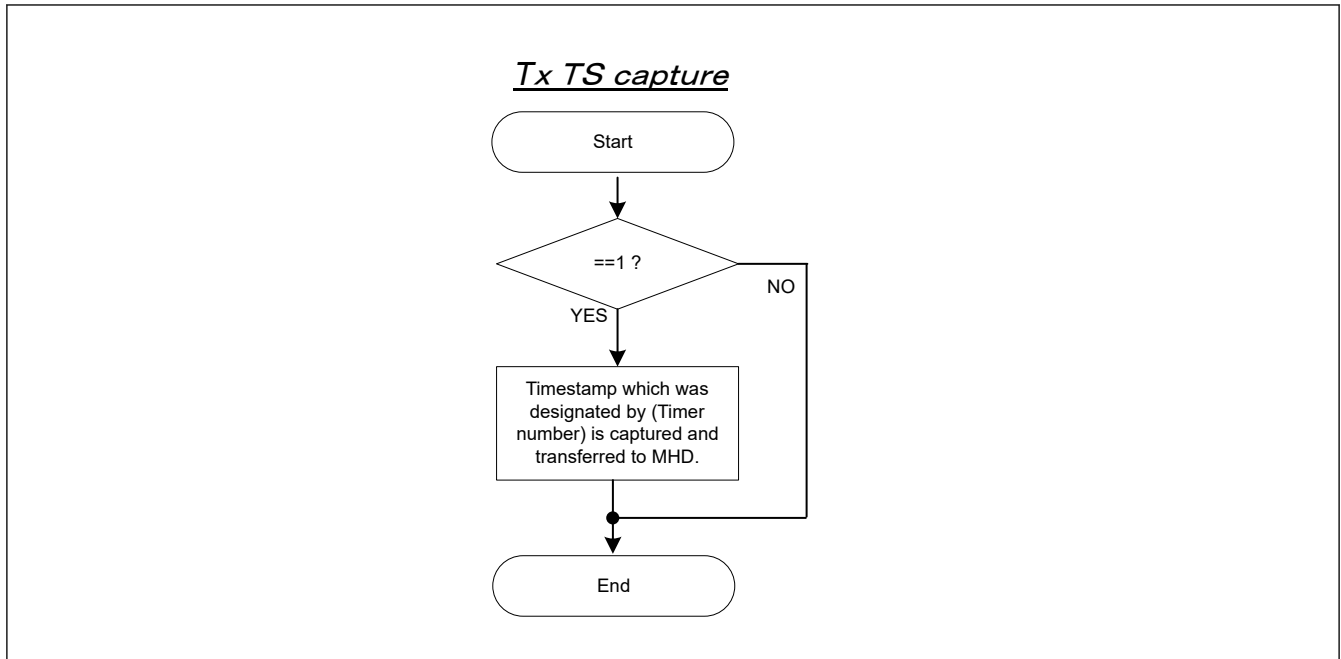


Figure 34.23 Tx Timestamp capture flow

Note: On the TX Side, the timestamp capture function by this IP and PHY can be enabled at same time.

[Timestamp capture on RX side]

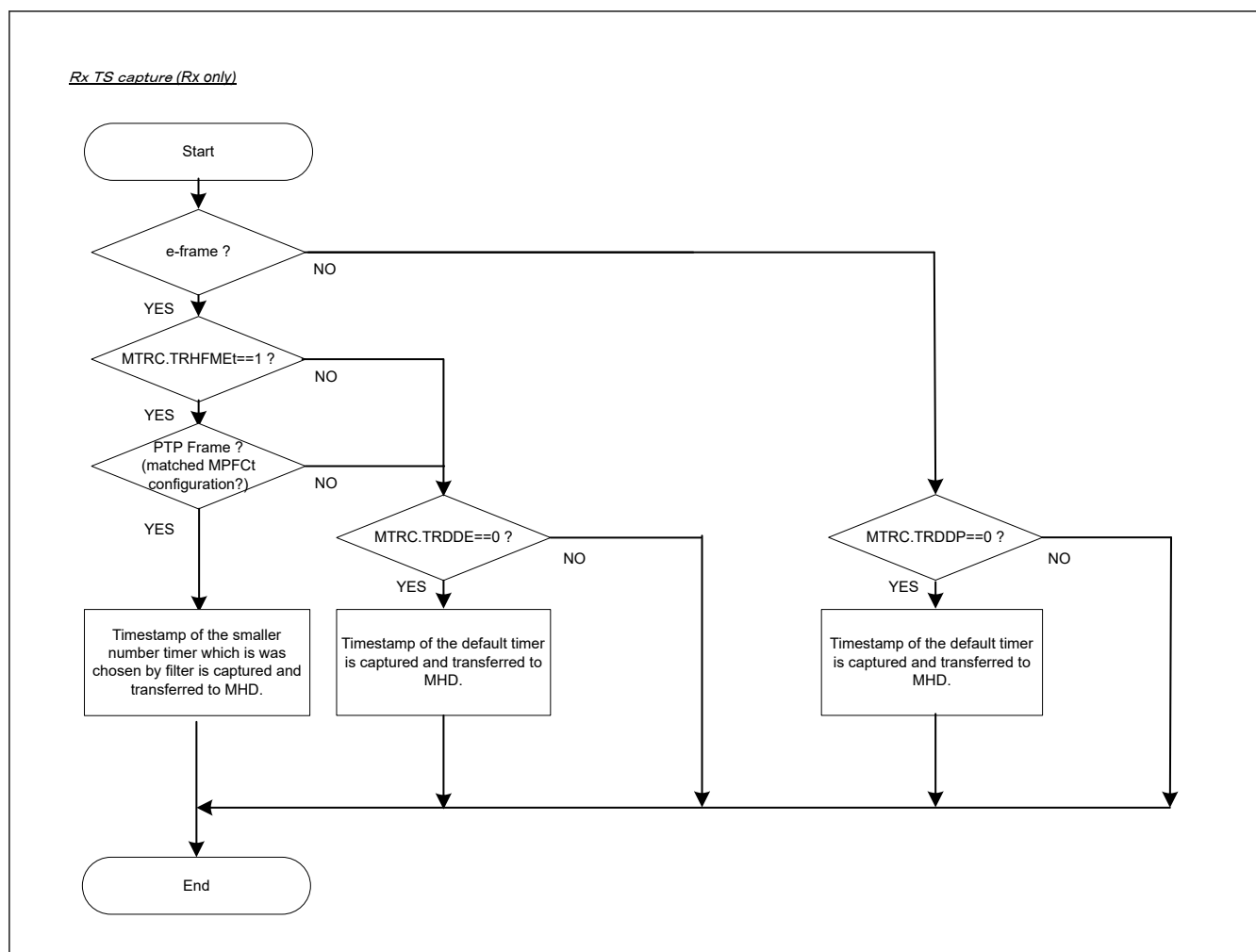
MTRC.TRDDE, MTRC.TRDDP, and MTRC.TRHFME_n control this function.

The configuration of the timestamp capture is following.

- MTRC.TRDDE controls this function for e-frames.
 - MTRC.TRDDE = 0
The default timer configured by MTRC.DTN is captured for a received e-frame.
- MTRC.TRDDP controls this function for p-frames.
 - MTRC.TRDDP = 0
the default timer configured by MTRC.DTN is captured for a received p-frame.
- MTRC.TRHFME_n controls this function for e-frames using the HW PTP message filtering.
The bit[0] is for Timer0 and bit[1] is for Timer1.
See [Figure 34.26](#) to use the HW PTP message filter.
 - TRHFME_n = x1 and the received e-frame matched the HW PTP messaged filter for Timer0
The Timer0 is captured for that received e-frames.
 - TRHFME_n = 10 and the received e-frame matched the HW PTP messaged filter for Timer1
The Timer1 is captured for that received e-frames.
 - TRHFME_n = 11 and the received e-frame matches the HW PTP messaged filter for Timer0 and 1
The Timer0 is captured for that received e-frames.

A captured timestamp is outputted from this IP with the received frame.

[Figure 34.25](#) shows Rx Timestamp capture flow.

**Figure 34.24 Rx TS capture (Rx only)**

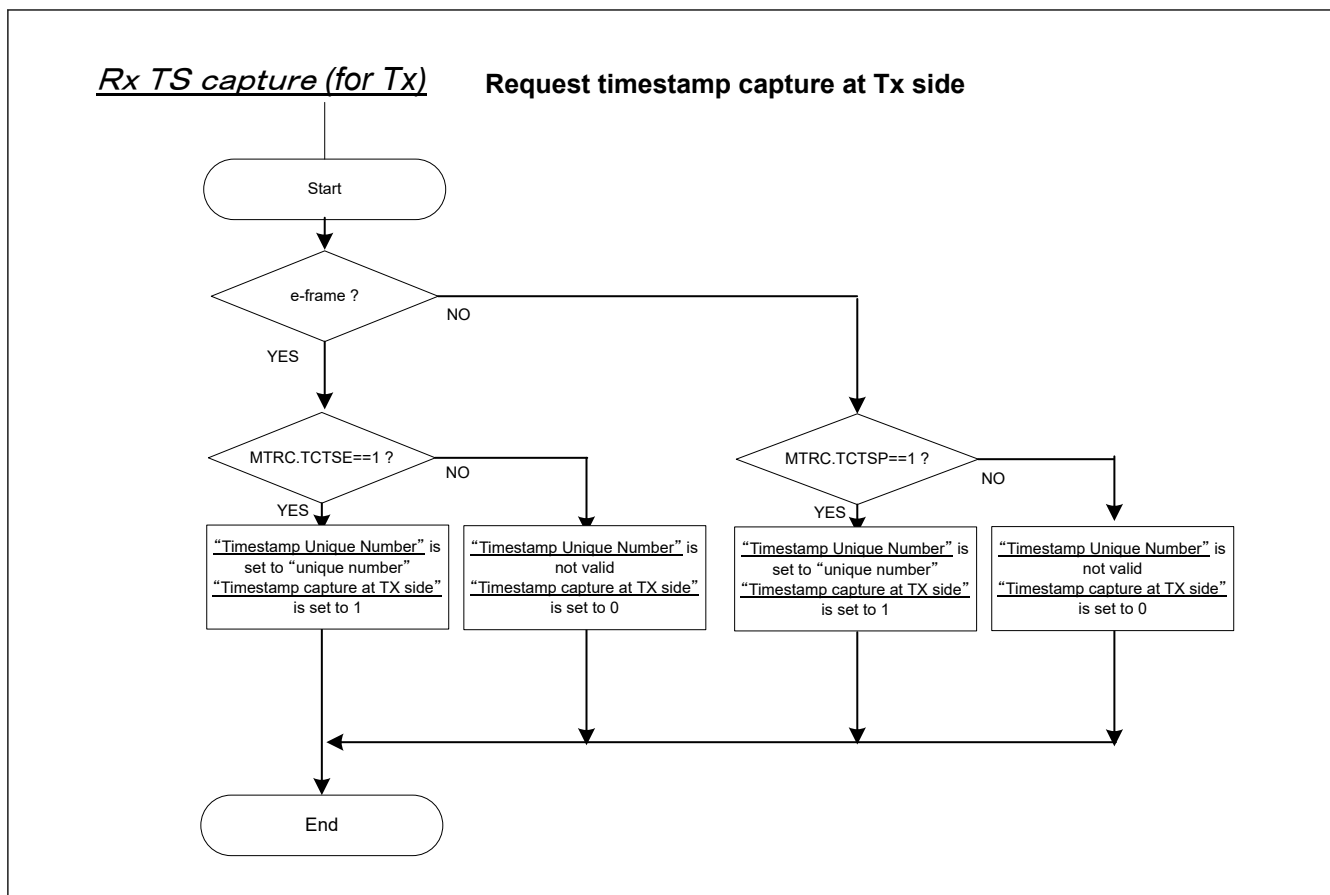


Figure 34.25 Rx timestamp capture flow

[Configuration of HW PTP Filter]

HW PTP message filter is configured by using MPFC0-15. The [Figure 34.26](#) shows an example configuration of PTP Message Filter. This is a case of MTRC.TRHFME_n = 11.

In this case:

MPFC0-7 are used as the filters for both Timer0 and Timer1. Because TEF field is set to 0x03.

MPFC8 is used as the filter for Timer0. Because TEF field is set to 0x01.

MPFC9 is used as the filter for Timer1. Because TEF field is set to 0x02.

MPFC10-15 are not used. Because TEF field is set to 0x00.

If an e-frame matches all MPFC0-7 and MPFC8, it is judged as a frame of Timer0 domain.

If an e-frame matches all MPFC0-7 and MPFC9, it is judged as a frame of Timer1 domain.

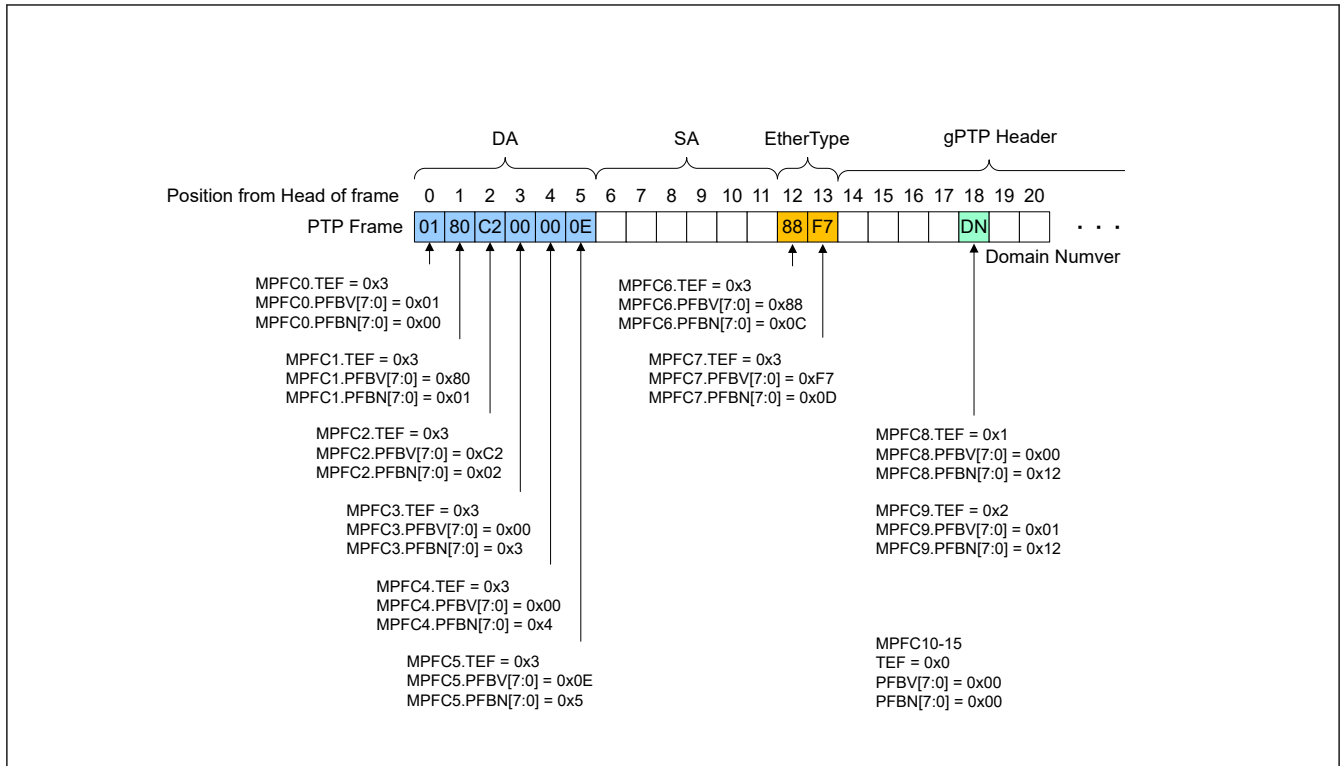


Figure 34.26 Example configuration of PTP message filter

- Note:
- Depending on the configuration of the filter, a frame may match both filters. In this case, Timer0 is selected.
 - PFBN should not select a position that exceeds the end of frame. Therefore, this value less than or equal to the minimum frame size is recommended.

34.6.4 Station Management

The Station Management controls the access to Management registers in PHY.

rmc_mdi, rmc_mdo, rmc_mde, rmc_mdc and rmc_mdio_int are used for this function.

34.6.4.1 Management Register Access

This IP has two types of access methods, one is specified by clause 22 the other is specified by clause 45 in IEEE802.3.

34.6.4.1.1 Access Method Specified in Clause 22

To write or read of Management register of PHY which supports this access method using management frame format, execute the following steps.

[write access]

Set MPSM as following.

PRD[15:0]="Write Data"

POP[1:0]=01

PRA[4:0]="Register Address"

PDA[4:0]="PHY Address"

MFF=0

PSME=1

After setting MPSM.PSME is kept to 1 during the access. Wait an interrupt is detected (MMIS1.PWACS=1) or MPSM.PSME turns to 0 as a sign of an access completion.

At this time, when MMIE1.PWACE is set to 1, the interrupt appears on rmc_mdio_int.

Set MMIS1.PWACS to 0 to clear the interrupt.

[Read access]

Set MPSM as following.

PRD[15:0]="0x00"

POP[1:0]=10

PRA[4:0]="Register Address"

PDA[4:0]="PHY Address"

MFF=0

PSME=1

After setting MPSM.PSME is kept to 1 during the access. Wait an interrupt is detected (MMIS1.PRACS=1) or MPSM.PSME turns to 0 as a sign of an access completion.

At this time, when MMIE1.PRACE is set to 1, the interrupt appears on rmc_mdio_int.

After the completion of read access, the read data is stored in MPSM.PRD[15:0]. So read it.

Set MMIS1.PRACS to 0 to clear the interrupt.

Note: When MPSM.PSME is set to 1, this IP outputs preamble code of fixed length specified in standard on rmc_mdo before write or read access. IEEE802.3 clause 22 allows skipping preamble code. When the preamble code is not needed and PHY also supports to skip the preamble code, this IP does not output the preamble code by setting MPIC.PSMDB to 1.

34.6.4.1.2 Access Method Specified in Clause 45

To write or read Management register of PHY which supports this access method using extended management frame format, execute the following steps.

[write access]

1. Set MPSM as following. ("Register Address" is set to a PHY)

PRD[15:0] = "Register Address"

POP[1:0] = 00

PRA[4:0] = "Device Address"

PDA[4:0] = "Port Address"

MFF = 1

PSME = 1

After setting MPSM.PSME is kept to 1 during the access. Wait an interrupt is detected (MMIS1.PAACS = 1) or MPSM.PSME turns to 0 as a sign of an access completion.

At this time, when MMIE1.PAACE is set to 1, the interrupt appears on rmc_mdio_int.

Set MMIS1.PWACS to 0 to clear interrupt.

2. Set MPSM as following. ("Write Data" is set to the "Register Address" which was configured by step 1.)

PRD[15:0] = "Write Data"

POP[1:0] = 01

PRA[4:0] = "Device Address"

PDA[4:0] = "Port Address"

MFF = 1

PSME = 1

After setting MPSM.PSME is kept to 1 during the access. Wait an interrupt is detected (MMIS1.PWACS = 1) or MPSM.PSME turns to 0 as a sign of an access completion.

At this time, if MMIE1.PWACE is set to 1, the interrupt appears on rmc_mdio_int.

Set MMIS1.PWACS to 0 to clear interrupt.

[Read access]

1. Set MPSM as following. ("Register Address" is set to a PHY)

PRD[15:0] = "Register Address"

POP[1:0] = 00

PRA[4:0] = "Device Address"

PDA[4:0] = "Port Address"

MFF = 1

PSME = 1

After setting MPSM.PSME is kept to 1 during the access. Wait an interrupt is detected (MMIS1.PAACS = 1) or MPSM.PSME turns to 0 as a sign of an access completion.

At this time, when MMIE1.PAACE is set to 1, the interrupt appears on rmc_mdio_int.

Set MMIS1.PWACS to 0 to clear interrupt.

2. Set MPSM as following. (Read access of the Register Address which was pointed by previous step)

PRD[15:0] = 0x00

POP[1:0] = 11

PRA[4:0] = "Device Address"

PDA[4:0] = "Port Address"

MFF = 1

PSME = 1

After setting MPSM.PSME is kept to 1 during the access. Wait an interrupt is detected (MMIS1.PRACS = 1) or MPSM.PSME turns to 0 as a sign of an access completion.

At this time, when MMIE1.PRACE is set to 1, the interrupt appears on rmc_mdio_int.

After the completion of read access, the read data is stored in MPSM.PRD[15:0]. So read it.

Set MMIS1.PRACS to 0 to clear interrupt.

[Post read incremental access]

Read access needs setting "Register Address" every time. Only when continuous Register Address fields are read continuously, using this access (POP[1:0] = 10) is efficient.

1. Set MPSM as following. ("Register Address" is set to a PHY)

PRD[15:0] = "Register Address"

POP[1:0] = 00

PRA[4:0] = "Device Address "

PDA[4:0] = "Port Address"

MFF = 1

PSME = 1

After setting MPSM.PSME is kept to 1 during the access. Wait an interrupt is detected (MMIS1.PAACS = 1) or MPSM.PSME turns to 0 as a sign of an access completion.

At this time, when MMIE1.PAACE is set to 1, the interrupt appears on rmc_mdio_int.

Set MMIS1.PWACS to 0 to clear the interrupt.

2. Set MPSM as following. (Read access of the Register Address which was pointed by previous step)

PRD[15:0] = 0x00

POP[1:0] = 10

PRA[4:0] = "Device Address "

PDA[4:0] = "Port Address"

MFF = 1

PSME = 1

After setting MPSM.PSME is kept to 1 during the access. Wait an interrupt is detected (MMIS1.PPRACS = 1) or MPSM.PSME turns to 0 as a sign of an access completion.

At this time, when MMIE1.PPRACE is set to 1, the interrupt appears on rmc_mdio_int.

After the completion of read access, the read data is stored in MPSM.PRD[15:0]. So read it. And "Register Address" set by step 1 is automatically changed in the PHY to point a next register address field. So don't need to re-configuration of "Register Address" for next read access.

Set MMIS1.PPRACS to 0 to clear the interrupt.

3. Repeat step 2 for the read access from the Next Register Address field.

Note: When MPSM.PSME set to 1, this IP outputs preamble code of fixed length specified in standard on rmc_mdo before write or read access. IEEE802.3 clause 45 does not allow skipping the preamble code. But skipping preamble code is also supported in this access method by setting MPIC.PSMDP to 1. To use this function, check a PHY specification.

34.6.4.2 Management Data Clock (MDC) Setting

To access Management registers in PHY, this IP outputs MDC. This is generated by dividing CHI clock (clk) in this IP and outputted from rmc_mdc. The division ratio can be set by the MPIC.PSMCS.

The following table shows the programmed MDC frequency depending on MPIC.PSMCS. MDC frequency is calculated by $\text{clk}[\text{MHz}] / ((\text{MPIC.PSMCS} + 1) * 2)$.

Table 34.7 Programming example for MPIC.PSMCS

clk frequency	MPIC.PSMCS	MDC frequency
100MHz	0x13	2.5MHz
200MHz	0x27	2.5MHz

Note: IEEE 802.3 specifies as the clock should be slower than 2.5MHz. But some PHY could operate with a higher frequency. See the PHY specification for detail.

34.6.4.3 Management Data Timing Adjustment

This IP can adjust the drive timing of MDO (rmc_mdo and rmc_mde) and the capture timing of MDI (rmc_mdi). It may be used for the propagation delay on the PCB.

The drive timing could be adjusted by PHY Station Management Hold Time Adjustment (MPIC.PSMHT). The capture timing could be adjusted by PHY Station Management Capture Time Adjustment (MPIC.PSMCT). The figure below is an example shifted by 2clk cycles.

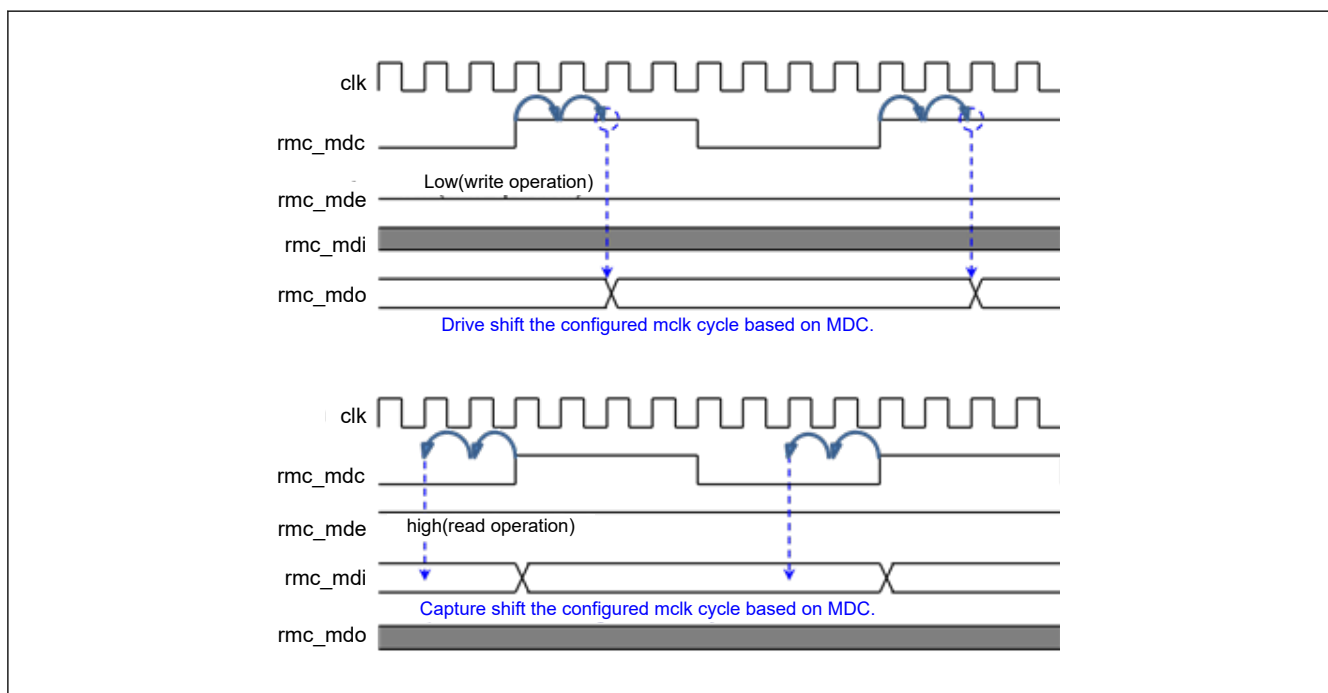


Figure 34.27 Management data timing adjustment

34.6.5 Energy Efficient Ethernet

This IP supports Energy Efficient Ethernet (EEE) which is standardized in IEEE 802.3. Tx-LPI mode and Rx-LPI mode can be activated independently.

34.6.5.1 LPI Control

This IP can request LPI mode for Tx port to the PHY when the LPI request bit (MEEEC.LPITR) is asserted. LPI mode is released when the bit is negated. The operation flow is shown in [section 34.5.2.5. LPI Transition for Tx Port](#) in detail.

The LPI mode for Rx port can be recognized by the reception LPI interrupt status bit (MMIS2.LPIAIS). When the bit (MMIS2.LPIDIS) is asserted, it means RMAC detected deassertion of LPI.

Table 34.8 and Table 34.9 show LPI specification on each xMII interface. The specification for MII or GMII is specified by IEEE802.3.

Table 34.8 Rx LPI specification on each xMII interface

PHY IF	RX_DV	RX_ER	RXD
MI	rmc_rx_valid_gmii = 0	rmc_rx_err_gmii = 1	rmc_rx_data_gmii[3:0] = 0001
GMII	rmc_rx_valid_gmii = 0	rmc_rx_err_gmii = 1	rmc_rx_data_gmii[7:0] = 00000001

Table 34.9 Tx LPI specification on each xMII interface

PHY IF	TX_EN	TX_ER	TXD
MI	rmc_tx_valid_gmii = 0	rmc_tx_err_gmii = 1	rmc_tx_data_gmii [3:0] = 0001
GMII	rmc_tx_valid_gmii = 0	rmc_tx_err_gmii = 1	rmc_tx_data_gmii [7:0] = 00000001

34.6.6 MAC Address Filtering on Reception

This IP supports the frame reception in a full-duplex operation mode.

This IP can pass all received Ethernet frames data to an upper layer via xMII Interface.

This IP strips off automatically SFDs/SMDs, preambles, FRAG_COUNTs and mCRCs from all the frames.

The FCS reception by the upper layer can be set though MRGC.RCPT register.

This IP identifies the frames with the special format and informs to the upper layer. It is indicated with the dedicated flag in status information followed payload data. The table below indicates the frames with the special format.

Table 34.10 Operation for special format frame

Frame	Format	Operation
Broadcast frame	DA: FF_FF_FF_FF_FF_FF	Filtering possible.
Multicast frame	DA: B'xxxx_xxx1_xxxx_xxxx_xxxx_xxxx_xxxx_xxxx_xxxx_xxxx_xxxx	Filtering possible.
Pause frame	DA: 01_80_C2_00_00_01 TYPE: 88_08 OPCODE: 00_01	This IP handles Flow Control, refer to section 8.2. Pause information is not sent to upper layer.
PFC frame	DA: 01_80_C2_00_00_01 TYPE: 88_08 OPCODE: 01_01	This IP handles Flow Control, refer to section 8.2. Pause information is not sent to upper layer.
Magic packet	When the MAC address (MRMAC.MA) is 00-11-22-33-44-55h, this IP would scan for the data sequence: Destination-Address Source-AddressMISC..... FF FF FF FF FF FF 00 11 22 33 44 55 00 11 22 33 44 55 00 11 22 33 44 55 00 11 22 33 44 55 00 11 22 33 44 55 00 11 22 33 44 55 00 11 22 33 44 55 00 11 22 33 44 55 00 11 22 33 44 55 00 11 22 33 44 55 00 11 22 33 44 55 00 11 22 33 44 55 00 11 22 33 44 55 00 11 22 33 44 55 00 11 22 33 44 55 00 11 22 33 44 55MISC..... FCS This sequence can be located anywhere in the Magic Packet payload. The Magic Packet is formed with a standard Ethernet header and FCS.	Interrupt. Provide information to upper layer. Provide magic packet interrupt (MMIS2.MPDIS). Magic packets are only detected as E-frame.

This IP supports the MAC address filtering for Unicast, Multicast, Broadcast, Broadcast storm (continuous Broadcast frame reception) and Multicast storm (continuous Multicast frame reception). It uses the destination address (DA) and source Address(SA) of a received frame for the address filtering. When the filter condition is satisfied, the received frames are transferred to an upper layer. If not satisfied, the received frames are transferred to the upper layer with the error flag of "Frame filtered" in [ETHA].

The filter condition is configured in Reception filter configuration register (MRAFC). It can be individually configured for Unicast address, Multicast address, and Broadcast address.

And this IP also supports Promiscuous mode.

When MRAFC.UCEN*, MCEN*, BCEN*, NDARE*, SDSFRE*, NSARE* and MSARE* are set to 1, all frames are transferred to the upper layer without error flag of “Frame filtered”.

Note : Regardless of MRAFC configuration, Verify and Respond are not transferred to the upper layer.

The flow of [Figure 34.28](#) shows the MAC Address filtering.

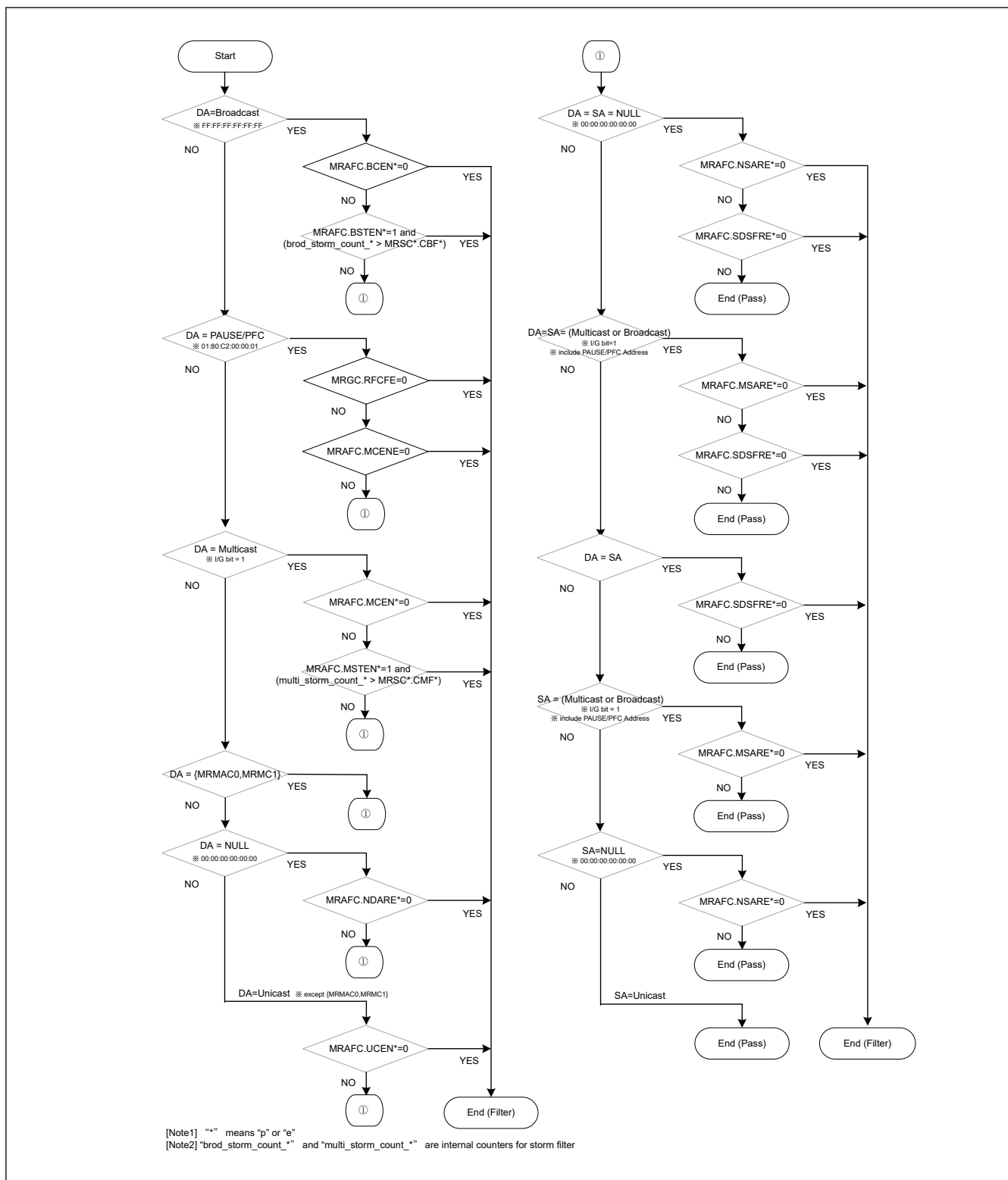


Figure 34.28 MAC address filtering